

Financial Management

Module 1: Scope of Financial Management

Module 1: Scope of Financial Management

Topic 001: Functions of Financial Management

Module 1: Scope of Financial Management

Functions of Financial Management

- Financing Activities
- Investing Activities
- Asset Management (Working Capital Management)
- Financial Analysis

Module 1: Scope of Financial Management

Topic 2: Financing Activities-I

- Sources of Financing
 - External Financing
 - Internal Financing
- Types of Financing
 - Equity Financing
 - Debt Financing
- Time Period of Financing
 - Long Term Financing
 - Equity
- Cost of Financing

Module 1: Introduction to Financial Management

Topic 3: Financing Activities-II

- Equity Financing
 - Common Shares
 - Features and Rights
 - Preferred Shares?
 - Types and Features
 - Dividend
 - Nature, needs and types
 - Profit (Internal Financing)

Module 1: Scope of Financial Management

Topic 4: Financing Activities-III

- Debt Financing
 - Types of Debts
 - Public Issue
 - Private Loans and Financing
 - Types of Interest and its Base
 - Fixed verses Floating Rate
 - KIBOR or Prime Rate
 - Financial Leverage

Module 1: Scope of Financial Management

Topic 5: Investing Activities-I

- Term of Investments
 - Long Term Investments
 - Short Term Investments
- Rate of Return
 - How much should we earn?
- Investing outside the Business
 - Stock Markets
 - Other Markets
 - Lending and other options

Module 1: Scope of Financial Management

Topic 6: Investing Activities-II

- Long Term Investments
 - Capital Budgeting
 - Discounted Cashflow Methods
- Marketable Investments
 - Capital Markets www.psx.com.pk
 - Money Markets
- Indirect Investment
 - Mutual Funds www.mufap.com.pk

Module 1: Introduction to Financial Management

Topic 7: Asset Management (Working Capital Management)-I

- Why Working Capital is Required?
- Major Components of Working Capital
 - Inventories/Stocks
 - Receivables
 - Cash
- Gross and Net Working Capital
- Working Capital Ratios
 - Current Ratio/Acid Test Ratio
 - Cash Cycle/Credit Cycle

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Topic 8: Asset Management (Working Capital Management)-II

- Inventory Management
 - Types of Inventories
 - Stock in Trade
 - Stores and Spares
 - Types of Stock in Trade
 - Raw Material
 - Work in Process
 - Finished Goods
 - Cost of Inventories
 - Carrying Cost
 - Ordering Cost

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Topic 9: Asset Management (Working Capital Management)-III

- Receivables Management
- Credit Policy
 - Credit Period
 - Cash Discount
 - Discount Period
 - Bad Debts
 - Collection Cost
 - Factoring Receivables
- Matching Receivables and Payables

Module 1: Scope of Financial Management

Topic 10: Asset Management (Working Capital Management)-IV

- Cash Management
- Cash Budget
 - Cash Inflows and Outflows
 - Surplus and Deficit
 - Financing and Lending
- Technical Insolvency
- Cash Flow Statement
- Cash Ratios

Module 1: Scope of Financial Management

Topic 11: Financial Analysis-I

- Why Financial Analysis?
- When it is done?
- Types of Financial Analysis
 - Absolute verses Relative Analysis
 - Comparative Analysis
 - Index Analysis
- Cost, Volume and Profit Analysis
- Risk Analysis

Module 1: Scope of Financial Management

Topic 12: Financial Analysis-II

- Ratio Analysis: Why Ratios?
 - Liquidity Analysis
 - Turnover or Activity Analysis
 - Leverage or Gearing Analysis
 - Coverage Analysis
 - Profitability Analysis
 - Market Analysis

END

Financial Management (Module 2)

Topic 013: Introduction to Financial Environment

Financial Environment

Introduction of Financial Environment

- Financial Environment: Where do Businesses Operate?
- Regulators
 - SECP
 - State Bank of Pakistan-SBP
 - Federal Board of Revenue-FBR
- Credit Providers
 - Commercial Banks
 - Investment Banks
 - Non-banking Financial Institutions
- Financial Markets

Module:2 (Financial Environment)

Topic 014: Central Bank

Central Bank

State Bank of Pakistan

- State Bank of Pakistan-SBP
 - Bank of Issue
 - Banker, Agent and Advisor to Government
 - Custodian of Cash Reserves
 - Custodian of Foreign Balances
 - Lender of Last Resort
 - Clearing House
 - Controller of Credit
 - Protection of Depositor's Interest

Module:2 (Financial Environment)

Topic 015: Securities and Exchange Commission of Pakistan

Securities and Exchange Commission of Pakistan

SECP

- Regulates Corporate Sector and Capital Markets
 - Support Investment
 - Economic Growth
 - Prosperity
- Regulatory Functions of SECP
 - Registration and Licensing
 - Supervision (Prudential and Regulatory)
 - Appellate Bench
 - Investors' Awareness and Education
 - Administration of Professionals (CAs, CMAs, CSs etc.)

Module:2 (Financial Environment)

Topic 016: Federal Board of Revenue (FBR)

Federal Board of Revenue

FBR

- Functions
 - Fiscal Policy
 - Taxation
 - Government Spending
 - Transfer Payments
 - Levy and Collect Taxes
 - Direct and Indirect Taxes
 - Appellate Tribunal

Module:2 (Financial Environment)

Topic 17: Pakistan Stock Exchange (PSX)

Pakistan Stock Exchange

PSX

- Capital and Money Markets
- Primary and Secondary Markets
- Economic Indicator
- Foreign Investment
- Pakistan Stock Exchange-PSX
 - Listing of Companies
 - Trading of Securities
 - Stock Market Indices
 - Data Portal
 - Guidance of Investors

Module:2 (Financial Environment)

Topic 18: Commercial Banks



Commercial Banks

Banking System

- Deposits: Checking and Saving Accounts
- Lending: Short Term and Long Term
- Financial Services: Transfer, Payments, Insurance, etc.
- Financial Products: CDs, Mortgages, Unsecured Financing, etc.
- Governed by State Bank of Pakistan

Module:2 (Financial Environment)

Topic 19: Development Finance Institutions

Development Finance Institutions

DFIs

- Funds for Economic Development Projects
- Developing and Emerging Economies
- Non-commercial Basis
- High Risk Loans, Equity Positions, Risk Guarantees etc.
- Types of DFIs
 - Multilateral Development Banks
 - Bilateral Development Institutions
 - National Development Institutions
 - Community Development Institutions
 - Microfinance Institutions

Module:2 (Financial Environment)

Topic 20: Non-Banking Financial Companies (NBFCs)

Non-Banking Financial Companies

NBFCs

- Not a Bank and Not Regulated like Banks
- Non-Banking Finance
 - Commercial Loans
 - Social and Community Lending
 - Joint Ventures
 - Partnerships
 - Factoring
 - Equity Financing
 - Crowd Funding
 - Government Funding

Non-Banking Finance Companies

- Major Forms of NBFIs
 - Development finance institutions
 - Insurance companies
 - Leasing companies
 - Investment companies
 - Modaraba companies
 - House finance companies
 - Venture capital companies
 - Currency exchanges
 - Microloan providers etc.

Module:2 (Financial Environment)

Topic 021: Islamic Banks



Islamic Banks

Islamic Banking

- Shariah Compliance
- Shariah Boards
- Profit and Loss Sharing and Prohibition of Riba
- Indexation
- Islamic Banking Instruments
 - Musharika
 - Modaraba
 - Ijara
 - Murabaha etc.

Module:2 (Financial Environment)

Topic 022: Foreign Banks



Foreign Banks

- What is a Foreign Bank?
- Governance of Foreign Banks
- Impact on Host Country
- Advantages
 - Efficiency of local banks
 - Sophisticated financial services
 - Support weak local banks
 - Foreign investment
 - Access to international capital markets
- Disadvantages
 - Growth of local banks
 - Loss of Control

Module 4: Valuation of Long-term Securities

Topic 053: Long Term and Short Term Securities

Long Term and Short Term Securities

Securities Definition

- The word security is used in different manners.
- In financial management it means, “A financial instrument that is interchangeable and has a monetary value”.
- ‘Financial Instrument’ is a term used for assets, which are a part of the Equity Capital or Debt Capital of company and are tradable.

Long Term and Short Term Securities

Securities

- In simple words a financial instrument is a piece of paper that has some monetary value.
- This piece of paper shows the claim of an owner on the assets of the company.

Long Term and Short Term Securities

Term of Securities

- A security is issued by an authority or a corporation.
- With this reference these are called 'Government Securities' and 'Corporate Securities'.
- Some securities are issued for less than one year time period called 'Short-term Securities', those for more than one year are called Long-term securities.

Long Term and Short Term Securities

Long Term Securities: Equity

- Stocks or Shares are equity securities and these are issued for long term i.e. more than one year. They get share of profit called Dividend.
- Stocks are further categorized as Common Stocks and Preferred Stocks.
- Stock repurchased by a company are called Treasury Stocks.

Long Term and Short Term Securities

Long Term Securities: Debt

- Bonds are the debt securities, which earn interest for the investor.
- Bond is a generic name and there are various other names and types of bonds.
- For example, Debentures, Term Finance Certificates, Participation Term Certificates, Sukuk etc.

Long Term and Short Term Securities

Short Term Securities

- All short-term securities are debt securities.
- These are the securities having a maturity value less than one year.
- Short term securities are mostly not accessible to individual investors.

Long Term and Short Term Securities

Term of Securities: Examples

- Treasury Bills (T-Bills)
- Promissory Notes
- Certificates of Deposit-CDs
- Bankers' Acceptance
- Repurchase Agreements-Repurchase

Module 4: Valuation of Long-term Securities

Topic 054: Concept of Valuation of Securities

Concept of Valuation

Valuation

- Oscar Wilde said, “What is a cynic? A person who knows the price of everything and value of nothing”.
- The term valuation refers to calculating the value of a security.
- Values are of different types, which are calculated from various viewpoints and reasons.

Concept of Valuation

Why to Value a Security?

- Securities are valued to make financial decisions.
- Authorities and Corporations when issuing securities need to know their value to set issue price (Book Building).
- Investors need to know the true value to make a good investment decision.

Concept of Valuation

Why to Value a Security?

- To assign a small value to the portions of Equity and Debt Capital.
- To determine the claim of an investor on the assets of the issuing company.
- So that a particular security is not 'Over Bought' or 'Over Sold'.
- To determine the growth potential of a company.

END

Module 4: Valuation of Long-term Securities

Topic 055: Various Values of Securities



Various Values of Securities

Values of Securities

- Face Value
- Par Value
- Market Value
- Book Value/Written Down Value
- Liquidation Value
- Going Concern Value
- Break-up Value
- Intrinsic Value

Various Values of Securities

Face Value

- The value written at the face of a security. This may be the value of one security if the certificate is issued for more than one security, then it is the value of all.

Various Values of Securities

Par Value

- When we divide the total capital into smaller pieces, the value which one piece is called Par Value.

Various Values of Securities

Market Value

- Market Value is the value of a security in the stock exchange or security market. It is also called the market price.

Various Values of Securities

Book Value

- The term Book Value or Written Down Value is also used with reference to Fixed Assets, meaning Cost minus Accumulated Depreciation.
- In case of stocks Book Value is the value of one share in accounting books calculated by dividing the Shareholder Equity with the number of outstanding shares.

Various Values of Securities

Liquidation Value

- Liquidation is a process of winding up a company. When the process starts the value of its securities falls drastically.
- The value of security when a company is going into liquidation is called Liquidation Value.

Various Values of Securities

Going Concern Value

- Going Concern means that the business is not likely to discontinue in the foreseeable future.
- The Value of securities, which is a Going Concern is called Going Concern Value and it is always more than the Liquidation Value.

Various Values of Securities

Break-up Value

- Break-up Value is the value of segments of a large company.
- When a company is not performing well it is desired that the company is divided into segments, so that the value profitable segments go up.

Various Values of Securities

Intrinsic Value

- The real value, the value on the basis of internal factors, the true value is called the Intrinsic Value.
- This value is calculated to make a good investment decision by comparing it with the market price.

END

Module 4: Valuation of Long-term Securities

Topic 056: Dividend Discount Model and its Assumptions

Dividend Discount Model and its Assumptions

Dividend Discount Model

- Dividends
- Discounting Dividends-PVs
- Determining Intrinsic Values on the basis of discount models
- Required rate of return (Opportunity Cost)

Dividend Discount Model and its Assumptions

Assumptions of Dividend Discount Model

- External factors are constant
- The earning power of a company remains constant
- Growth is the function of **Retention Ratio** only
- Dividend remains constant if all profit is paid as dividend
- Going Concern Assumption
- Required rate is always greater than the growth rate

END

Module 4: Valuation of Long-term Securities

Topic 057: Bond Valuation

Bond Valuation

Bonds: Fixed Income Securities

- Definition
- Meaning of Fixed Income
- Face Value/Par Value
- Maturity Period
- Maturity Value (Variations)
- Coupon (Interest Rate on Bonds)
- Required Rate (Discounting Rate)

Bond Valuation

Bonds Valuation

- How interest amount is calculated?
- Pattern of interest payment
- Discounting interest amounts
- Discounting maturity value
- Intrinsic value of Bonds

END

Module 4: Valuation of Long-term Securities

Topic 058: Perpetual Bond Valuation

Perpetual Bond Valuation

Perpetual Bonds

- Definition
- Face value of a perpetual bonds
- No maturity period
- Coupon rate
- Intrinsic value is the discounting of interest payments only

Perpetual Bond Valuation

Perpetual Bonds' Valuation Formula

- Present Value formula of a Perpetuity is used
- The Formula:
 - $PV = \text{Installment} / \text{Required Rate}$
 - $PV = R / r$

Time Line



Perpetual Bond Valuation

Example of Perpetual Bonds' Valuation

- A bond is issued forever, with a coupon rate of 10% p.a. and a face value of Rs. 5,000. The required rate is 15% p.a. Calculate the intrinsic value?
- Solution:
 - Interest = $5,000 \times 10\% = 500$
 - Value = Interest/Required Rate
 - $V = 500/15\%$
 - $V = \text{Rs. } 3,333.33$

Perpetual Bond Valuation

Investment Decision

- Intrinsic Value = Rs. 3,333.33
- Should be buy this bond at this price?
- Compare the Intrinsic Value and Market Price
 - Market price = Rs. 4,000 (No)
 - Market price = Rs. 3,000 (Yes)

END

Module 4: Valuation of Long-term Securities

Topic 059: Zero-Coupon Bond Valuation

Zero-Coupon Bond Valuation

Zero-Coupon Bonds

- Definition
- What does 'zero-coupon' mean
- Does it have a Maturity Period?
- Does it have a Coupon Rate?
- What do we discount to get the intrinsic value? (the MV)

Time Line



Zero-Coupon Bond Valuation

PV Formula to Calculate the Intrinsic Value

- Present Value formula of a Future Amount
- Why do we use this formula?
- The Formula:
 - $PV = FV / (1+r)^n$

Zero-Coupon Bond Valuation

Modified Formula to Calculate the Intrinsic Value

- The modified form of the formula:
 - $V = MV/(1+r)^n$
 - Where:
 - 'MV' is the maturity value of the bond
 - 'V' is the intrinsic value
 - 'r' is the discounting rate
 - 'n' is the number of years to maturity

Zero-Coupon Bond Valuation

Example of Zero-Coupon Bond Valuation

- A Zero-Coupon bond is issued at with a maturity value (face value) of Rs. 5,000 for 5-years. What is the intrinsic value of the bond is discounting rate is 12% p.a.?
 - $V = MV/(1+r)^n$
 - $V = 5,000/(1+12\%)^5$
 - $V = \text{Rs. } 2,837.13$

Zero-Coupon Bond Valuation

Investment Decision

- Intrinsic Value = Rs. 2,837.13
- Should be buy this bond at this price?
- Compare the Intrinsic Value and Market Price
 - Market price = Rs. 3,000 (No)
 - Market price = Rs. 2,500 (Yes)
 - Market price = Rs. 2,837.13 (?)

END

Module 4: Valuation of Long-term Securities

Topic 060: Bonds with a Definite Maturity: Coupon Bonds Valuation

Bonds with a Definite Maturity: Coupon Bonds Valuation

Time Line



Coupon Bonds

- Definition
- Coupon Bonds have a Coupon Rate
- Coupon Bonds have a definite Maturity
- To get the Intrinsic Value we discount both interest and the maturity value of the bond.

Bonds with a Definite Maturity: Coupon Bonds Valuation

PV Formulas to Calculate the Intrinsic Value

- Two Present Value formulas are needed:
 1. For interest
 2. For maturity value
- The formulas:
 - $PV = FV / (1+r)^n$ (for MV)
 - $PV = R \times \frac{1 - (1+r)^{-n}}{r}$ (for Int.)

Bonds with a Definite Maturity: Coupon Bonds Valuation

Example of PV of Coupon Bonds

- What is the intrinsic value of a coupon bond with Rs. 5,000 face value (maturity value), 10% coupon rate, and 5-year maturity? The discounting rate is 13% p.a.

Bonds with a Definite Maturity: Coupon Bonds Valuation

Example of PV of Coupon Bonds

- Solution: We have to separately calculate the two present values and add them up.
 - Interest = $5000 \times 10\% = 500$
 - $V = 5,000 / (1 + 13\%)^5$
 - $V = \text{Rs. } 2,713.80$
 - $V = 500 \times \frac{[1 - (1 + 13\%)^{-5}]}{13\%}$
 - $V = \text{Rs. } 1,758.62$
- Bond Value = $2713.8 + 1758.62$
- $V = \text{Rs. } 4472.42$

Bonds with a Definite Maturity: Coupon Bonds Valuation

Investment Decision

- Intrinsic Value = Rs. 4,472.42
- At what price we should buy this bond?
- Compare with the Market Price
 - Market price = Rs. 5,000 (No)
 - Market price = Rs. 4,000 (Yes)
 - Market price = Rs. 4472.42 (?)

END

Module 4: Valuation of Long-term Securities

Topic 061: Common and Preferred Stocks

Common and Preferred Stocks

Common Stocks

- Stocks issued to Common Shareholders
- Issued forever
- Treasury stock
- True owners: Equity holders
- Bear the risk of business
- Voting Rights
- First right to buy new shares
- Receive dividend
- Have last Right on the assets in case of Liquidation

Common and Preferred Stocks

Preferred Stocks

- Stocks issued with some preferences
- First right on dividend over common stocks
- Right of receipt on case of liquidation over common stocks
- Fixed rate of dividend
- Different from Bonds
- Usually do not have Voting Rights

END

Module 4: Valuation of Long-term Securities

Module 062: Types of Preferred Stocks-1

Types of Preferred Stocks-1

Features of Preferred Stocks

- Distinguishing features
 - Cumulation
 - Participation
 - Conversion
 - Redemption

Types of Preferred Stocks-1

Cumulative Preferred Stocks

- Dividend with fixed rate
- Is dividend payment to Preferred Shareholders a must?
- Dividend is contingent on earning
- Cumulation of unpaid dividend

Types of Preferred Stocks-1

Example of Cumulation of dividend of Preferred Stocks

- Preferred stocks of Rs. 100,000 issued @ 10%. In the year 2018 earned profit is Rs. 8,000.
- Solution:
 - Dividend of 2018 = Rs.10,000
 - Maximum profit available = Rs. 8,000
 - Amount carried forward to 2019 = Rs. 2,000
 - Dividend of 2019 = 10,000 + 2,000 = 12,000

Types of Preferred Stocks-1

Participative Preferred Stocks

- Participation with common shareholders in remaining profit
- Rate of participation
- Conditions of participation

Types of Preferred Stocks-1

Example of Participation of Preferred Stocks

- Preferred stocks of Rs. 100,000 issued @ 10%, earned profit is Rs. 40,000. If the company's profit goes above Rs. 30,000, preferred shareholders will get 2% extra dividend.

Types of Preferred Stocks-1

Example of Participation of Preferred Stocks

- Solution:
 - Normal dividend = Rs. 10,000
 - Extra dividend = $100,000 \times 2\% = 2,000$
 - Total dividend = 12,000

END

Module 4: Valuation of Long-term Securities

Topic 063: Types of Preferred Stocks-2

Types of Preferred Stocks-2

Convertible Preferred Stocks

- Convertible to common shares
- Terms of conversion
 - Time, ratio, option, etc.
- Advantage of conversion
 - Risk level, ownership rights etc.
- **Conversion Ratio** (No. of common shares against a preferred share)
 - E.g. 5 common shares for one preferred share

Types of Preferred Stocks-2

Convertible Preferred Stocks: Example

- Market conversion price
 - Purchase Price of the PS/Conversion Ratio
 - $\text{Rs. } 100/5 = \text{Rs. } 20$
- If common share price is Rs. 15,
loss on conversion = Rs. - 5
- If common share price is Rs. 23,
gain on conversion = Rs. 3

Types of Preferred Stocks-2

Convertible Preferred Stocks Premium

- Premium is the probable financial gain due to Convertibility
- Premium increases the market price of preferred stocks

Types of Preferred Stocks-2

Redeemable/Callable Preferred Stocks

- Redemption/Call
- Terms of redemption
 - Time, price, choice/option etc.
- Advantages of redemption to issuer and shareholder
- Should the price of callable preferred shares be less?
- Some PS are mandatorily redeemable PS

END

Module 4: Valuation of Long-term Securities

Topic 064: Preferred Stocks Valuation

Preferred Stocks Valuation

Concept of Valuation

- Intrinsic value-IV of preferred stocks
- IV is the present value of future cash inflows
- Assumptions of valuation:
 - Non-redeemable
 - Non-convertible
 - Non-participating
 - Sufficient earnings

Preferred Stocks Valuation

The Formula

- Dividends of PS are constant and infinite
- Present Value of Perpetuity formula is used
 - $PV = \text{Installment} / \text{Rate}$
 - $V = \text{Dividend} / \text{Rate}$
 - $V = D / r$
- Required rate of return

Preferred Stocks Valuation

Example of PS Valuation

- Par value = Rs. 100
- Market Price = Rs. 80
- Dividend rate = 13% per annum
- Required rate = 20% per annum
- Solution:
 - Dividend = $100 \times 13\% = 13$
 - $V = D/r$
 - $V = 13/20\%$
 - $V = \text{Rs. } 65$

Preferred Stocks Valuation

Analysis and Decision

- Market Price-MP is higher
- Required rate is more than dividend rate
- IV is equal to MP if:
 - Dividend Rate = Required Rate
- **Decision:** As IV is less than MP so do NOT BUY

END

Module 4: Valuation of Long-term Securities

Topic 065: Common Stocks Valuation Models

Common Stocks Valuation Models

The Models of Valuation

- Dividend Discount Model-DDM
- Three Models of Valuation:
 - No Growth Model
 - Constant Growth Model
 - Growth in Stages Model

Common Stocks Valuation Models

What Different Models of Valuation?

- Valuation is based on the cash inflow (i.e. Dividend)
- Three possibilities:
 - All earnings are paid as dividend
 - Retention ratio is constant
 - Retention ratio changes

END

Module 4: Valuation of Long-term Securities

Topic 066: No Growth Model

No Growth Model

No Growth Model (Constant Dividend)

- Assumptions:
 - All earnings are paid as dividend
 - External factors remain constant
 - Earning capacity remains constant
 - Growth is the function of retention only

No Growth Model

No Growth Model

- Dividends remains constant
- Dividends are paid till infinity
- PV formula of Perpetuity
 - $PV = \text{Installment} / \text{Rate}$
- IV formula:
 - $V = D / r$

No Growth Model

Example of No Growth Model

- Par value = Rs. 10
- Market Price = Rs. 11
- Last Dividend D_0 = Rs. 2 per share
- Required rate = 17% per annum
- Solution:
 - $V = D/r$
 - $V = 2/17\%$
 - $V = \text{Rs. } 11.75$

No Growth Model

Analysis and Decision

- Last dividend will be the future dividend
- Market Price-MP is higher
- **Decision:** As MP is higher than the IV so do NOT BUY

END

Module 4: Valuation of Long-term Securities

Topic 067: Constant Growth Model

Constant Growth Model

Constant Growth Model (Gordon's Growth Model)

- Assumptions:
 - Going Concern
 - DPO remains constant
 - External factors remain constant
 - ROI remains constant
 - Growth is the function of retention only

Constant Growth Model

Constant Growth Model

- The dividend grows with a constant rate
- Example:
 - Year 1 = Rs. 1
 - Year 2 = Rs. 1.1 (+10%)
 - Year 3 = Rs. 1.21 (+10%)
 - Year 4 = Rs. 1.331 (+10%)
 - Year 5 = Rs. 1.4641 (+10%)

Constant Growth Model

Formula: Constant Growth Model

- Reduced form of PVs
 - $V = D_1 / (r - g)$
 - Where D_1 = next year's dividend
 - $D_1 = D_0(1+g)$
 - g = growth rate
 - r = required rate of return
 - $r > g$

Constant Growth Model

Example Constant Growth Model

- Last year's dividend was Rs. 10, growth rate of dividend is 7%, and required rate is 11%. What is the intrinsic value?
- Solution:
 - $V = D_1 / r - g$
 - $D_1 = D_0(1+g) = 10(1+7\%) = 10.7$
 - $V = 10.7 / 11\% - 7\%$
 - $V = \text{Rs. } 267.5$

Constant Growth Model

Analysis and Decision

- Purchase at fair or better price
- Purchase stock if:
 - Market price = Rs. 267.5
 - Market price less than Rs. 267.5
- Do not purchase if price is greater than Rs. 267.5

END

Module 4: Valuation of Long-term Securities

Topic 068: Two-Stage Growth Model-I

Two-Stage Growth Model-I

Two-Stage Growth Model

- Assumptions:
 - All assumptions of DDM
 - Going Concern
 - DPO changes once
 - External factors remain constant
 - ROI remains constant
 - Growth is the function of retention only

Two-Stage Growth Model-I

Two-Stage Growth Model Calculations

- V_1 = Value of Stage-I
- V_2 = Value of Stage-II
- $V = V_1 + V_2$
- Stage-I = Individual growth formula
- Stage-II = Constant Growth formula

Two-Stage Growth Model-I

Stage One

- The dividend grows with a **constant rate 'g₁'** e.g. $g_1 = 10\%$
- The stage is **finite** e.g. 5,6,...10 years
- Example:
 - Year 1 = Rs. 10
 - Year 2 = Rs. 11 (+10%)
 - Year 3 = Rs. 12.1 (+10%)
 - Year 4 = Rs. 13.31 (+10%)
 - Year 5 = Rs. 14.641 (+10%)

Two-Stage Growth Model-I

Stage Two

- Growth rate changes to ' g_2 ' e.g. 9%
- The second stage starts after the first stage
- The second stage goes till infinity
- Constant Growth formula is applied:
 - $V_2 = D_{n+1}/r - g_2$
- The Value ' V_2 ' is further discounted

END

Module 4: Valuation of Long-term Securities

Topic 069: Two-Stage Growth Model-II

Two-Stage Growth Model-II

Example Two-Stage Growth Model

- $D_0 = \text{Rs. } 5$
- $g_1 = 7\%$ per annum (year 1-5)
- $g_2 = 6\%$ per annum (year 5- ∞)
- $r = 10\%$ per annum
- Value = ?

Two-Stage Growth Model-II

Example Two-Stage Growth Model

- Stage-I (Dividend Calculations $D_1, D_2, D_3, D_4 \dots$)
 - Year 1=Rs.5 (1+7%) = 5.35
 - Year 2=Rs. 5.35(1+7%)=5.73*
 - Year 3=Rs. 5.73(1+7%) = 6.13
 - Year 4=Rs. 6.13(1+7%) = 6.56
 - Year 5=Rs. 6.56(1+7%) = 7.02
- * Rounded to nearest Rupee

Two-Stage Growth Model-II

Example Two-Stage Growth Model

- Stage-I (PVs of Dividends)
 - Year 1 = $5.35 / (1 + 10\%) = 4.86$
 - Year 2 = $5.73 / (1 + 10\%)^2 = 4.74$
 - Year 3 = $6.13 / (1 + 10\%)^3 = 4.61$
 - Year 4 = $6.56 / (1 + 10\%)^4 = 4.48$
 - Year 5 = $7.02 / (1 + 10\%)^5 = 4.36$
- $V_1 = [4.86 + 4.74 + 4.61 + 4.48 + 4.36]$
- $V_1 = \text{Rs. } 23.02$

Two-Stage Growth Model-II

Example Two-Stage Growth Model

- Stage-II
 - $V_2 = D_{5+1}/r - g_2$
 - $V_2 = D_6/10\% - 6\%$
 - $D_6 = D_5(1+g_2) = 7.02(1+6\%) = \text{Rs. } 7.44$
 - $V_2 = 7.44/4\%$
 - $V_2 = \text{Rs. } 186/(1+10\%)^5 = \mathbf{115.49}$
- The Value ' V_2 ' is further discounted

Two-Stage Growth Model-II

Example Two-Stage Growth Model

- $V = V_1 + V_2$
- $V = 23.02 + 115.49$
- $V = \text{Rs. } 138.51$

Two-Stage Growth Model-II

Analysis and Decision

- Purchase at fair or better price
- Purchase stock if:
 - Market price = **Rs. 138.51**
 - Market price less than **Rs. 138.51**
- Do not purchase if price is greater than **Rs. 138.51**

END

Module 4: Valuation of Long-term Securities

Topic 070: Three-Stage Growth Model-I

Three-Stage Growth Model-I

Three-Stage Growth Model

- Assumptions:
 - All assumptions of DDM
 - Going Concern
 - **Plough Back-PB Ratio** changes **twice**
 - External factors remain constant
 - Earning power remains constant
 - Growth is the function of PB

Three-Stage Growth Model-I

Three-Stage Growth Model Calculations

- V_1 = Value of Stage-I
- V_2 = Value of Stage-II
- V_3 = Value of Stage-III
- $V = V_1 + V_2 + V_3$
- Three growth rates

Three-Stage Growth Model-I

Stage One

- The same as Two-stage model;
constant rate 'g₁' e.g. $g_1 = 11\%$
- The stage is **finite**
- Example:
 - Year 1 = Rs. 10
 - Year 2 = Rs. 11.10 (+11%)
 - Year 3 = Rs. 12.32 (+11%)

Three-Stage Growth Model-I

Stage Two

- Growth rate changes to ' g_2 ' e.g. 10%
- The second stage starts after the first stage
- The second stage is also **finite**
- Individual Growth formula:
 - Year 3 = Rs. 12.32 (+11%)
 - Year 4 = Rs. 13.55 (+10%)
 - Year 5 = Rs. 14.91 (+10%)

Three-Stage Growth Model-I

Stage Three

- Growth rate changes to 'g₂' e.g. 8%
- The third stage starts after the second stage
- The **third** stage goes till **infinity**
- Constant Growth formula is applied:
 - $V_3 = D_{n+1}/r - g_3$
- The Value 'V₃' is further discounted

END

Module 4: Valuation of Long-term Securities

Topic 072: Three-Stage Growth Model-II

Three-Stage Growth Model-II

Example Three-Stage Growth Model

- $D_0 = \text{Rs. } 5$
- $g_1 = 7\%$ per annum (year 1-3)
- $g_2 = 6\%$ per annum (Year 4-5)
- $g_3 = 5\%$ per annum (year 5- ∞)
- $r = 10\%$ per annum
- Value = ?

Three-Stage Growth Model-II

Example Two-Stage Growth Model

- Stage-I (Dividend Calculations for 3-years)
 - $D_1 = \text{Rs. } 5 (1+7\%) = 5.35$
 - $D_2 = \text{Rs. } 5.35(1+7\%) = 5.73$
 - $D_3 = \text{Rs. } 5.73(1+7\%) = 6.13$

Three-Stage Growth Model-II

Example Three-Stage Growth Model

- Stage-I (PVs of Dividends)
 - $PV-D_1 = 5.35 / (1+10\%) = 4.86$
 - $PV-D_2 = 5.73 / (1+10\%)^2 = 4.74$
 - $PV-D_3 = 6.13 / (1+10\%)^3 = 4.61$
- $V_1 = [4.86 + 4.74 + 4.61]$
- $V_1 = \text{Rs. } 14.21$

Three-Stage Growth Model-II

Example Two-Stage Growth Model

- Stage-II (Dividend Calculations for 2-years)
 - $D_4 = \text{Rs. } 6.13(1+6\%) = 6.50$
 - $D_5 = \text{Rs. } 6.50(1+6\%) = 6.89$

Three-Stage Growth Model-II

Example Three-Stage Growth Model

- Stage-II (PVs of Dividends)
 - $PV-D_4 = 6.50/(1+10\%)^4 = \mathbf{4.44}$
 - $PV-D_5 = 6.89/(1+10\%)^5 = \mathbf{4.28}$
- $V_2 = [4.44+4.28]$
- $V_2 = \mathbf{Rs. 8.72}$

Three-Stage Growth Model-II

Example Two-Stage Growth Model

- Stage-III
 - $V_3 = D_{5+1}/r - g_3$
 - $V_3 = D_6/10\% - 5\%$
 - $D_6 = D_5(1+g_2) = 6.89(1+5\%) = \text{Rs. } 7.24$
 - $V_3 = 7.24/5\%$
 - $V_3 = \text{Rs. } 144.8/(1+10\%)^5 = \mathbf{89.91}$
- The Value ' V_3 ' is further discounted

Three-Stage Growth Model-II

Example Two-Stage Growth Model

- $V = V_1 + V_2 + V_3$
- $V = 14.21 + 8.72 + 89.91$
- $V = \text{Rs. } 112.84$

Three-Stage Growth Model-II

Analysis and Decision

- Purchase at fair or better price
- Purchase stock if:
 - Market price = **Rs. 112.84**
 - Market price less than Rs. **Rs. 112.84**
- Do not purchase if price is greater than **Rs. 112.84**

END

Module 5: Financial Analysis Tools

Topic 073: Introduction to Financial Statements-1

Introduction to Financial Statements-1

Financial Statements

1. Statement of Financial Position
2. Statement of Comprehensive Income
3. Statement of Cashflows
4. Statement of Changes in Owners' Equity
5. Notes to the Accounts

Introduction to Financial Statements-1

Statement of Financial Position

1. Assets

- I. Operating Fixed Assets
- II. Non-operating Fixed Assets
- III. Non-Current Assets
- IV. Current Asset

2. Liabilities

- I. Long term liabilities
- II. Current liabilities

Introduction to Financial Statements-1

Statement of Financial Position

3. Owner's Equity

- I. Issued, Subscribed, and Paid-up Capital
- II. Addition Capital
- III. Reserves
- IV. Retained Earnings

Module 5: Financial Analysis Tools

Topic 074: Introduction to Financial Statements-2

Introduction to Financial Statements-2

Statement of Comprehensive Income

1. Revenue/Income/Sales
2. Cost of Sales
3. Gross Profit
4. Operating Expenses
5. Net Operating Profit
6. Extraordinary and Other Income

Introduction to Financial Statements-2

Statement of Comprehensive Income

1. Income Tax
2. Earnings before Tax-EBT
3. Earnings after Tax-EAT
4. Earnings before Interest and Tax-EBIT
5. Earnings before Interest Tax and Depreciation-EBITD

Module 5: Financial Analysis Tools

Topic 075: Introduction to Financial Statements-3

Introduction to Financial Statements-3

Statement of Cashflows

1. Cashflow and Profit
2. Cashflow from Operating Activities
3. Cashflow from Investing Activities
4. Cashflow from Financing Activities

Introduction to Financial Statements-3

Statement of Changes in Owners' Equity

1. Parts of Shareholders' Equity
 1. Issued, Subscribed, and Paid up Capital
 2. Additional Capital (Share Premium)
 3. Reserves (General and Specific)
 4. Retained Profits
2. Dividend Declared
3. Profit Earned
4. Adjustments

Module 5: Financial Analysis Tools

Topic 076: Financial Analysis Introduction

Financial Analysis Introduction

Financial Analysis

1. Definition
2. Data for Financial Analysis
3. Absolute Analysis
4. Relative Analysis

Financial Analysis Introduction

Relative Analysis

1. Ratio Analysis
2. Index Analysis
3. Differential Analysis
4. Common Size Analysis
5. Trend Analysis
6. Risk Analysis

Module 5: Financial Analysis Tools

Topic 077: Relative Analysis using Ratios

Financial Analysis Introduction

Ratios

1. Ratios Definition
2. Financial Ratios
3. How Ratios Help?
4. Why Relative Analysis?
5. Who Should do Ratio Analysis?

Module 5: Financial Analysis Tools

Topic 78: Types of Ratios



Types of Ratios

Areas of Financial Analysis

1. Liquidity
2. Activity or Turnover
3. Leverage or Gearing
4. Coverage
5. Profitability
6. Market Analysis

Module 5: Financial Analysis Tools

Topic 79: Benchmarking the Ratios



Benchmarking the Ratios

Users of Financial Ratios

1. Company's Management
2. Lenders and Creditors
3. Shareholders
4. Regulatory Bodies
5. Potential Business Partners
6. Customers
7. Analysts

Benchmarking the Ratios

Benchmarks for Ratios

1. Past performance/Trends
2. Company's average
3. Industry's average
4. General standards
5. Standards of covenants
6. Target ratios:
 1. Mergers
 2. Financing
 3. Spin offs/Divestiture
 4. Going Public

Module 5: Financial Analysis Tools

Topic 80: Liquidity Analysis-1

Liquidity Analysis-1

Liquidity

1. Definition
2. What are liquid assets?
3. How to determine liquidity?
4. Why liquidity is required?
5. Technical Insolvency
6. How to improve liquidity?
7. Liquidity for various sectors
8. From where do we get data?
9. Liquidation

Liquidity Analysis-1

Who are Concerned about Liquidity?

1. Management
2. Lenders/Financers
3. Creditors

Module 5: Financial Analysis Tools

Topic 81: Liquidity Analysis-2



Liquidity Analysis-2

Liquidity Ratios and Measures

1. Current Ratio
2. Quick Ratio
3. Cash Ratio
4. Gross Working Capital
5. Net Working Capital
6. Cash Conversion Cycle
7. Trade/Operating Cycle

Liquidity Analysis-2

Liquidity Ratios and Measures

1. Current Ratio

$$\frac{\text{Current Assets}}{\text{Current Liabilities}}$$

Liquidity Analysis-2

Liquidity Ratios and Measures

1. Quick Ratio

$$\frac{\text{Quick Assets}}{\text{Current Liabilities}}$$

- Quick Assets = Current Assets – Inventory and Prepayments

Liquidity Analysis-2

Liquidity Ratios and Measures

1. Cash Ratio

$$\frac{\text{Cash and Cash Equivalents}}{\text{Current Liabilities}}$$

Liquidity Analysis-2

Liquidity Ratios and Measures

1. Gross Working Capital

All the current assets i.e. Cash, Accounts Receivable, Inventory, and Prepayments

Liquidity Analysis-2

Liquidity Ratios and Measures

1. Net Working Capital

Current Assets - Current Liabilities

Liquidity Analysis-2

Liquidity Ratios and Measures

1. Cash Conversion Cycle-CCC

$$CCC = RTD + ITD - PTD$$

RTD = Receivables Turnover in days

ITD = Inventory Turnover in days

PTD = Payables Turnover in days

Liquidity Analysis-2

Liquidity Ratios and Measures

1. Trade/Operating Cycle

$$OC = RTD + ITD$$

RTD = Receivables Turnover in days

ITD = Inventory Turnover in days

Module 5: Financial Analysis Tools

Topic 82- Liquidity Analysis 3



Liquidity Analysis-3

Current Ratio

1. Current Ratio

$$\frac{\text{Current Assets}}{\text{Current Liabilities}}$$

$$\text{CR} = 250,000/100,000$$

$$\text{CR} = 2.5 \text{ Times}$$

Liquidity Analysis-3

Current Ratio Standard

- Standard Current Ratio
 - General Standard 1.5-2.0
 - Other Standards

Liquidity Analysis-3

Quick Ratio

1. Quick Ratio

Quick Assets

Current Liabilities

$$QR = 120,000 / 100,000$$

$$QR = 1.2$$

QR = Current Assets – Inventory
and Prepayments

$$QR = 250,000 - 100,000 - 30,000$$

Liquidity Analysis-3

Quick Ratio Standard

- Standard Quick Ratio
 - General Standard 1.0
 - Other Standards

Liquidity Analysis-3

Cash Ratio

1. Cash Ratio

Cash and Equivalents

Current Liabilities

$$CR = 30,000/100,000$$

$$CR = 0.3$$

Cash Equivalents = cash in hand,
cash at bank, cash convertibles

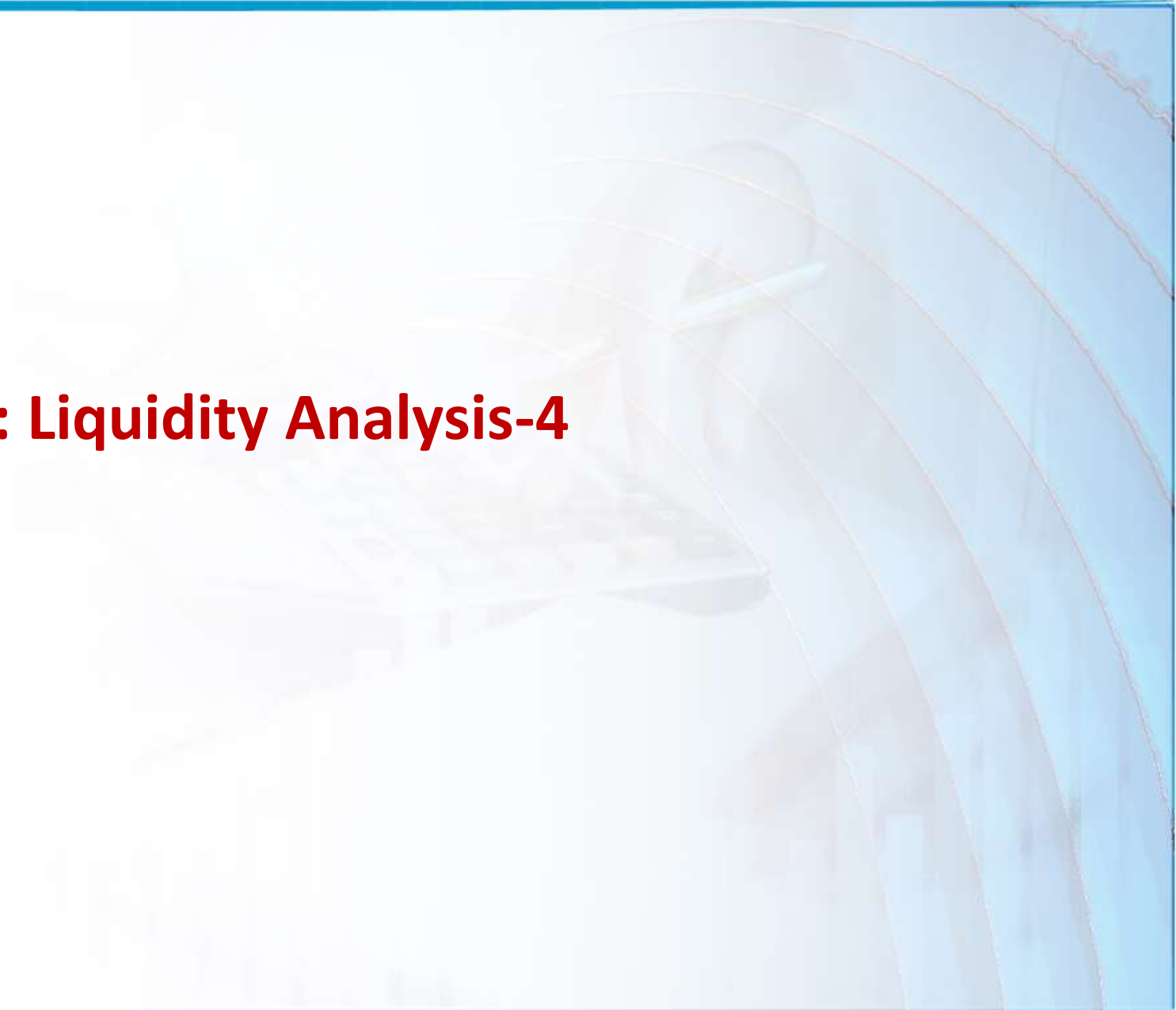
Liquidity Analysis-2

Cash Ratio Standard

1. Nature of Industry
2. Past results
3. Regulatory bodies' requirements
4. Smooth functioning

Module 5: Financial Analysis Tools

Topic 83: Liquidity Analysis-4



Liquidity Analysis-4

Gross Working Capital

- All Current Assets
 - Stock in Trade
 - Raw material, Work in process, and Finished goods
 - Stores and Spares
 - Trade debts
 - Advances, prepayments and other receivables
 - Cash and cash equivalents

Liquidity Analysis-4

Net Working Capital-NWC

- $NWC = \text{Current Assets} - \text{Current Liabilities}$
- $NWC = 40,000 - 28,000$
- $NWC = \text{Rs. } 12,000$

Liquidity Analysis-4

Cash Conversion Cycle

Cash Conversion Cycle-CCC

$$CCC = RTD + ITD - PTD$$

The number of days in which economic activity is converted to cash

Liquidity Analysis-4

Cash Conversion Cycle

Cash Conversion Cycle-CCC

$$CCC = 55 + 60 - 45$$

$$CCC = 70 \text{ days}$$

$$\text{RTD} = 55 \text{ days}$$

$$\text{ITD} = 60 \text{ days}$$

$$\text{PTD} = 45 \text{ days}$$

Liquidity Analysis-4

Trade Cycle

Trade Cycle

$$\text{Trade Cycle} = \text{RTD} + \text{ITD}$$

The number of days for which cash is blocked in trade cycle

Liquidity Analysis-4

Trade Cycle

Trade Cycle

$$\begin{aligned}\text{Trade Cycle} &= 55 + 60 \\ \text{Trade Cycle} &= 115 \text{ days}\end{aligned}$$

Module 5: Financial Analysis Tools

Topic 84: Turnover of Activity Analysis-1

Turnover of Activity Analysis-1

What is Turnover or Activity?

- Turnover
- Turnover in times
- Turnover in days
- Why turnover is important?
- How do we set benchmark for turnover?
- From where we get data?
- For whom it is important?

Turnover of Activity Analysis-1

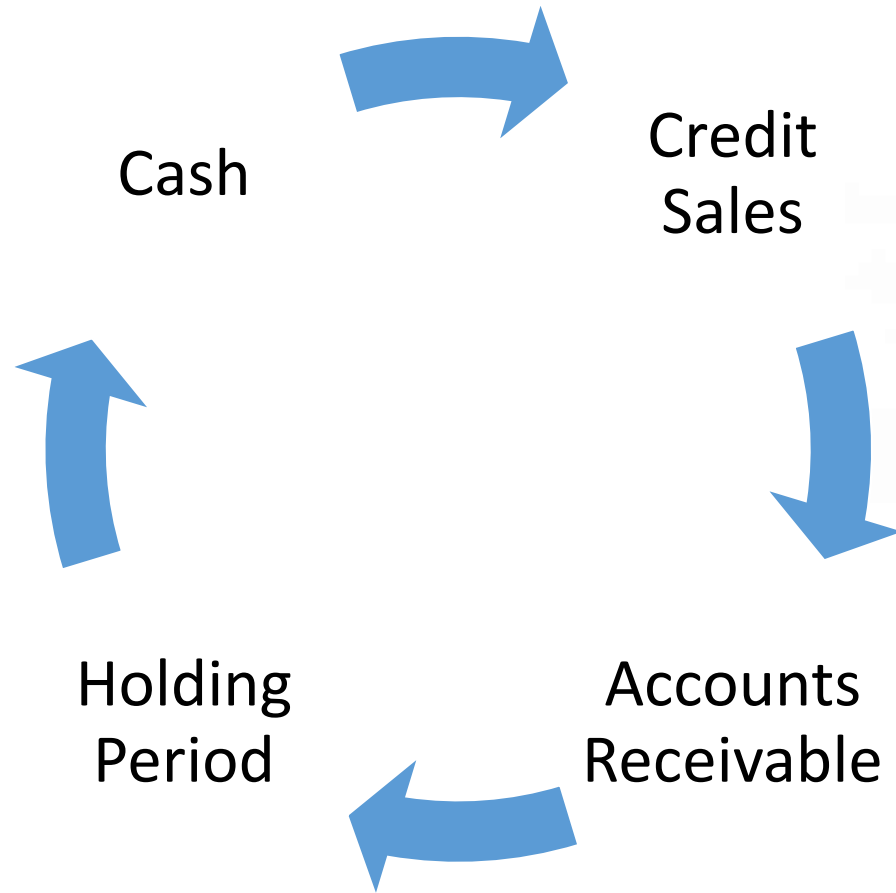
Turnover or Activity Ratios

- Receivables Turnover
- Inventory Turnover
- Payables Turnover
- Asset Turnover

Module 5: Financial Analysis Tools

Topic 85: Turnover of Activity Analysis-2

Turnover of Activity Analysis-2



Receivables Turnover

- Receivables Turnover in times
- Receivables Turnover in days
- What does turnover show?
- Ageing Accounts Receivables
- Turnover and profit
- Data sources
- Benchmarking
- For whom it is important?

Turnover of Activity Analysis-2

Receivables Turnover

- Receivables Turnover in Times
 - Average Receivables
 - Credit Sales
- $$\text{RTO} = \frac{\text{Credit Sales}}{\text{Average Receivables}}$$
- $\text{RTO} = 500,000 / 50,000$
- $\text{RTO} = 10 \text{ Times}$

Turnover of Activity Analysis-2

Receivables Turnover

- Receivables Turnover in Days
- Holding Period of Receivables
 - Receivables Turnover Ratio
 - Days in a year
- $\text{RTO in days} = \frac{\text{Days in a year}}{\text{RTO Ratio}}$
- $\text{RTO in days} = 365/10$
- $\text{RTO} = 36.5 \text{ days}$

Turnover of Activity Analysis-2

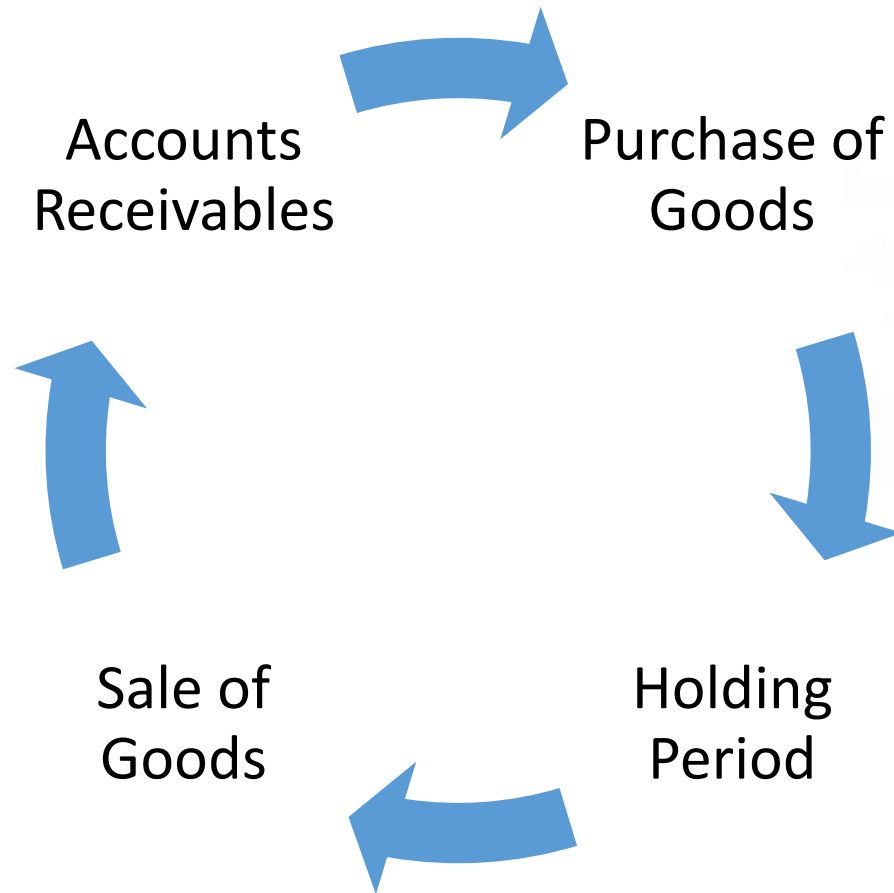
Receivables Turnover

- RTO in times be more or less?
- RTO in days be more or less?
- How do we compare it with the industry?

Module 5: Financial Analysis Tools

Topic 86: Turnover of Activity Analysis-3

Turnover of Activity Analysis-3



Inventory Turnover

- Which Inventory?
- Inventory Turnover in times
- Inventory Turnover in days
- What does inventory turnover ratio show?
- Holding or Carrying cost
- Data sources
- Benchmarking
- Inventory Management

Turnover of Activity Analysis-3

Inventory Turnover

- Inventory Turnover in Times
 - Average Inventory
 - Cost of Goods Sold
- $$\text{ITO} = \frac{\text{Cost of Goods Sold}}{\text{Average Inventory}}$$
- $$\text{ITO} = 400,000 / 50,000$$
- $$\text{ITO} = 8 \text{ Times}$$

Turnover of Activity Analysis-3

Inventory Turnover

- Inventory Turnover in Days
- Holding Period of Inventory
 - Inventory Turnover Ratio
 - Days in a year
- ITO in days = $\frac{\text{Days in a year}}{\text{ITO Ratio}}$
- ITO in days = $365/8$
- ITO = 45.6 days

Turnover of Activity Analysis-3

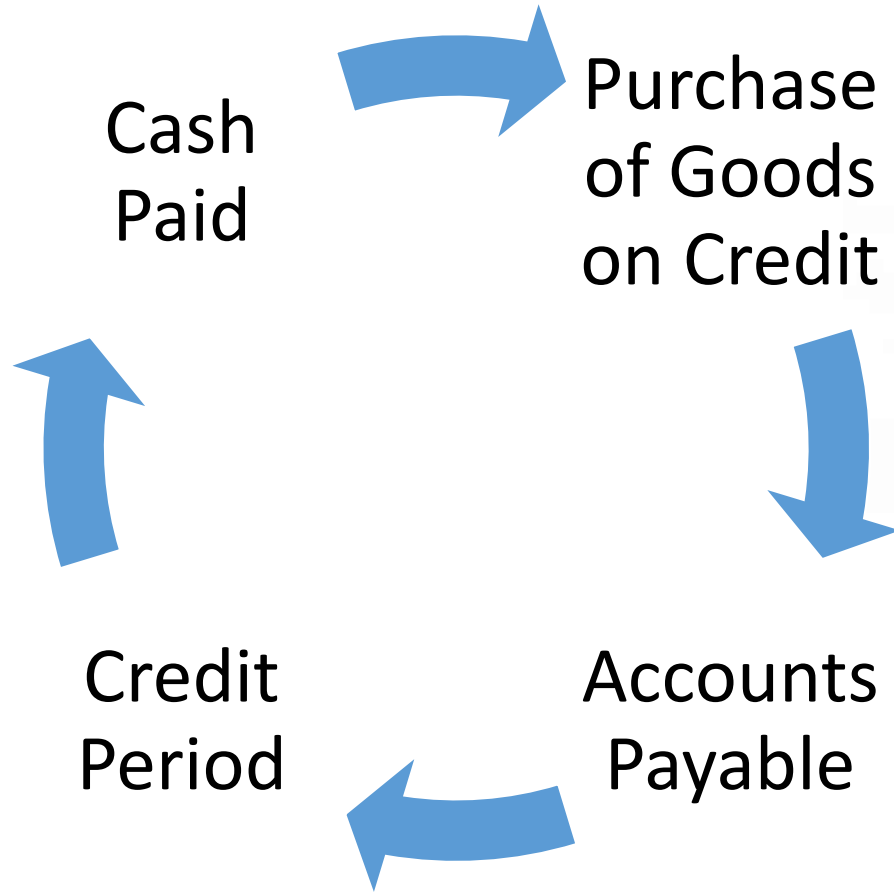
Inventory Turnover

- ITO in times be more or less?
- ITO in days be more or less?
- What if it is higher than competitors?

Module 5: Financial Analysis Tools

Topic 87: Turnover of Activity Analysis-4

Turnover of Activity Analysis-4



Payables Turnover

- What are Accounts Payable?
- Payable Turnover in Times
- Payable Turnover in Days
- What does payables turnover ratio show?
- Credit Period/Payment Period
- Data sources
- Benchmarking the Ratio
- Stretching the Payables

Turnover of Activity Analysis-4

Stretching Payables

- Who will be concerned?
- Costs of Stretching the Payables
 - Fines and Penalties
 - Lost of Cash Discount
 - Loss of Credibility

Turnover of Activity Analysis-4

Payables Turnover

- Payables Turnover in Times
 - Average Payables
 - Credit Purchases
- $PTO = \frac{\text{Credit Purchases}}{\text{Average Payables}}$
- $PTO = 300,000/60,000$
- $PTO = 5 \text{ Times}$

Turnover of Activity Analysis-4

Payables Turnover

- Payables Turnover in Days
- Payment Period of Purchases
 - Payables Turnover Ratio
 - Days in a year
- $\text{PTO in days} = \frac{\text{Days in a year}}{\text{PTO Ratio}}$
- $\text{PTO in days} = 365/5$
- $\text{PTO} = 73 \text{ days}$

Turnover of Activity Analysis-4

Payables Turnover

- PTO in times be more or less?
- PTO in days be more or less?
- What if others have a higher payables turnover?

Module 5: Financial Analysis Tools

Topic 88: Turnover of Activity Analysis-5

Turnover of Activity Analysis-5

Asset Turnover

- Sales and Assets
- Capital Turnover
- Asset Turnover in Times
- What does asset turnover ratio show?
- Source of data
- Benchmarking
- For whom it is important?

Turnover of Activity Analysis-5

Asset Turnover

- Asset Turnover in Times
 - Annual Sales
 - Total Assets
- $ATO = \frac{\text{Annual Sales}}{\text{Total Assets}}$
- $ATO = 1,000,000/750,000$
- $ATO = 1.33$ Times

Turnover of Activity Analysis-5

Asset Turnover Comparison

- Should it be more or less?
- With the industry
- With other industries
- With the past results
- Effectiveness of the management

Module 5: Financial Analysis Tools

Topic 89: Coverage Analysis-1

Coverage Analysis-1

Coverage Ratios

- What is Coverage?
- What does a coverage ratio show?
- Source of data
- Benchmarking
- For whom these are important?

Coverage Analysis-1

Interest Coverage Ratio

- Interest and Financial Charges
- Earnings before Interest and Taxes-EBIT
- Interest Coverage Ratio-ICR
- $$\text{ICR} = \frac{\text{EBIT}}{\text{Interest}}$$
- $\text{ICR} = 500,000 / 200,000$
- $\text{ATO} = 2.5 \text{ Times}$

Coverage Analysis-1

Interest Coverage Ratio

- Is EBIT given in annual reports?
- How to calculate EBIT?
 - $\text{EBIT} = \text{Profit before Tax} + \text{Interest}$

Coverage Analysis-1

Interest Coverage Ratio

- Sources of Data
- ICR and Capital Structure
- ICR higher or lower, why?
- Who will be concerned, why?
- Does it show the performance of management?
- Why Profit after Tax is not used in the ratio?

Module 5: Financial Analysis Tools

Topic 90: Coverage Analysis-2

Coverage Analysis-2

Debt Service Coverage Ratio

- Debt Servicing
- Amortization Schedules
- Tax Implications
- Debt Service Coverage Ratio-
DSCR
- $$\text{DSCR} = \frac{\text{EBIT}}{\text{Interest} + [\text{Principal}/(1-t)]}$$
- t = Income Tax Rate

Coverage Analysis-2

Debt Service Coverage Ratio

- EBIT = Rs. 500,000
- Interest = Rs. 100,000
- Principal amount per Installment of Loan = Rs. 110,000
- Tax rate = 40%
- $DSCR = \frac{500,000}{100,000 + [110,000/1-0.4]}$
- $DSCR = 500,000/283,333.33$
- DSCR = 1.76 Times

Coverage Analysis-2

Debt Service Coverage Ratio

- Tax implication on principal repayment
- Who is concerned?
- What is the standard?
- Is EBIT given in annual reports?
- How to calculate EBIT?
- $\text{EBIT} = \text{Profit before Tax} + \text{Interest}$

Module 5: Financial Analysis Tools

Topic 91: Leverage Analysis-1

Leverage Analysis-1

Financial Leverage

- Debt and Equity
- Short term and Long term Debt
- Cost of Debt and Equity
- Fixed Financial Cost
- Data Sources

Leverage Analysis-1

Financial Leverage

- How does Financial Leverage work?
- What is Gearing?
- Financial Leverage and Risk
- What is the Limit of Leverage?
- Who defines the limit and why?

Leverage Analysis-1

Users of Financial Leverage Ratios

- Shareholders
- Banks/Lenders
- Regulatory Authorities

Module 5: Financial Analysis Tools

Topic 92: Leverage Analysis-2

Leverage Analysis-2

How does Financial Leverage Work?

- Fixed and Flexible Cost of Debt
- Earning Power of a Business
- Return on Investment-ROI
- Return on Equity-ROE

Leverage Analysis-2

Example of Financial Leverage

- Cost of debt-CD: 10%
- ROI: 15%
- Equity: Rs. 100,000
- Debt: Rs. 50,000
- Profit: Rs. 30,000

Leverage Analysis-2

Example of Financial Leverage

Solution:

- Total Investment=150,000
- $ROI = 30000/150000=20\%$
- $CD=50000 \times 10\%=5000$
- $Equity= 30000-5000=25000$
- $ROE = 25000/100000=25\%$

Module 5: Financial Analysis Tools

Topic 93: Leverage Analysis-3



Leverage Analysis-3

Common Leverage Ratios

- Debt to Equity Ratio
 - Total Debt/Equity
- Debt to Total Assets Ratio
 - $\text{Total Debt/Total Assets}$
- Long Term Debt to Total Assets Ratio
 - LTD/Total Assets

Leverage Analysis-3

Debt to Equity Ratio

- Total Debt/Equity
- Example:
 - Total Debt = 600,000
 - Equity = 400,000
- $D/E \text{ Ratio} = 600000/400000$
- $D/E \text{ Ratio} = 1.5 \text{ Times}$

Leverage Analysis-3

Debt to Total Assets Ratio

- Total Debt/Total Assets
- Example:
 - Total Debt = 600,000
 - Equity = 400,000
 - Total Assets = 1,000,000
- $D/TA \text{ Ratio} = 600000/1000000$
- $D/TA \text{ Ratio} = 0.6 \text{ Times OR } 60\%$

Leverage Analysis-3

Long Term Debt to Total Assets

- Long Term Debt/Total Assets
- Example:
 - Long Term Debt = 400,000
 - Total Assets = 1,000,000
- $LTD/TA \text{ Ratio} = 400000 / 1,000000$
- LTD/TA Ratio = 1.0 Times

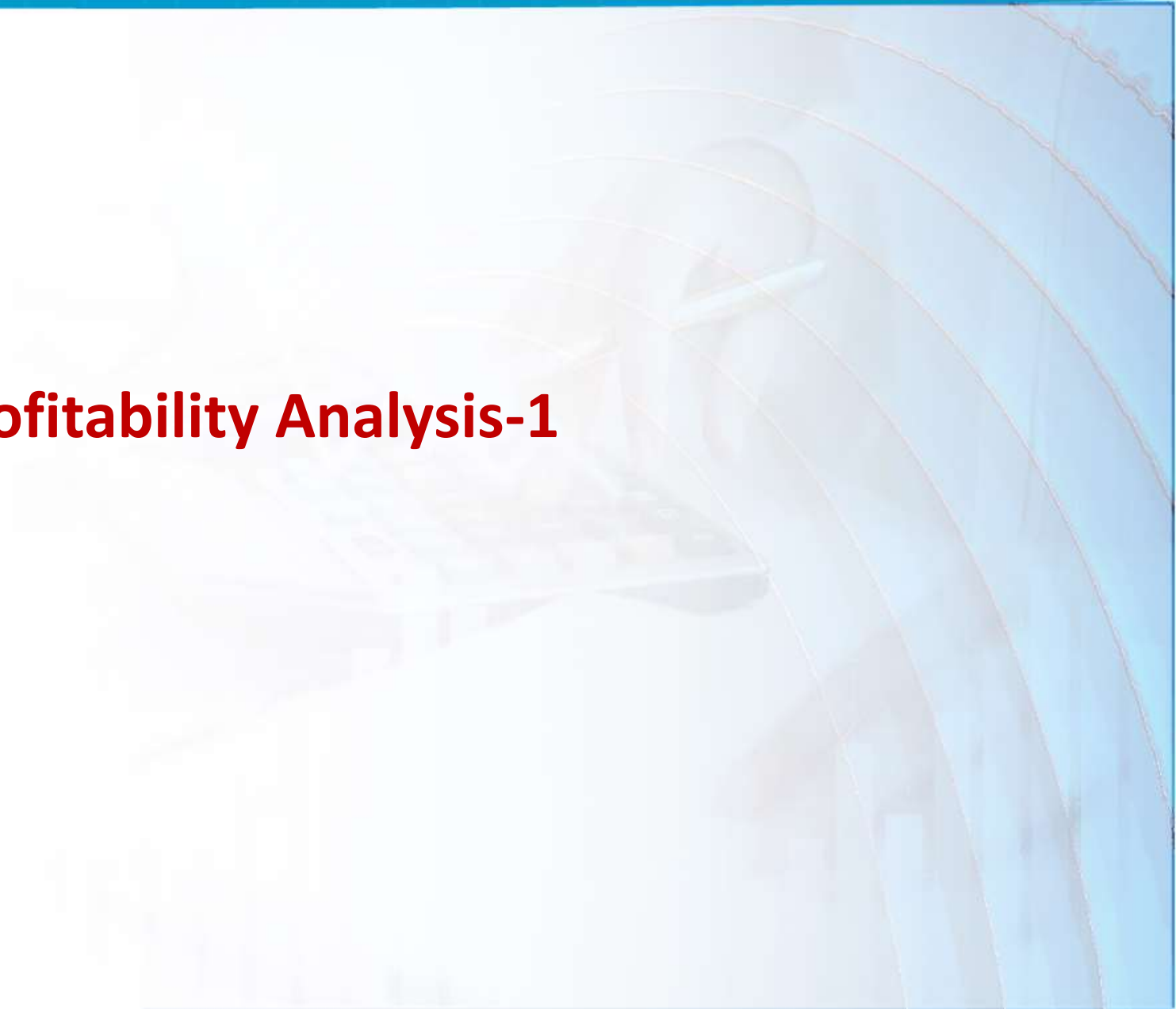
Leverage Analysis-3

Leverage Ratios Analysis

- For shareholders
- For lenders
- Regulatory authorities
- Negative effect of leverage

Module 5: Financial Analysis Tools

Topic 94: Profitability Analysis-1



Profitability Analysis-1

Profitability

- Profit and Profitability
- Revenue/Income
- Cost and Expenses
- Users of profitability analysis
- Various Profits
 - Gross Profit, Net Operating Profit, Profit before/after Tax, EBIT, and EBITDA

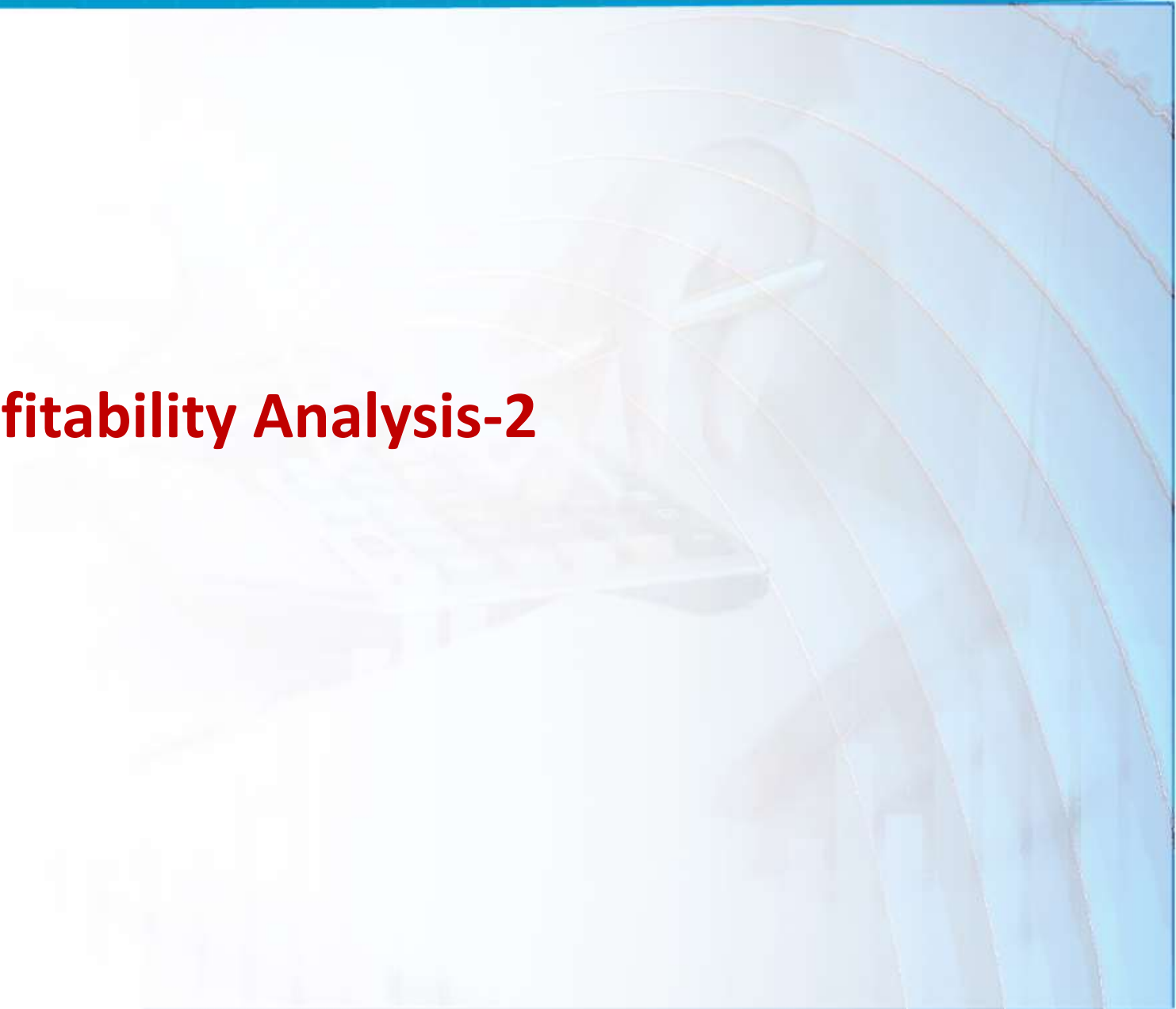
Profitability Analysis-1

Benchmarking Profitability

- Cost of Financing
- Market Competition
 - Inside the Sector
 - With other Sectors
- Profit and Cashflows
- Total and Marginal Profits

Module 5: Financial Analysis Tools

Topic 95: Profitability Analysis-2



Profitability Leverage Analysis-2

Profitability Ratios

- Margin and Return Ratios
- Margin Ratios:
 - Gross Profit Margin
 - Net Profit Margin
 - EBITDA Margin
- Return Ratios:
 - Return on Assets
 - Return on Equity
 - Return on Capital Employed

Profitability Leverage Analysis-2

Profitability Ratios

- Gross Profit Margin
 - $\text{Gross Profit/Sales}$
- Net Profit Margin
 - PAT/Sales
- EBITDA Margin
 - EBITDA/Sales

Profitability Leverage Analysis-2

Profitability Ratios

- Return on Assets
 - $\text{PAT} / \text{Total Assets}$
- Return on Equity
 - $\text{PAT} / \text{Equity}$
- Return on Capital Employed
 - $(\text{PBT} + \text{Interest}) / \text{Total Investment}$

Module 5: Financial Analysis Tools

Topic 96: Profitability Analysis-3

Profitability Leverage Analysis-3

Profitability Ratios

- Example:
 - Sales = Rs. 100,000
 - Gross Profit = Rs. 40,000
 - PBT = Rs. 10,000
 - Income Tax Rate = 30%
 - PAT = Rs. 7,000
 - Interest = Rs. 5,000
 - Depreciation and Amortization = Rs. 11,000
 - Equity = Rs. 80,000
 - Total Assets = Rs. 120,000

Profitability Leverage Analysis-3

Profitability Ratios

- Gross Profit Margin
 - Gross Profit/Sales
 - 40,000/100,000
 - 0.4 or 40%
- Net Profit Margin
 - PAT/Sales
 - 7,000/100,000
 - 0.07 or 7%

Profitability Leverage Analysis-3

Profitability Ratios

- EBITDA Margin
 - EBITDA/Sales
 - EBITDA = PBT + Interest + Depreciation and Amortization
 - EBITDA = 10000 + 5000 + 11000
 - EBITDA = 26,000
- EBITDA Margin = $26000/100000$
- EBITDA Margin = 0.26 or 26%

Profitability Leverage Analysis-3

Profitability Ratios

- Return on Assets
 - PAT/Total Assets
 - $7000/120000$
 - 0.058333 or 5.833%
- Return on Equity
 - PAT/Equity
 - $7000/80000$
 - 0.0875 or 8.75%

Profitability Leverage Analysis-3

Profitability Ratios

- Return on Capital Employed
 - $(\text{PBT} + \text{Interest}) / \text{Total Investment}$
 - $(10000 + 5000) / 120000$
 - $15000 / 120000$
 - 0.125 or 12.5%

Module 5: Financial Analysis Tools

Topic 97: Market Ratios-1

Market Ratios-1

Market Analysis

- Stock Exchange
- Market Price
- Factors that affect Market Price
 - Book Value
 - Growth
 - External Factors
- Earnings
- Dividend
- Yield

Market Ratios-1

Market Analysis

- Earnings per Share
- Dividend per Share
- Dividend Payout
- Retention Ratio
- Earning Yield
- Dividend Yield
- Market Value
- Market Value per Share
- Book Value
- Book Value per Share
- Price Earning Ratio

Module 5: Financial Analysis Tools

Topic 98: Market Ratios-2

Market Ratios-2

Earnings per Share

- Earnings per Share-EPS
- Outstanding Shares
- Formula
 - $\text{PAT/No. of Shares-NOS}$
- Example
 - PAT = Rs. 100,000
 - NOS = 200,000 shares
 - $\text{EPS} = 100000/200000$
 - EPS = Rs. 0.5 per share

Market Ratios-2

Dividend per Share

- Dividend per Share-DPS
- Dividend Declared
- Dividend Paid
- Formula
 - $\text{Dividend Declared} / \text{NOS}$
- Example
 - Dividend Declared = Rs. 500,000
 - NOS = 200,000 shares
 - $\text{DPS} = 500000 / 200000$
 - DPS = Rs. 2.5 per share

Market Ratios-2

Dividend Payout Ratio-DPO

- Dividend per Share-DPS
- Earnings per Share-EPS
- Formula
 - DPS/EPS
- Example
 - $DPS = \text{Rs. } 2.5$
 - $EPS = \text{Rs. } 5.0$
 - $DPO = 2.5/5.0$
 - $DPO = 0.5 \text{ time or } 50\%$

Market Ratios-2

Retention/Plough back Ratio-RR

- Earnings per Share-EPS
- Retention per share = $\text{EPS} - \text{DPS}$
- Formula
 - $\text{Retention} / \text{EPS}$
- Example
 - $\text{DPS} = \text{Rs. } 2.0$
 - $\text{EPS} = \text{Rs. } 5.0$
 - $\text{Retention} = 5 - 2 = 3$
 - $\text{RR} = 3/5 = 0.6 \text{ or } 60\%$

Module 5: Financial Analysis Tools

Topic 99: Market Ratios-3

Market Ratios-3

Earning Yield-EY

- Concept of Yield
- Earnings per Share-EPS
- Stock Price per Share
- Sources of Data
- Formula:

$$EY = (EPS/Price \text{ per Share}) \times 100$$

Market Ratios-3

Earning Yield-EY

Example

- Co. A: EPS = Rs. 10
- Co. A: Price per Share = Rs. 50
- Co. B: EPS = Rs. 15
- Co. B: Price per Share = Rs. 100

Solution

$$\text{EY Co. A} = 10/50 = 0.2 \text{ or } 20\%$$

$$\text{EY Co. B} = 15/100 = 0.15 \text{ or } 15\%$$

Market Ratios-3

Dividend Yield-DY

- Dividend declared
- Dividend per share-DPS
- Stock price per share
- Sources of data
- Formula:

$$DY = (DPS/Price \text{ per Share}) \times 100$$

Market Ratios-3

Dividend Yield-DY

Example

- Co. A: DPS = Rs. 5
- Co. A: Price per Share = Rs. 100
- Co. B: DPS = Rs. 10
- Co. B: Price per Share = Rs. 100

Solution

DY Co. A = $5/100 = 0.05$ or 5%

DY Co. B = $10/100 = 0.10$ or 10%

END

Module 5: Financial Analysis Tools

Topic 100: Market Ratios-4

Market Ratios-4

Book Value-BV

- Concept
- Relevance to Market Price
- Shareholders' Equity
- Outstanding Shares
- Treasury Stock

Market Ratios-4

Book Value-BV

Formula:

$$BV = (\text{Shareholders' Equity} / \text{Number of Outstanding Shares})$$

Market Ratios-4

Book Value-BV

Example

- Co. A: Equity = Rs. 100,000
- Co. A: No. of Shares = 5,000
- Co. B: Equity = Rs. 200,000
- Co. B: No. of Shares = 8,000

Solution

$$\text{BV Co. A} = 100000 / 5000 = \text{Rs. } 20$$

$$\text{BV Co. B} = 200000 / 8000 = \text{Rs. } 25$$

Market Ratios-4

Price Earnings Ratio

- Price per Share
- Earnings per Share
- What does it show?
- Why does it change?

Formula:

$$\text{P/E Ratio} = \text{Price/EPS}$$

Market Ratios-4

Price Earnings Ratio

Example

- Co. A: Price = Rs. 100
- Co. A: EPS = Rs. 10
- Co. B: Price = Rs. 200
- Co. B: EPS = Rs. 10

Solution

$P/E \text{ Co. A} = 100/10 = 10 \text{ times}$

$P/E \text{ Co. B} = 200/10 = 20 \text{ times}$

END

Module 5: Financial Analysis Tools

Topic 101: Index Analysis-1



Index Analysis-1

Index Analysis

- Index Numbers
- Current Value
- Base Value
- Current and Base Years
- Converting to Base-100

Index Analysis-1

Index Analysis

- Profit and Loss Account
- Balance Sheet

Index Analysis-1

Formula of Indexing

Formula:

- Converting to Base-100 = {Each Item of Current Year / Same Item of Base Year} x 100

Index Analysis-1

Indexing a Value

Example:

Base Year = 2010

Current Year = 2019

Sales of Base Year = Rs. 400,000

Sales of Current Year = Rs. 560,000

- **$\{\text{Sales of 2019} / \text{Sales of 2010}\} \times 100$**
- **$\{560,000 / 400,000\} \times 100 = 140$**

Index Analysis-1

Indexing a Value

- Sales
- Cost of Goods Sold
- Gross Profit
- Operating Expenses
- Extraordinary Items
- Profit before Tax
- Profit after Tax

Index Analysis-1

	2010	2019
Sales	165,000	186,000
COGS	94,000	121,000
Gross Profit	71,000	65,000
Operating Expenses	38,000	41,000
PBT	33,000	24,000
Income Tax	9,900	7,200
PAT	23,100	16,800

Profit & Loss Account

Indexed Profit & Loss
Account

	2010	2019
Sales	100	112.73
COGS	100	128.72
Gross Profit	100	91.55
Operating Expenses	100	107.89
PBT	100	72.73
Income Tax	100	72.73
PAT	100	72.73

Module 5: Financial Analysis Tools

Topic 102: Index Analysis-2



Index Analysis-2

Index Analysis

- Horizontal Analysis
- Vertical Analysis

Index Analysis-2

Formula Index Analysis: Balance Sheet

- Converting to Base-100

{Each Item of Balance Sheet of Current Year/ Each Item of Balance Sheet of Base Year} x 100

Index Analysis-2

Indexing Balance Sheet Items

Example:

Base Year = 2010

Current Year = 2019

Stock Base Year = Rs. 400,000

Stock Current Year = Rs. 560,000

Solution:

- **$\{\text{Stock of 2019} / \text{Stock of 2010}\} \times 100$**
- **$\{400,000 / 450,000\} \times 100 = 88.88$**

Index Analysis-2

Indexing a Value

- Total Assets/Investment
- Operating Fixed Assets
- Current Assets
- Long term Liabilities
- Current Liabilities
- Shareholders' Equity

Index Analysis-2

	2010	2019
Inventories	1,000	1,400
Accounts Receivable	2,000	2,400
Prepayments	500	600
Bank	800	600
Fixed Assets	15,000	16,000
Accumulated Depreciation	3,000	4,200

Assets Balance Sheet

Indexed Balance Sheet Assets

	2010	2019
Inventories	100	140
Accounts Receivable	100	120
Prepayments	100	120
Bank	100	75
Fixed Assets	100	107
Accumulated Depreciation	100	140

Module 5: Financial Analysis Tools

Topic 103: Common Size Analysis of Balance Sheet-1

Common Size Analysis of Balance Sheet-1

Common Size Analysis

- Percentage of a Base Value
- Base Value of P & L Account
- Base Value of Balance Sheet
- Vertical Analysis
- Why Perform Common Size Analysis?

Common Size Analysis of Balance Sheet-1

Formula Common Size Analysis: Balance Sheet

- Base of Balance Sheet
- Converting to Base as Percentage

$\{\text{Each Asset} / \text{Total Assets}\} \times 100$

Common Size Analysis of Balance Sheet-1

The Formula

- **$\{\text{Each Asset} / \text{Total Assets}\} \times 100$**
- **$\{\text{Each Liability} / \text{Total Assets}\} \times 100$**
- **$\{\text{Each Item of Equity} / \text{Total Assets}\} \times 100$**

END

Module 5: Financial Analysis Tools

Topic 104: Common Size Analysis of Balance Sheet-2

Common Size Analysis of Balance Sheet-2

Using Commons Size Analysis

- Comparison
- Trending the Statements

Common Size Analysis of Balance Sheet-2

Application of Common Size Analysis

- Profit and Loss Account
- Balance Sheet
- Cash Flow Statement
- Owners' Equity

Common Size Analysis of Balance Sheet-2

	2,019	%
Inventories	1,000	4
Accounts Receivable	2,000	8
Prepayments	500	2
Bank	800	3
Fixed Assets	15,000	62
Accumulated Depreciation	3,000	12
Total	24,310	100

Common Size Analysis of Balance Sheet-2

	2,019	%
Accounts Payable	500	2%
Notes Payable	1,000	4%
Accrued Expenses	1,300	5%
Loan Payable	3,000	12%
Bonds Payable	4,000	17%
Lease Payable	3,000	12%
Total (Assets)	24,310	

Common Size Analysis of Balance Sheet-2

	2,019	%
Common Share Capital	5,000	20%
Share Premium Account	1,000	4%
Retained Earnings	4,300	18%
Reserves	1,210	5%
Total (Assets)	24,310	

Module 5: Financial Analysis Tools

Module 105: Common Size Analysis of Profit and Loss Account-1

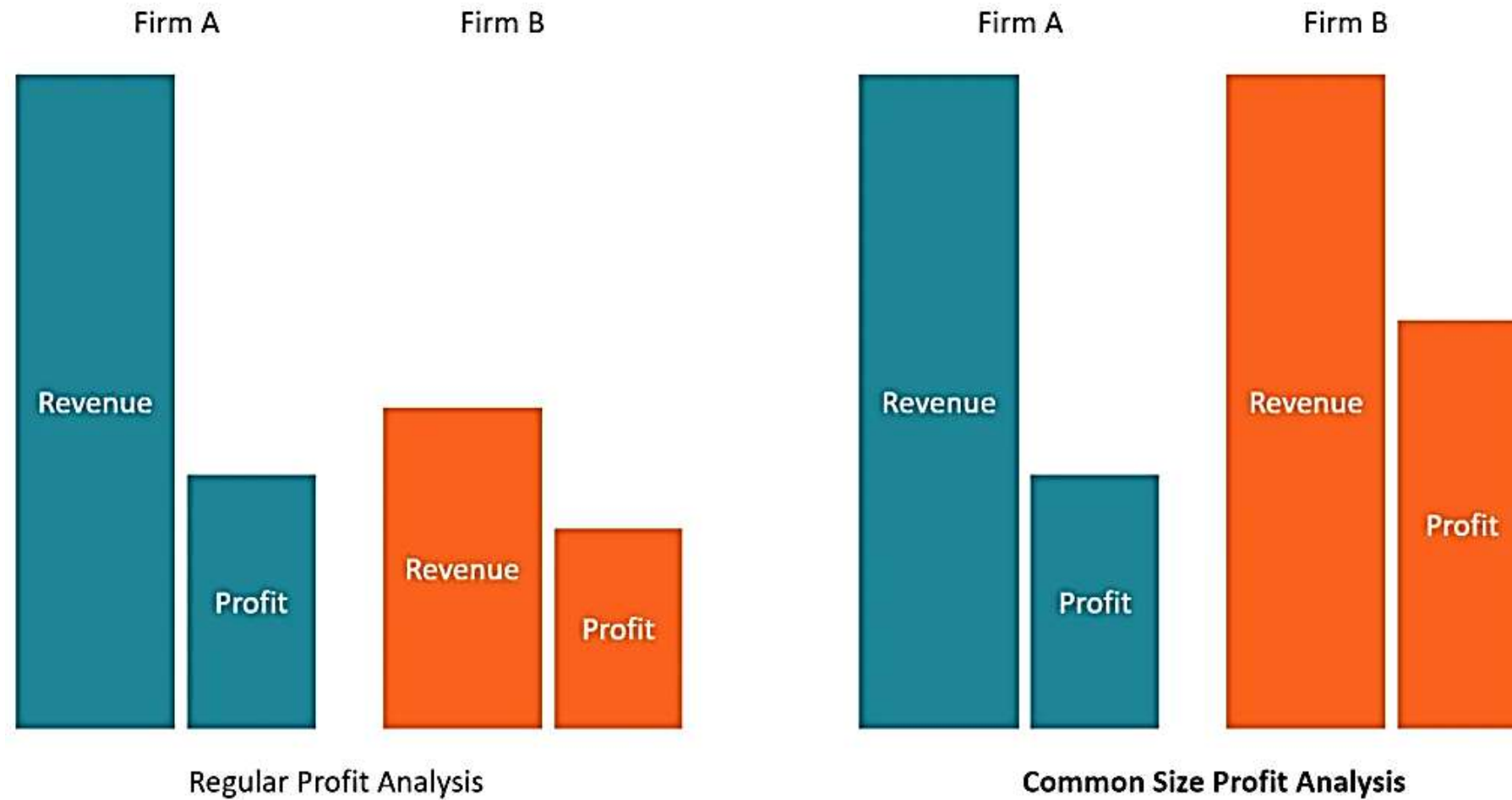
Common Size Analysis of Profit and Loss Account-1

Formula Common Size Analysis: P&L Account

- Base of Profit and Loss Account
- Converting to Base as Percentage

$\{\text{Each Item of P\&L Account} / \text{Sales}\} \times 100$

Common Size Analysis of Profit and Loss Account-1



Common Size Analysis of Profit and Loss Account-1

	2010	%
Sales	165,000	100
COGS	94,000	57
Gross Profit	71,000	43
Operating Expenses	38,000	23
PBT	33,000	20
Income Tax	9,900	6
PAT	23,100	14

Module 5: Financial Analysis Tools

Topic 106: Common Size Analysis of Profit and Loss Account-2

Common Size Analysis of Profit and Loss Account-2

Formula Common Size Analysis: P&L Account

- Base of Profit and Loss Account
- Converting to Base as Percentage

$\{\text{Each Item of P\&L Account/Sales}\} \times 100$

Common Size Analysis of Profit and Loss Account-2

Common Size Analysis COGS

	2010	%
Sales	350,000	100.00
Opening Inventory	40,000	11.43
Purchases	180,000	51.43
Purchases Returns	10,000	2.86
Closing Inventory	10,000	2.86
COGS	200,000	57.14

Common Size Analysis of Profit and Loss Account-2

	2010	%
Sales	350,000	100
Operating Expenses	145,000	57.14
Marketing Expenses	60,000	22.86
Admin. Expenses	45,000	17.14
General Expenses	10,000	5.71
Financial Expenses	30,000	11.43

Common Size Analysis of Profit and Loss Account-2

	2010	%
Sales	350,000	100
Profit before Tax	200,000	57.14
Income Tax	80,000	22.86
Profit after Tax	60,000	17.14

Module 5: Financial Analysis Tools

Topic 107: Trend Analysis-1



Trend Analysis-1

Trend Analysis

- Trends
- Graphs
- Time Period of Trends
- Upward and Downward Trends
- Methods of Performing Trend Analysis

Trend Analysis-1

Types of Trend Analysis

- Trend Analysis of Ratios
- Trend Analysis of Indexed Statements
- Trend Analysis of Financial Statements
- Trends of Common Sized Statements

END

Module 5: Financial Analysis Tools

Topic 108: Trend Analysis-2



Trend Analysis-2

Performing Trend Analysis

- Tables
- Graphs
- Forecasting/Predicting/
Estimating
- Forecasted Financial Statement
Values
- Budgets

Trend Analysis-2

	Year 1	Year 2	Year 3	Year 4	Year 5
Ratio	1.2	1.3	1.4	1.5	1.2
Sales	51 m	52m	54m	53m	52m
Indexed Amount	100	102	103	103	102
Common Sized Amount (Profit)	45%	46%	47%	48%	50%

Module 5: Financial Analysis Tools

Topic 109: Trend Analysis-3



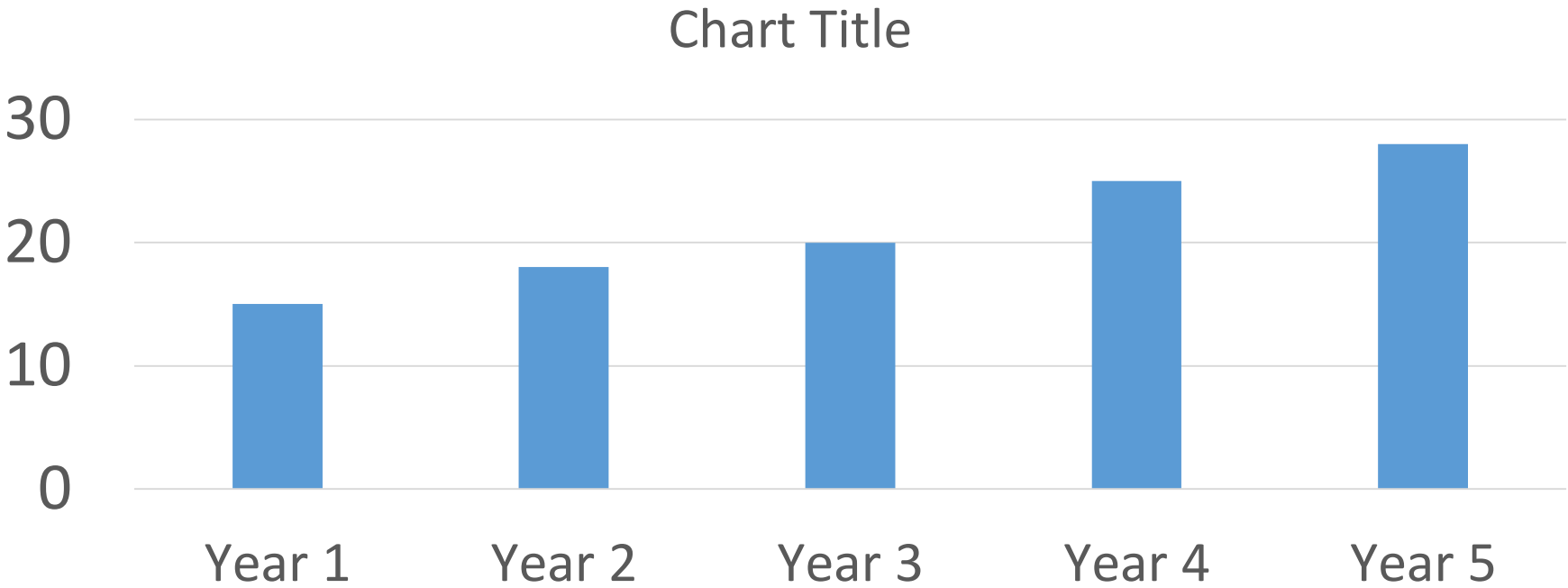
Trend Analysis-2

Performing Trend Analysis

- Trend Analysis in Stock Market
- Types of Graphs
 - Bar Charts
 - Line Charts
 - Other Charts
 - Candle Stick Charts

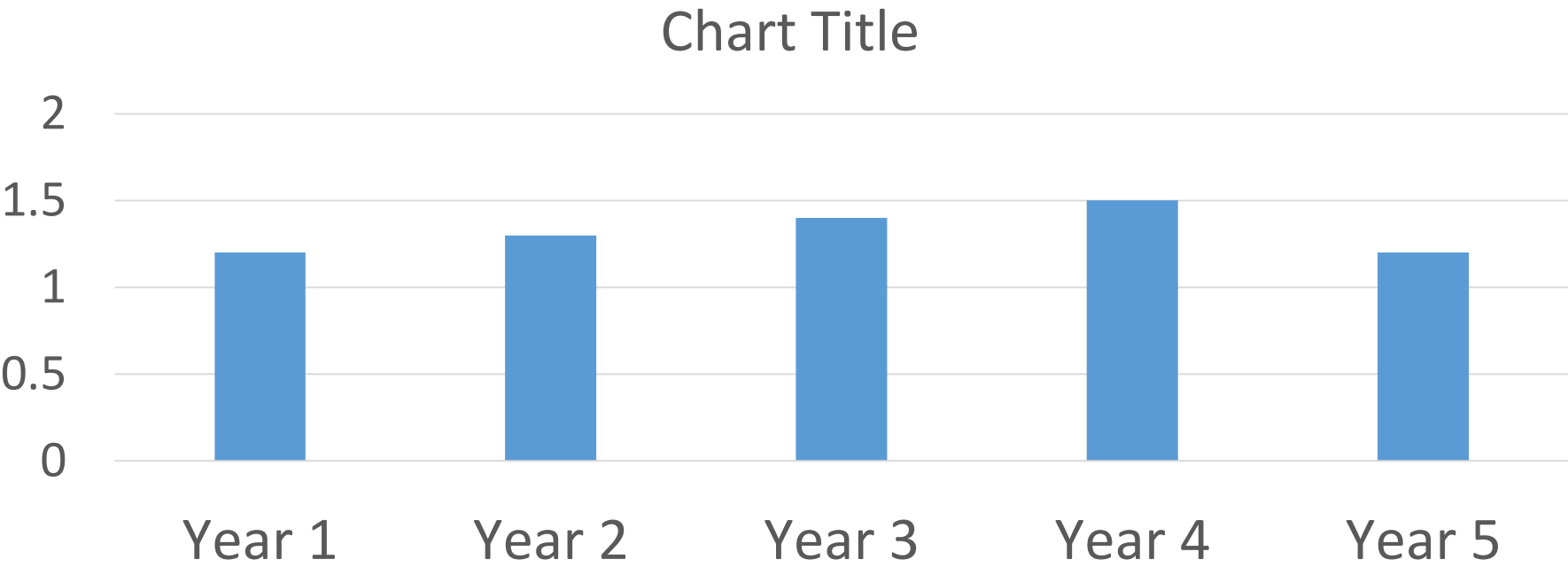
Trend Analysis-2

Sales in Million Rupees				
Year 1	Year 2	Year 3	Year 4	Year 5
15	18	20	25	28



Trend Analysis-2

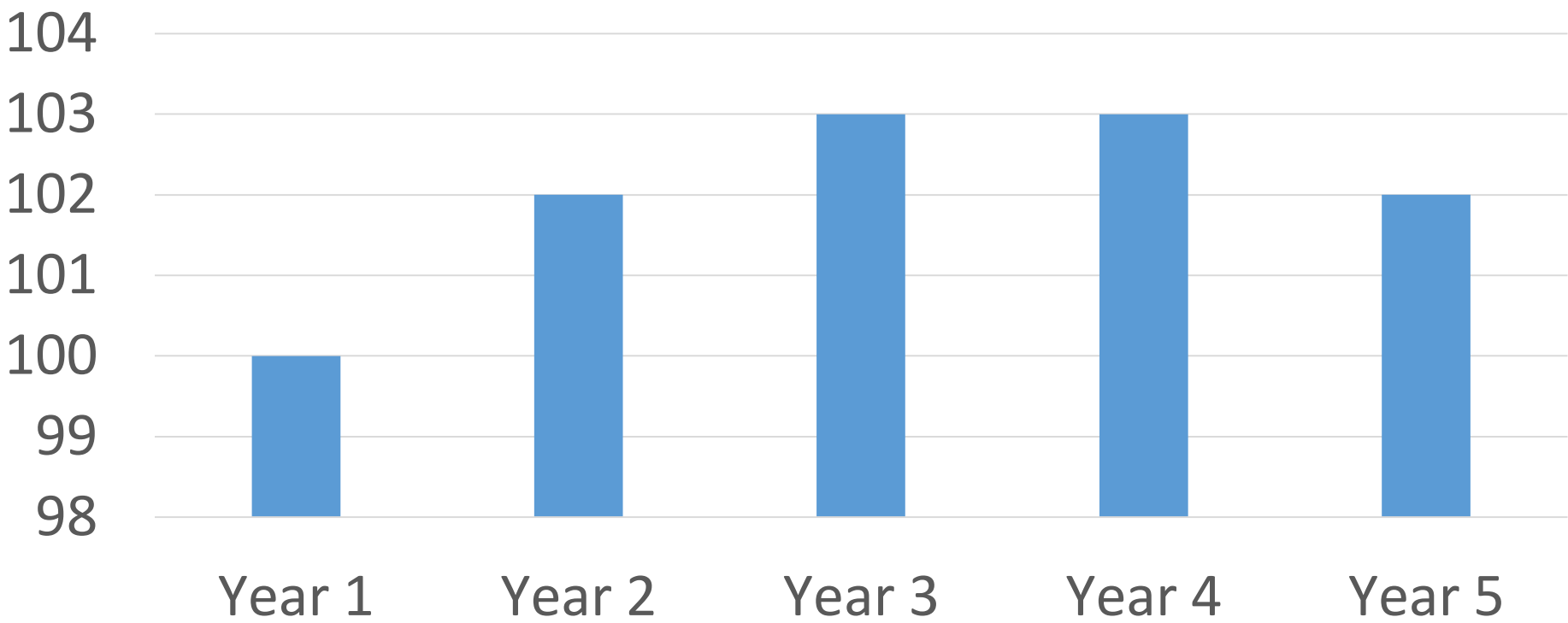
Ratio				
Year 1	Year 2	Year 3	Year 4	Year 5
1.2	1.3	1.4	1.5	1.2



Trend Analysis-2

Indexed Values				
Year 1	Year 2	Year 3	Year 4	Year 5
100	102	103	103	102

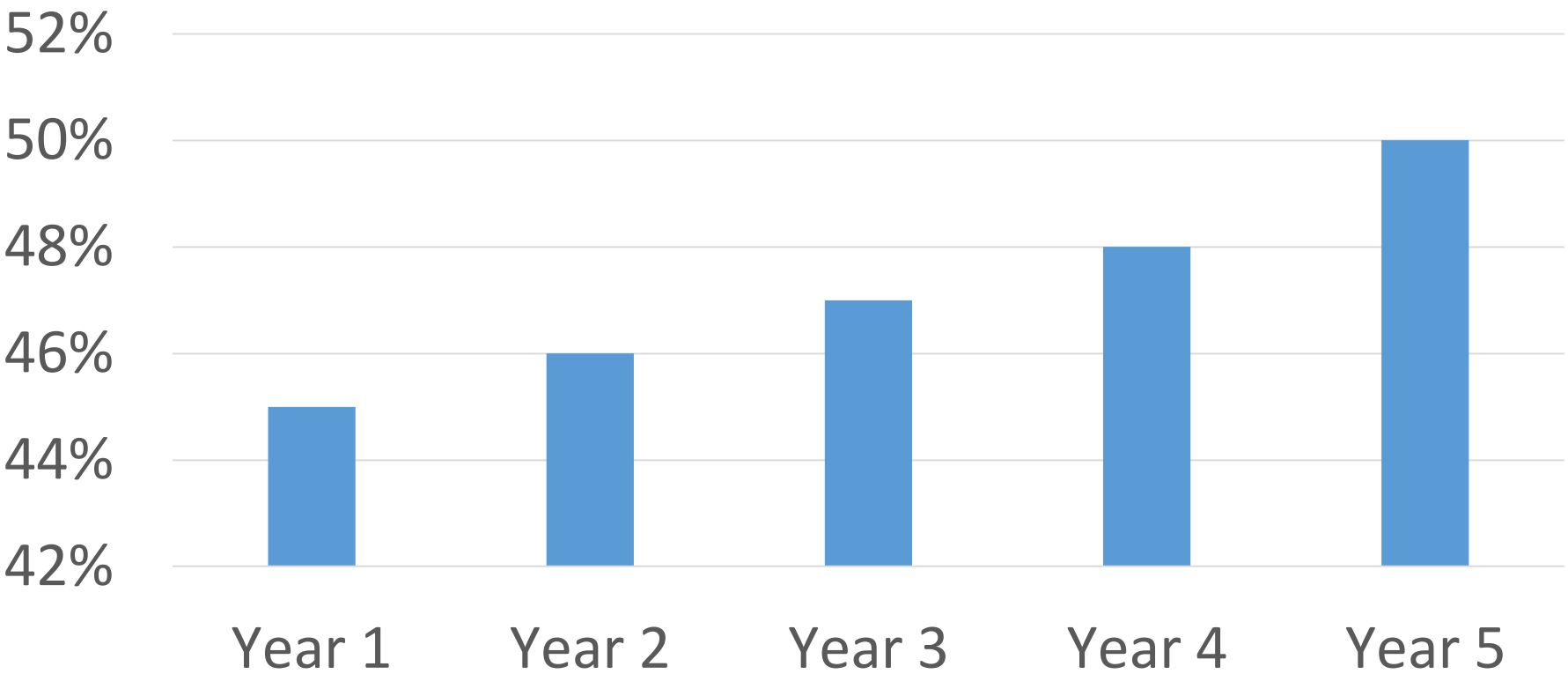
Chart Title



Trend Analysis-2

Common Sized Values				
Year 1	Year 2	Year 3	Year 4	Year 5
45%	46%	47%	48%	50%

Chart Title



Module 5: Financial Analysis Tools

Topic 110: Differential Analysis-1

Differential Analysis-1

Differential Analysis

- Definition and Basic Concept
- Uses of Differential Analysis
 - Drop a production line
 - Start a new production line
 - Borrow cash for investment
 - Invest in a given project
 - Compare two investment options
 - Financial Statements etc.

Differential Analysis-1

Relevant Costs

- Relevant Costs
- Underlying Concept
 - Determining
 - Analysis
 - Evaluation
- Examples of relevant costs:
 - Opportunity costs
 - Out-of-pocket costs
 - Future cash flow

Differential Analysis-1

Non-Relevant Costs

- Non-Relevant Costs
 - Sunk costs
 - General overheads
 - Depreciation

END

Module 5: Financial Analysis Tools

Topic 111: Differential Analysis-2



Differential Analysis-2

Differential Financial Statements

- Profit and Loss Account
- Balance Sheet
- Cash Flow Statement
- Statement of Owners' Equity

Differential Analysis-2

Differential P & L Account

- Comparing Output (Sales)
- Comparing Revenue
- Comparing Costs
- Comparing Profits
- Comparing EPS

Differential Analysis-2

	2018	2019	% Change
	Rs.	Rs.	
Sales	10000	15000	+ 50%
Cost of Goods Sold	6000	8500	+ 41.67%
Gross Profit	4000	6500	+ 62.5%
Operating Expenses	2500	2800	+ 12%
Profit before Tax	1500	3700	+ 147%
Income Tax 30%	450	1110	+ 147%
Profit after Tax	1050	2590	+ 147%

Differential Analysis-2

	2018	2019	% Change
	Rs.	Rs.	
Profit after Tax	1050	2590	+ 147%
No. of Shares	1000 sh.	1000 sh.	
EPS	1.05	2.59	+ 147%

Module 5: Financial Analysis Tools

Topic 112: Differential Analysis-3

Differential Analysis-3

Differential Balance Sheet

- Comparing Current Assets
- Comparing Fixed Assets
- Comparing Current Liabilities
- Comparing Long Term Liabilities
- Comparing Shareholders' Equity
- Comparing Earnings per Share

Differential Analysis-3

	2018	2019	% Change
	Rs.	Rs.	
Current Assets	10000	14000	+ 40%
Fixed Assets	80000	120000	+ 50%
Current Liabilities	8000	13000	+ 62.5%
Long Term Liabilities	30000	45000	+ 50%
Shareholders' Equity	52000	76000	+ 46.2%

Differential Analysis-3

	2018	2019	% Change
	Rs.	Rs.	
Shareholders' Equity	52000	76000	+ 46.2%
Share Capital	35000	35000	0%
Retained Earnings	15000	32000	+ 113.33%
Reserves	2000	9000	+ 350%

Module 6: Cash Flow Analysis

Topic 113: Cash Flow Analysis-Introduction

Cash Flow Analysis

Cash Flows

- Cash Inflows
 - Revenues
 - Borrowings
 - Equity
- Cash Outflows
 - Expenses
 - Repayments
 - Dividends

END

Module 6: Cash Flow Analysis

Topic 114: Statement of Cash Flows



Statement of Cash Flow

Cash Flows Statement Methods

- Indirect Method
- Direct Method

Statement of Cash Flow

Cash Flows from Operations

- Cash Flows from Operating Activities
 - Profit
 - Non-Cash Expenses
 - Changes in Receivables
 - Changes in Inventory
 - Changes in Prepayments
 - Changes in Current Liabilities

Statement of Cash Flow

Cash Flows from Investment

- Cash Flows from Investment
 - Fixed Assets Change
 - Long Term Investments Change
 - Intangible Assets Change

Statement of Cash Flow

Cash Flows from Financing

- Cash Flows from Financing Activities
 - Loans
 - Bonds
 - Share Capital
 - Dividend

END

Module 6: Cash Flow Analysis

Topic 115: Cash Flows and Profits



Cash Flows and Profits

Cash Flows Vs Profits

- Profits
 - Non-Cash Expenses
 - Changes in Methods of Valuation
 - Capitalizing Expenses
 - Receivables: Revenue
 - Payables: Costs and Expenses

Cash Flows and Profits

Cash and Profit

- Profit is not cash
- Profits but not cash!
- Cash but no profit!
- Cash flow is the actual movement of cash in and out
- Profit is the difference of revenue and expenses

END

Module 6: Cash Flow Analysis

Topic 116: Sources and Application of Cash



Sources and Application of Cash

Sources of Cash

- Revenue
- Collection from Receivables
- Sales of Assets
- Borrowing
 - Loans
 - Bonds
- Equity
 - Common Shares
 - Preferred Shares

Sources and Application of Cash

Application of Cash

- Expenses and Costs
- Purchasing Fixed Assets
- Payment to Payables
- Accrued Expenses
- Payment of Liabilities
- Dividend
- Treasury Stock

END

Module 6: Cash Flow Analysis

Topic 117: Cash Flow from Operating Activities-1

Cash Flows from Operating Activities-1

Operating Activities

- Three parts of a Cash Flow Statement
 - Operations
 - Investments
 - Financing
- Operating Activities
 - Profit and Loss Account
 - Current Assets
 - Current Liabilities

Cash Flows from Operating Activities-1

Operating Activities Format

- Net Income
 - Add Non-cash Expenses
 - Add Decrease in Current Assets
 - Subtract Increase in Current Assets
 - Add Increase in Current Liabilities
 - Subtract Decrease in Current Liabilities

END

Module 6: Cash Flow Analysis

Topic 118: Cash Flow from Operating Activities-2

Cash Flows from Operating Activities-2

Non-Cash Expenses

- Depreciation
- Amortization
- Writing-off of Assets
- Provision of Bad Debts
- Other Provisions for Expenses
- Effect on Net Profit

END

Module 6: Cash Flow Analysis

Topic 119: Cash Flow from Operating Activities-3

Cash Flows from Operating Activities-3

Impact of Current Assets' Change

- Opening and Closing Balances
- Changes in Current Assets
 - Accounts Receivable
 - Inventories
 - Prepayments
- Why not consider Cash?

Cash Flows from Operating Activities-3

Impact of Current Liabilities' Change

- Changes in Current Liabilities
 - Accounts Payable
 - Notes Payable
 - Accrued Expenses
- Short Term Financing?

END

Module 6: Cash Flow Analysis

Topic 120: Cash Flow from Operating Activities-4

Cash Flows from Operating Activities-4

Example of Operating Activities

- Net Income = Rs. 100,000
- Depreciation = 20,000
- Inc. in Acc. Rec. = 18,000
- Decrease in Inv. = 6,000
- Inc. in Acc. Pay. = 7,000
- Dec. in Accrued Exp. = 5,000

Cash Flows from Operating Activities-4

Cash Flow from Operating Activities	Rs.	Rs.
Net Income		100,000
Add: Deprecation	20,000	
Current Assets & Liabilities		
Increase in Accounts Receivable	(18,000)	
Decrease in Inventory	6,000	
Increase in Accounts Payable	7,000	
Decrease in Accrued Expenses	(5,000)	
		<u>10,000</u>
Net Cash Inflow from Operating Activities		110,000

Module 6: Cash Flow Analysis

Topic 121: Cash Flow from Investing Activities-1

Cash Flows from Investing Activities-1

Investing Activities

- Investing Activities
 - Purchasing a Fixed Asset
 - Acquiring a Non-Current Asset
 - Selling a Fixed or Non-Current Asset

Cash Flows from Investing Activities-1

Investing Activities

- Changes in Fixed Assets
- Changes in Accumulated Depreciation
- Example:

	2018	2019
• Net Asset	500	600
• Acc. Dep.	100	140
• Cash Outflow = 140*		

END

$$*600 - 500 + 40 = 140$$

Module 6: Cash Flow Analysis

Topic 122: Cash Flow from Investing Activities-2

Cash Flows from Investing Activities-2

Example of Investing Activities

- Purchase of land = Rs. 5,000
- Purchase of Build = 4,000
- Sale of Equip = 3,000
- Sale of Vehicles = 4,500
- Increase in Acc. Dep. = 4,000
- Buy Long Term Debt (Bond) of other company = 6,000
- Sale of Bonds of other Companies = 3,500

Cash Flows from Investing Activities-2

Cash Flow from Investing Activities	Rs.
Purchase of Land	(5,000)
Purchase of Building	(4,000)
Sale of Equipment	3,000
Sales of Vehicles	4,500
Increase in Accumulated Depreciation (all assets)	(4,000)
Buy Long Term Debt (Bond) of other company	(6,000)
Sale of Bonds of other Companies	<u>3,500</u>
Net Cash Outflow from Investing Activities	(8,000)

Module 6: Cash Flow Analysis

Topic 123: Cash Flow from Financing Activities-1

Cash Flows from Financing Activities-1

Financing Activities

- Sources of Finances
 - Equity
 - Debt
- Cost of Financing
 - Dividend
 - Interest

Cash Flows from Financing Activities-1

Financing Activities-Equity

- Issue of Common Shares
- Issue of Preferred Shares
- Additional Capital
- Treasury Stock
- Dividend
- Retained Earnings (Retained earnings/ net income are not part of financing activities because these have already been treated under operating activities)
- Reserves

Cash Flows from Financing Activities-1

Financing Activities-Debt

- Short Term Borrowing
- Long Term Borrowing
- Bonds
- Current Portion of LTD
- Lease Payments

Module 6: Cash Flow Analysis

Topic 124: Cash Flow from Financing Activities-2

Cash Flows from Financing Activities-2

Example of Financing Activities

- LT borrowing = Rs. 40,000
- Payment of LTD = 30,000
- ST borrowing = 15,000
- Issue of Bonds = 35,000
- Retirement of Bonds = 4,500
- Payment of Interest = 3,000
- Issue of Com. Shares = 20,000
- Purchase of Treasury S. = 5,000
- Dividend Paid = 10,000
- Profit Earned = 23,000

Cash Flows from Financing Activities-2

Cash Flow from Financing Activities	Rs.
Long Term Borrowing	40,000
Payment of Long Term Debt	(30,000)
Short Term Borrowing	15,000
Issue of Bonds	35,000
Retirement of Bonds	(4,500)
Payment of Interest	-
Issue of Common Shares	20,000
Purchase of Treasury Stock	(5,000)
Dividend Paid	(10,000)
Profit Earned	-
Net Cash Flow from Financing Activities	60,500

Module 6: Cash Flow Analysis

Topic 125: Cash Flow Statement: Direct Method-1

Cash Flows Statement: Direct Method-1

Example of Operating Activities

- Cash collected from customers
- Cash paid to suppliers
- Cash paid to employees
- Cash paid for Prepaid Expenses
- Interest received/Dividends received
- Interest paid
- Income taxes paid

Cash Flows Statement: Direct Method-1

Cash from Customers

- Cash Receipts from Customers
 - + Net Sales
 - + Opening Accounts Rec.
 - – Ending Accounts Rec.

Cash Flows Statement: Direct Method-2

Cash to Suppliers

- Cash Payments to Suppliers
 - + Purchases
 - + End. Stock – Open Stock
 - + Opening Accounts Payable
 - – Ending Accounts Payable.

Module 6: Cash Flow Analysis

Topic 126: Cash Flow Statement: Direct Method-2

Cash Flows Statement: Direct Method-2

Cash to Employees

- Cash Payments to Employees
 - + Beginning Salaries Payable
 - – Ending Salaries Payable
 - + Salaries Expense

Cash Flows Statement: Direct Method-2

Cash for Prepayments

- Cash Payments for Purchase of Prepaid Assets
 - + Ending Prepaid Rent, Prepaid Insurance etc.
 - + Expired Rent, Expired Insurance etc.
 - – Beginning Prepaid Rent, Prepaid Insurance etc.

Cash Flows Statement: Direct Method-2

Interest/Dividend Received

- Interest/Dividend Received
 - + Opening Interest/Dividend Receivable
 - – Ending Interest/Dividend Receivable
 - + Interest/Dividend Income

Cash Flows Statement: Direct Method-2

Cash for Interest Payment

- Interest Payments =
- + Beginning Interest Payable
- – Ending Interest Payable
- + Interest Expense

Cash Flows Statement: Direct Method-2

Cash for IT Payment

- Income Tax Payments
 - + Beginning Income Tax Payable
 - – Ending Income Tax Payable
 - + Income Tax Expense

Topic 127: Risk & Return Analysis

Module 7: Risk & Return

Return



- **Additional amount on investment**
- **Dividend**
- **Interest**

Risk

- **Uncertainty**
- **Expected Return/Mean**
- **Deviation for Mean/Std. Dev.**
- **Risk & Return are Proportional**



Topic 128: Systematic and Unsystematic Risk Risk

Module 7: Risk & Return

Market Risk

- **Undiversifiable Risk**
- **Unavoidable Risk**
- **Systematic Risk**
- **Volatility**



Unsystematic Risk

- **Avoidable Risk**
- **Diversifiable Risk**
- **Controllable Risk**



Topic 129: Individual Stock Risk Analysis-1

Module 7: Risk & Return

Individual Stock's Risk

- **Mean (Expected Return)**

- **Variance**

**STANDARD
DEVIATION**

- **Standard Deviation**



Formulas

- Mean
- Variance
- Standard Deviation

$$\bar{X} = \frac{\sum X}{N}$$



Standard
Deviation
Formula

$$= \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{N - 1}}$$



Individual Stock Risk-2

**Financial
Management
Module-130**

Individual Stock's Calculations

- **Calculation of Mean**
- **Calculation of SD**



Average Return: Mean

Return-R	Probability-P	Average: R x P
-0.1	0.05	-0.005
-0.02	0.1	-0.002
0.04	0.2	0.008
0.09	0.3	0.027
0.14	0.2	0.028
0.2	0.1	0.02
0.28	0.05	<u>0.014</u>
	Average	0.09

Risk: Standard Deviation		
Return-R	Probability-P	Var. = (R-Ave R) ² xP
-0.1	0.05	{(-0.1-0.09) ² }*0.05 = 0.001805
-0.02	0.1	0.00121
0.04	0.2	0.0005
0.09	0.3	0
0.14	0.2	0.0005
0.2	0.1	0.00121
0.28	0.05	<u>0.001805</u>
	Variance	0.00703

Coefficient of Variation

**Financial
Management
Module-131**

Coefficient of Variation

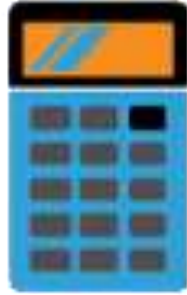
- Coefficient of Variation is a measure of relative dispersion (risk) i.e. **per unit of expected return**



Calculation CV



Formula



$$\text{Coefficient of Variation Formula} = \frac{\text{Standard Deviation}}{\text{Mean}}$$

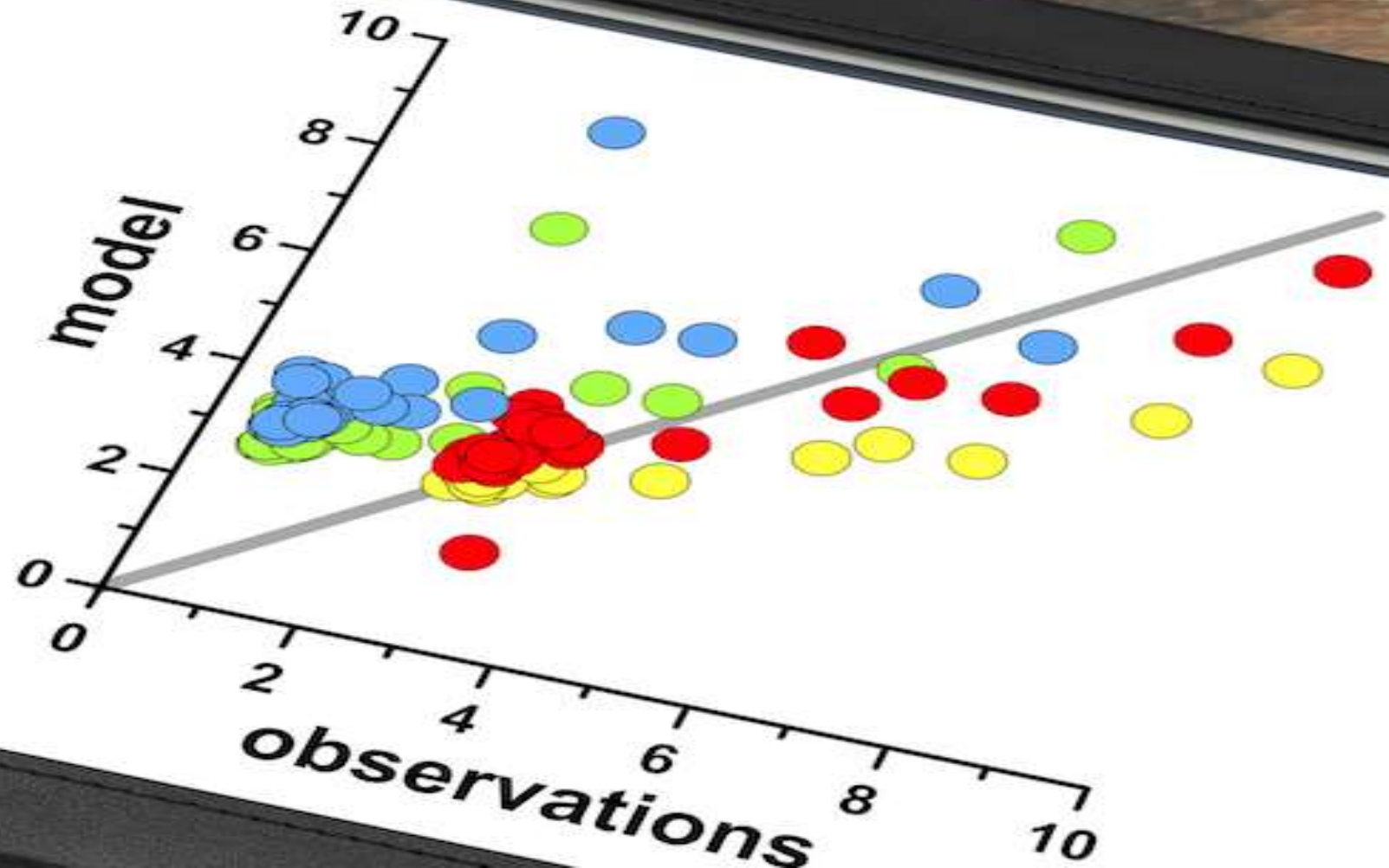


		Company A	Company B
Mean (also known as expected return of a stock)	Expected Return	10%	20%
Std. Dev.	Risk	8%	12%
CV	Relative Risk	80%	60%

**Portfolio risk
analysis-1
Internal Correlation
and Covariance**

**Financial
Management
Module-132**

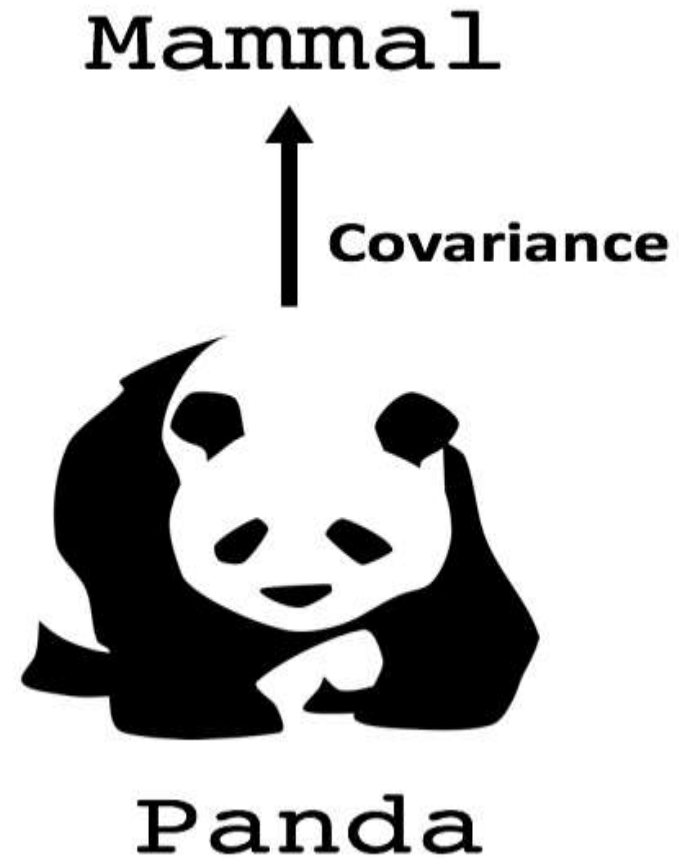
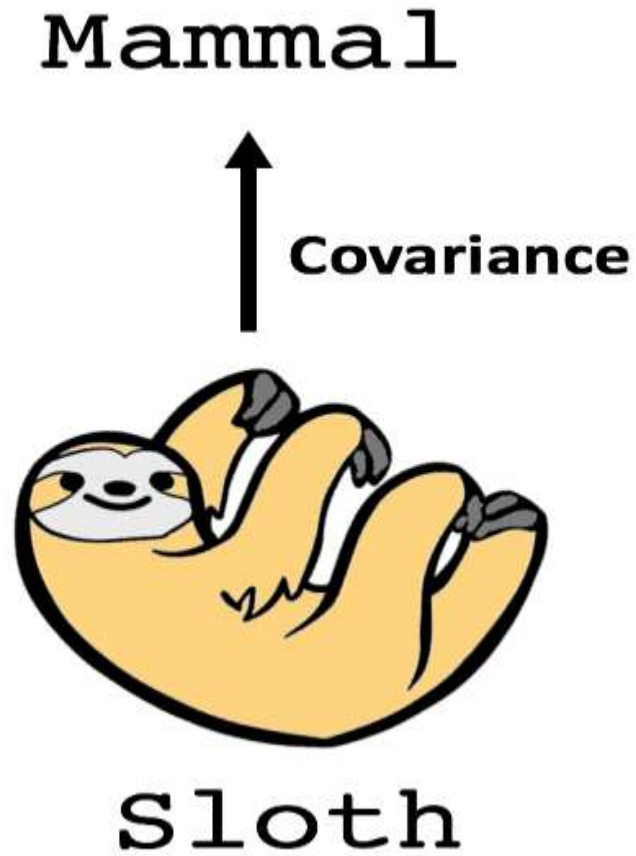
Correlation Is The Mutual Relationship Of Two Variables



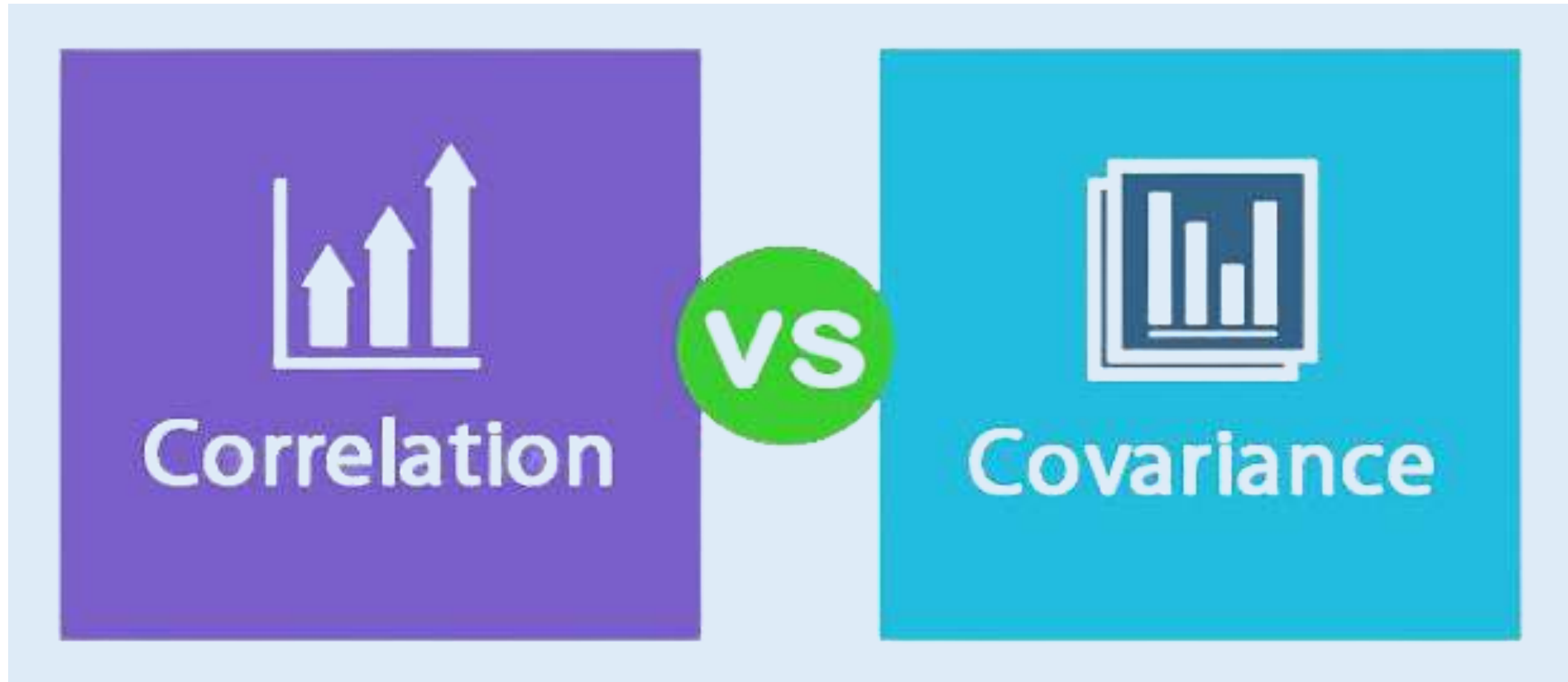
Correlation Coefficient - Rho

ρ

Covariance Measures The Directional Relationship Between The Returns On Two Assets.



How do we compare Correlation and Covariance?



Portfolio Risk Analysis-2

**Financial
Management
Module-133**

Portfolio's Risk

	Stock A	Stock B
Stock A	Covariance of A & A	Covariance of A & B
Stock B	Covariance of B & A	Covariance of B & B

Portfolio's Risk

	Stock A	Stock B
Stock A	$\sigma_a \sigma_a \rho_{aa} X_a X_a$	$\sigma_a \sigma_b \rho_{ab} X_a X_b$
Stock B	$\sigma_a \sigma_b \rho_{ab} X_a X_b$	$\sigma_b \sigma_b \rho_{bb} X_b X_b$

Equation

Risk of Portfolio

$$\sigma_a^2 X_a^2 + \sigma_b^2 X_b^2 + 2 (\sigma_a \sigma_b \rho_{ab} X_a X_b)$$

Portfolio Risk Analysis-3

**Financial
Management
Module-134**

	Stock A	Stock B	Stock C
Stock A	Cov. of A & A	Cov. of A & B	Cov. of A & C
Stock B	Cov. of B & A	Cov. of B & B	Cov. of B & C
Stock C	Cov. of C & A	Cov. of C & B	Cov. of C & C

	Stock A	Stock B	Stock C
Stock A	$\sigma_a \sigma_a \rho_{aa} X_a X_a$	$\sigma_a \sigma_b \rho_{ab} X_a X_b$	$\sigma_a \sigma_c \rho_{ac} X_a X_c$
Stock B	$\sigma_a \sigma_b \rho_{ab} X_a X_b$	$\sigma_b \sigma_b \rho_{bb} X_b X_b$	$\sigma_b \sigma_c \rho_{bc} X_b X_c$
Stock C	$\sigma_a \sigma_c \rho_{ac} X_a X_c$	$\sigma_b \sigma_c \rho_{bc} X_b X_c$	$\sigma_c \sigma_c \rho_{cc} X_c X_c$

Equation

Risk of Portfolio

$$\sigma_a^2 X_a^2 + \sigma_b^2 X_b^2 + \sigma_c^2 X_c^2 + 2 (\sigma_a \sigma_b \rho_{ab} X_a X_b) + 2 (\sigma_a \sigma_c \rho_{ac} X_a X_c) + 2 (\sigma_b \sigma_c \rho_{bc} X_b X_c)$$

Capital Asset Pricing Model CAPM-1

**Financial
Management
Module-135**



Capital asset pricing model

Components of CAPM

- R_f = Risk Free Rate of Return
- R_m = Market Rate of Return
- $R_m - R_f$ = Risk Premium
- β = Beta (Index of Systematic Risk)

Capital Asset Pricing Model CAPM-2

**Financial
Management
Module-136**

What is Security Market Line?



SML Equation

$$\text{Return} = R_f + (R_m - R_f) \beta$$

Fama and French Three Factor Model

**Financial
Management
Module-137**

F & F Three Factors

1. Market Risk

2. Size Risk

3. Value Risk



Market Risk = Beta

RISK



Size = Small Cap – Big Cap

XL
SIZE

XXS
SIZE

XS
SIZE

XXL
SIZE

L
SIZE

S
SIZE

XXXL
SIZE

XL
SIZE

M
SIZE

Value = High – Low
Book to Market Value



VALUE

F & F Three Factors Equation

$$\text{Return} = R_f + (R_m - R_f)\beta_1 + \text{Size}\beta_2 + \text{Value}\beta_3$$

OR

$$\text{Return} = R_f + RP\beta_1 + \text{SMB}\beta_2 + \text{HML}\beta_3$$

Topic 138: Four Factor Model

**Module 7: Risk &
Return**

Carhart Four Factor Model



- **Momentum**
- **Winners minus Losers (Ups minus Downs)**

Momentum Defined

“The tendency for the stock price to continue rising if it is going up and to continue declining if it is going down”



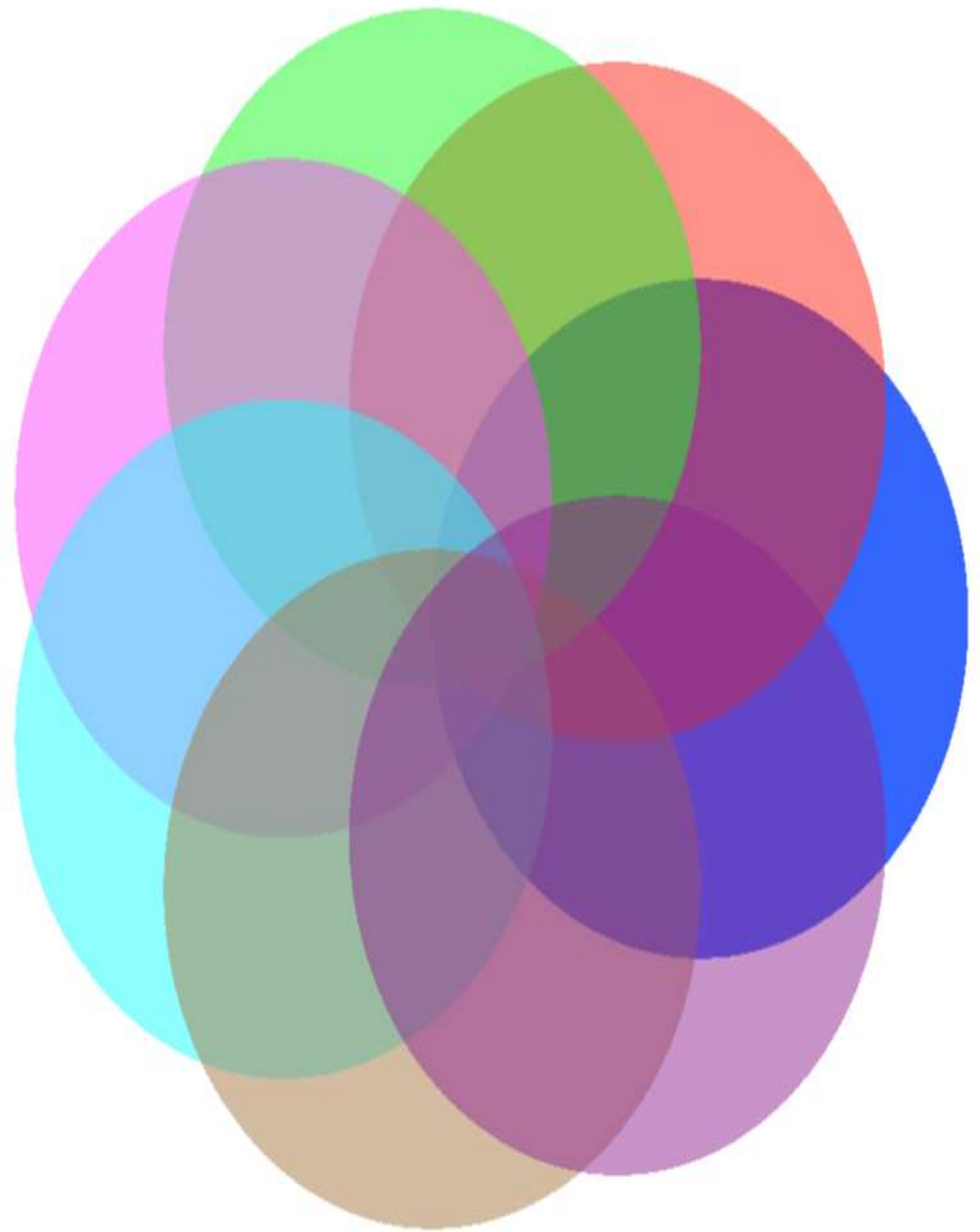
The Equation of Carhart's Four Factor Model

$$\text{Return} = R_f + (R_m - R_f)\beta_1 + \text{SMB} \beta_2 + \text{HML} \beta_3 + \text{UMD} \beta_4$$

Topic 139: Multiple Factors Risk Adjusted Models

Module 7: Risk & Return

Multiple Factor Models



Multiple Factors

- **Macroeconomic**
- **Fundamental**
- **Statistical**
- **Combination**

Hedge Fund Returns

Fung Hsieh 7-Factor Model

1. Bond Trend Following
2. Currency Trend Following
3. Commodity Trend
4. Equity Market
5. Equity Size
6. Bond Market
7. Bond Spread Size

The Equation of Multiple Factor Model

$$\text{Return} = R_f + (R_m - R_f) \beta_1 + \text{SMB } \beta_2 + \text{HML } \beta_3 + \text{UMD } \beta_4 + \dots + N \beta_n$$

Calculation of Beta using Characteristic Line Index of Systematic Risk

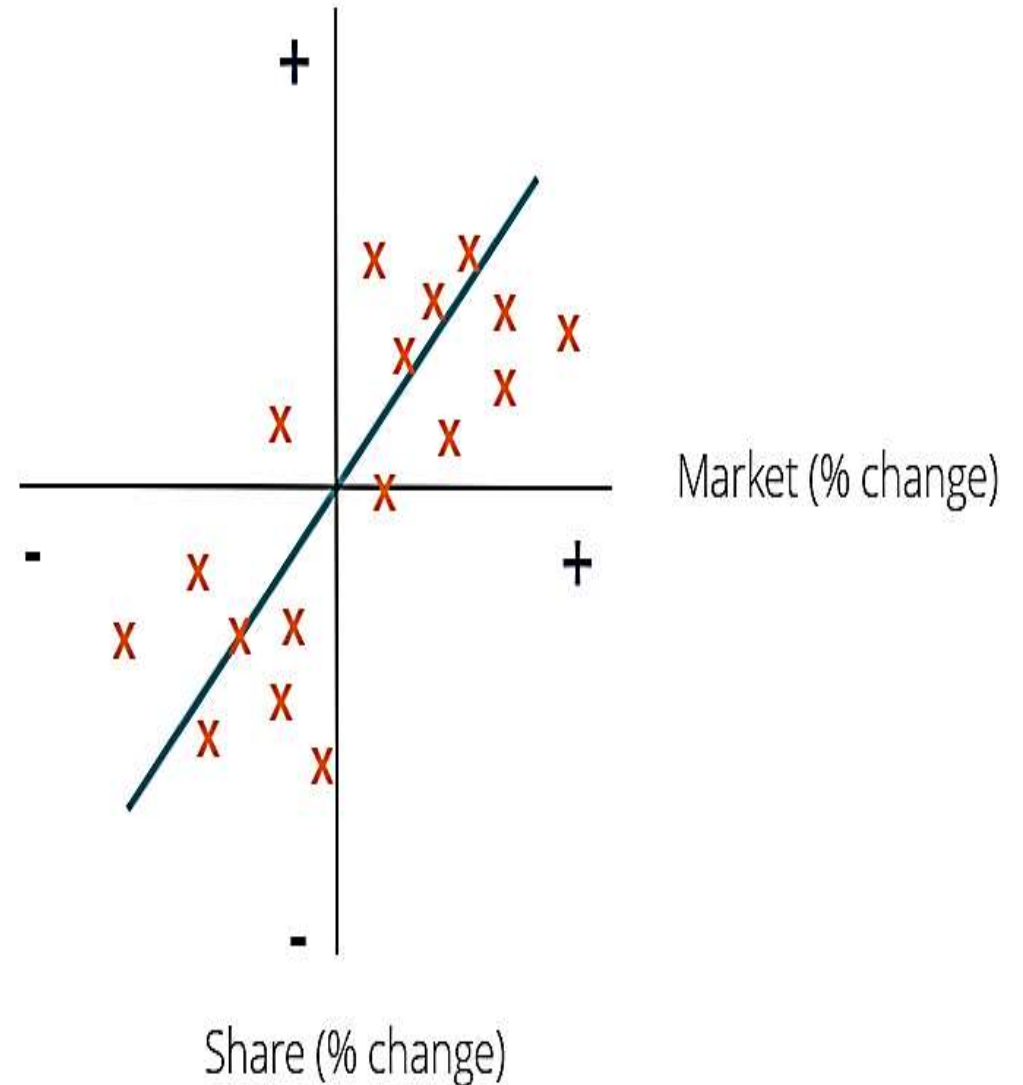
**Financial Management
Module-140**

Beta
Index of
Systematic Risk

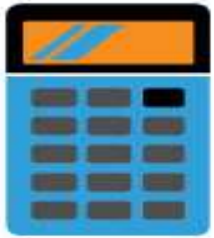


Characteristic Line

Beta = slope of the line



The Formula



**Beta
Formula**

=

$$\frac{\text{Covariance (Re, Rm)}}{\text{Variance (Rm)}}$$



Topic 141: Calculation of Portfolio Beta

Module 7: Risk & Return

Stock Beta Values

- **Beta Less than 1**
- **Beta Equal to 1**
- **Beta More than 1**



Portfolio Beta

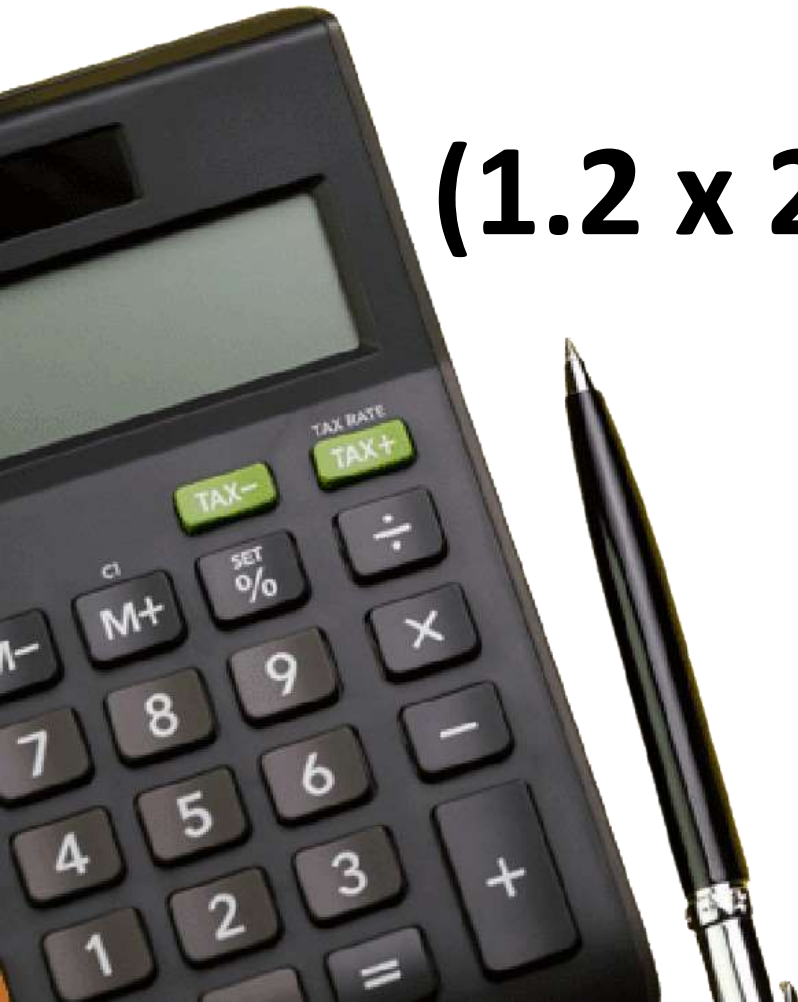


Company Beta Percentage of Investment		
A	1.2	20%
B	1.3	30%
C	1.1	40%
D	0.9	10%

Calculations

$$(1.2 \times 20\%) + (1.3 \times 30\%) + (1.1 \times 40\%) + (0.9 \times 10\%)$$

$$\beta = 1.16$$



Topic 142: Security Market Line

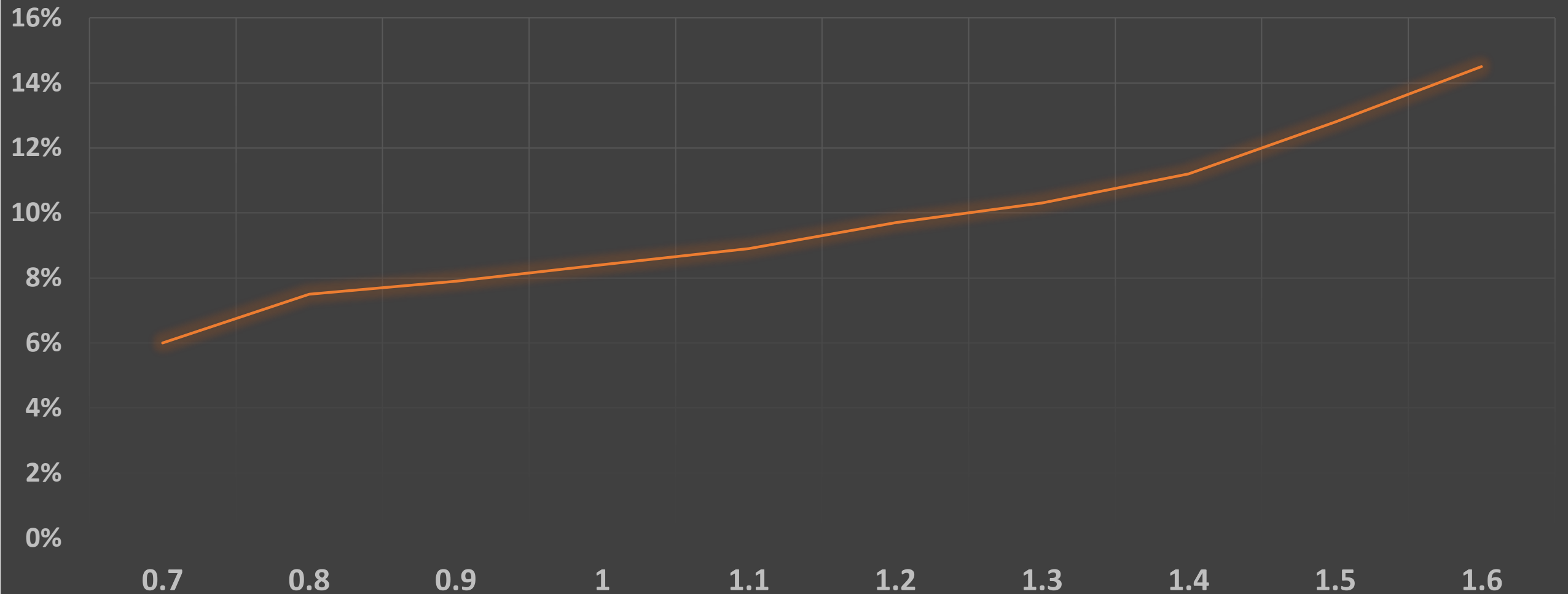
Module 7: Risk & Return

Returns at
Various
Values of
Betas

SML

Betas	Returns
0.7	6%
0.8	7.50%
0.9	7.90%
1	8.40%
1.1	8.90%
1.2	9.70%
1.3	10.30%
1.4	11.20%
1.5	12.80%

Security Market Line



Returns for Betas

Topic 143: Arbitrage Pricing Theory

Module 7: Risk & Return

Process of Arbitrage



Arbitrage Pricing Theory APT

ARBITRAGE
BUSINESS



Topic 144: Efficient Market Hypothesis

Module 7: Risk & Return

Efficient Markets



- **Weak Form**
- **Semi Strong Form**
- **Strong Form**



Be *strong*.

TOPIC 145-WORKING CAPITAL
ITS NEED IN BUSINESS

Working Capital Management
Module-8



- **Gross WC**
- **Net WC**

Why we need Working Capital?

- Inventory
- Receivables
- Prepayments
- Cash



TOPIC-146-APPROACHES OF WORKING CAPITAL MANAGEMENT

Working Capital Management Module-8



Hedging Approach



Conservative Approach

Aggressive Approach



TOPIC 147-HEDGING APPROACH OF WORKING CAPITAL MANAGEMENT

Working Capital Management Module-8

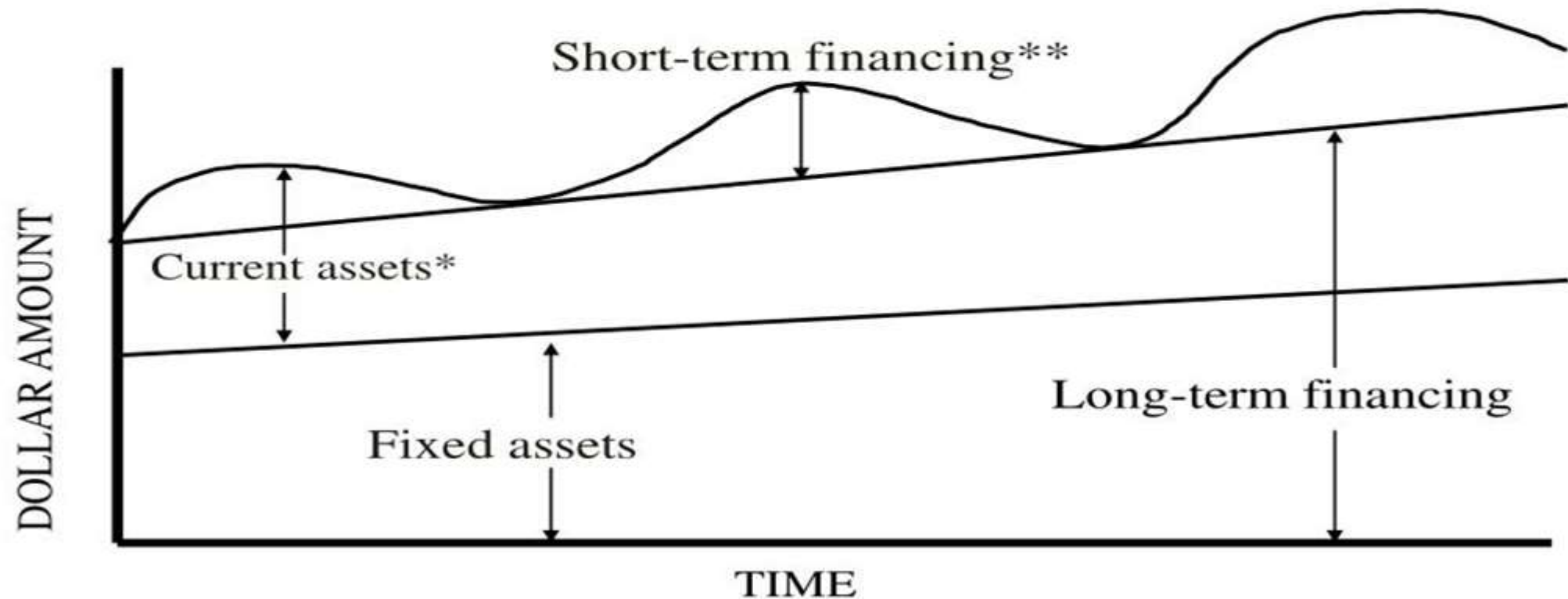
Hedging Approach

- Short Term Financing for Temporary Requirement
- Long Term Financing for Permanent Working Capital Requirement



Hedging or Maturity Matching Approach

A method of financing where each asset would be offset with a financing instrument of the same approximate maturity.

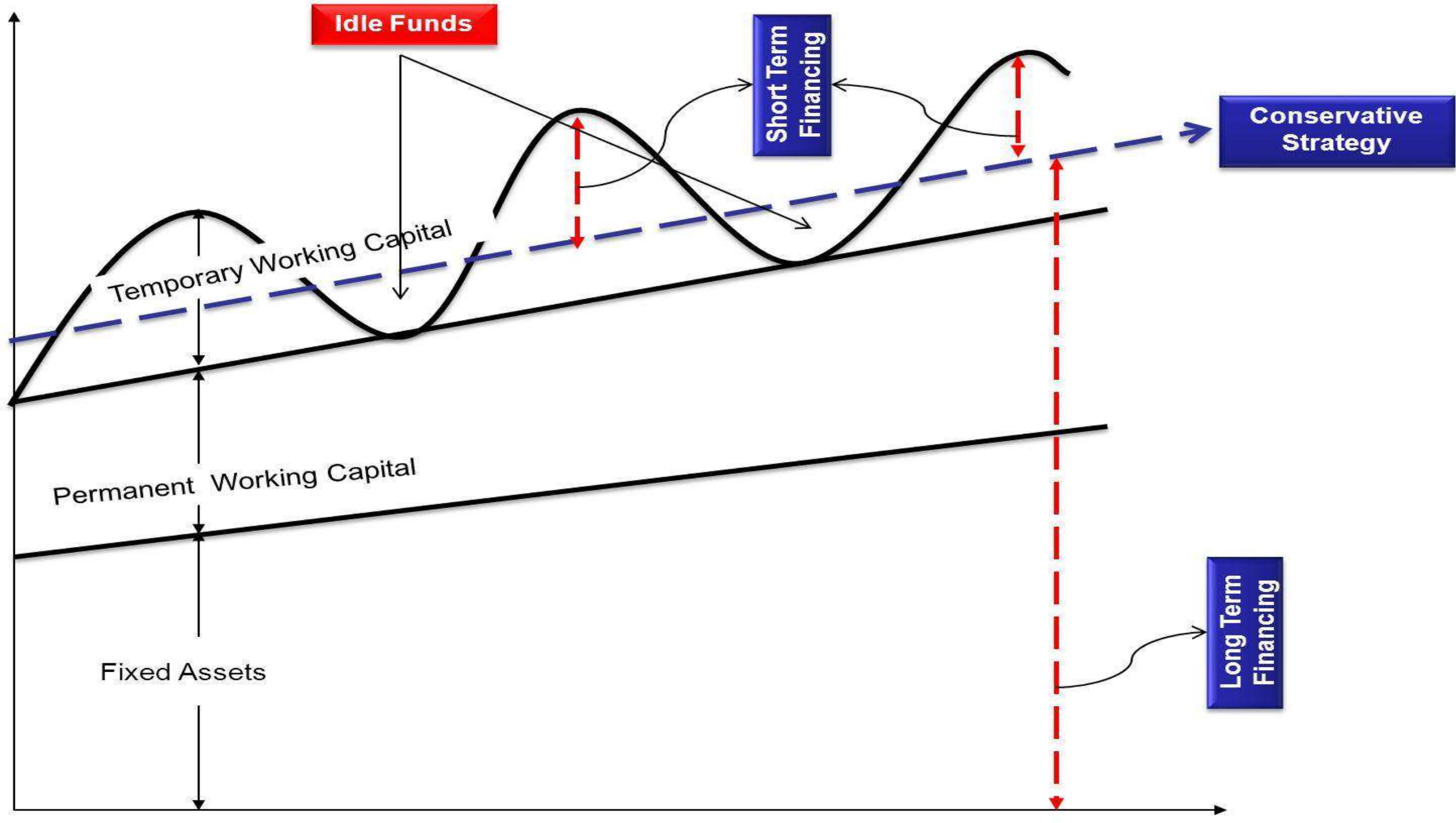


TOPIC 148-CONSERVATIVE APPROACH OF WORKING CAPITAL MANAGEMENT

Working Capital Management Module-8

Conservative Approach

Some part of temporary requirement of working capital is met by long term financing



In conservative approach the amount of profit is minimum and the cost of financing is minimum

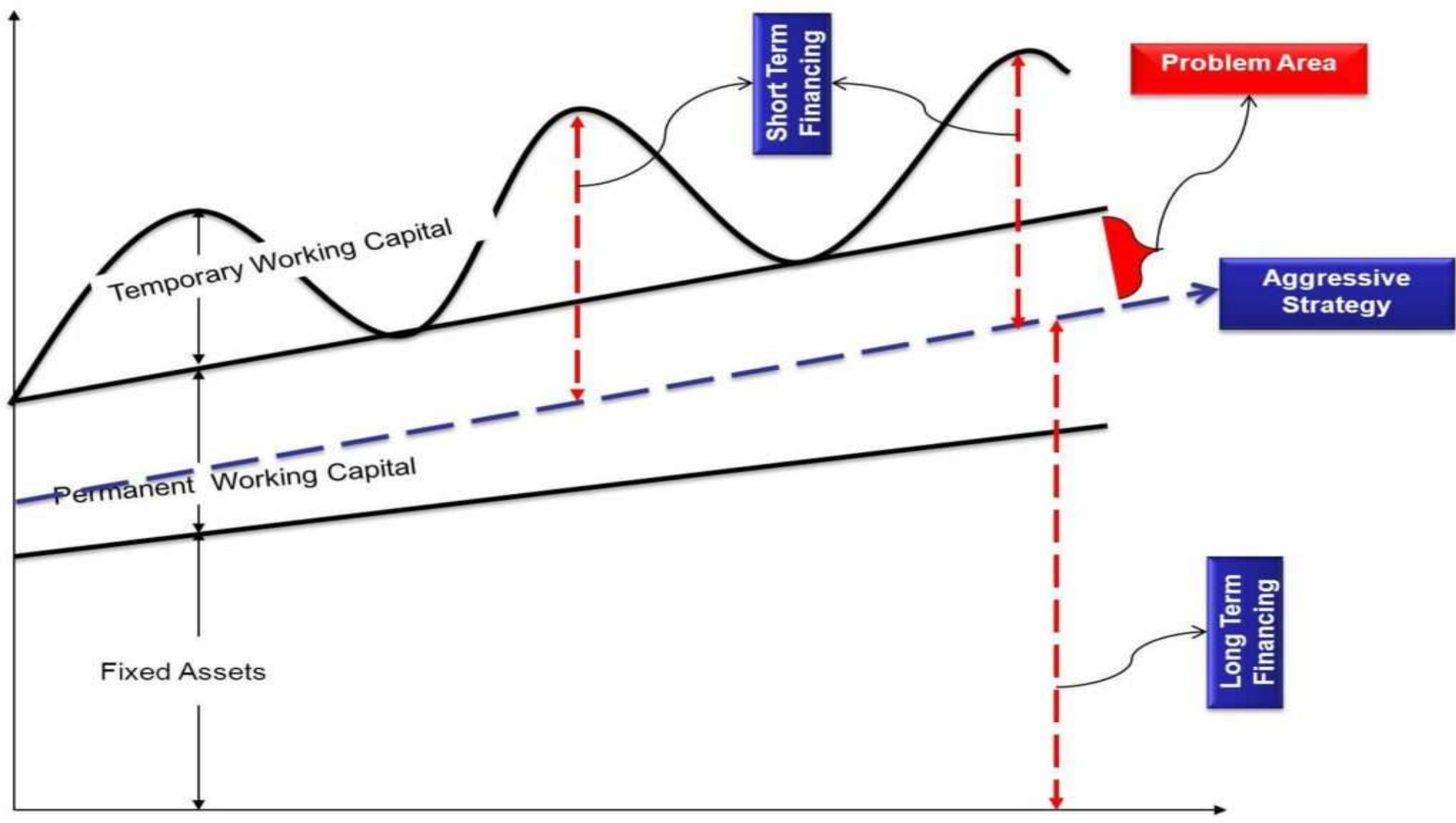


TOPIC 149–AGGRESSIVE APPROACH OF WORKING CAPITAL MANAGEMENT

Working Capital Management
Module-8

Aggressive Approach

Some part of permanent requirement of working capital is met by short-term financing



In aggressive approach profit is maximum as the cost of financing is minimum



TOPIC 150: ACCOUNTS
RECEIVABLE MANAGEMENT-1
CREDIT POLICY

Working Capital
Management
Module-8

Credit Policy

- Credit Period and Collection Period
- Credit Terms-2/10, n/30
- Cash Discount
- Maximum Amount of Credit
- Credibility



Credit Worthiness

- Credit Rating
- Standard & Poor, Moody's, PACRA
- Investment Grade and Junk

Credit Rating Scale



**TOPIC 151: ACCOUNTS
RECEIVABLE MANAGEMENT-2**

**Working Capital
Management
Module-8**



Fixed Cost



Variable Cost

Unit Contribution Margin

$$\text{Unit Contribution Margin Formula} = \text{Sales Per Unit} - \text{Total Variable Costs Per Unit}$$



Contribution Margin Ratio

$$\frac{\text{Contribution Margin}}{\text{Sales}}$$

Variable Cost Ratio

Variable Cost

Sales



opportunity cost

Opportunity cost represents the potential benefits an investor misses out on when choosing one alternative over another.

**TOPIC 152: ACCOUNTS
RECEIVABLE MANAGEMENT-3**

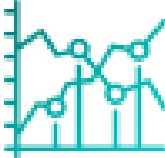
**Working Capital
Management
Module-8**

Credit Policy Decisions

Points to be Considered

- Addition Sales
- Additional Receivables
- Additional Margin
- Addition Cost

Accounts Receivable Turnover Ratio

$$\text{ART Ratio} = \frac{\text{Net Credit Sales} \text{  \text{ }{\text{Average Accounts Receivable} \text{ $$



Additional Receivables

Additional Sales

Receivables Turnover Ratio

**Topic 153: Accounts receivable
management-4**

**Working Capital
Management
Module-8**

Selling Price	Rs. 10 per unit
Variable Cost	Rs. 8 per unit
Contribution Margin	Rs. 2 per unit
Current Sales	Rs. 2,400,000
Current Credit Period	30-Days
New Credit Period	60-Days
Additional Sales = 25%	Rs. 600,000
Additional Units = $600,000/10$	60,000 Units
Opportunity Cost	20% per annum

Profitability of additional sales = (Contribution margin per unit) × (Additional units sold)	Rs. 2 × 60,000 units = Rs. 120,000
Additional receivables = (Additional sales revenue)/(Receivable turnover for new customers)	Rs. 600,000/6 = Rs. 100,000
Investment in additional receivables = (Variable cost per unit/Sales price per unit) × (Additional receivables)	0.80 × Rs. 100,000 = Rs. 80,000
Required before-tax return on additional investment = (Opportunity cost) × (Investment in additional receivables)	0.20 × Rs. 80,000 = Rs. 16,000

**Topic 154: Accounts Receivable
Management-5**

**Working Capital
Management
Module-8**

New Credit Policy Applicable to Current Sales



Selling Price	Rs. 10 per unit
Variable Cost	Rs. 8 per unit
Contribution Margin	Rs. 2 per unit
Current Sales	Rs. 2,400,000
Current Credit Period	30-Days
New Credit Period	60-Days
Additional Sales (assumed figure i.e. 15% of current sales)	Rs. 360,000
Additional Units = $360,000/10$	36,000 Units
Opportunity Cost	20% per annum

Profitability of additional sales = (Contribution margin per unit) × (Additional units sold)	Rs.2 × 36,000 units = Rs.72,000
Additional receivables of new sales = (New sales revenue)/(New receivable turnover for new customers)	Rs.360,000/6 = Rs.60,000
Investment in additional receivables of new sales = (Variable cost per unit/Sales price per unit) × (Additional receivables)	0.80 × Rs.60,000 = Rs.48,000
Level of receivables before credit period change = (Annual credit sales)/(Old receivables turnover)	Rs. 2,400,000/12 = Rs.200,000
New level of receivables associated with original sales = (Annual credit sales)/(New receivables turnover)	Rs. 2,400,000/6 = Rs. 400,000
Investment in additional receivable associated with original sales	Rs.400,000 - Rs.200,000 = Rs.200,000
Total investment in additional receivables associated with original sales	Rs.48,000 + Rs.200,000 = Rs.248,000
Required before-tax return on additional investment = (Opportunity cost) × (Total investment in additional receivables)	0.20 × Rs.248,000 = Rs.49,600

**Topic 155: Accounts Receivable
Management-6**

**Working Capital
Management
Module-8**

Bad Debts

Investing activities
Capital expenditures
Purchases of restaurant businesses
Cash used for investing activities

Financing activities
Long-term borrowing
Long-term financing issued
Common stock purchased
Preferred stock issued
Cash used for financing activities

Operating activities
Net income
Depreciation
Amortization
Losses on sale of assets
Gains on sale of assets
Cash used for operating activities

Cash

Selling Price	Rs. 10 per unit
Variable Cost	Rs. 8 per unit
Contribution Margin	Rs. 2 per unit
Current Sales	Rs. 2,400,000
Current Credit Period	30-Days
New Credit Period	60-Days
Additional Sales = 25%	Rs. 600,000
Additional Units = $600,000/10$	60,000 Units
Opportunity Cost	20% per annum
Bad Debts Expense	10%

Profitability of additional sales = (Contribution margin per unit) × (Additional units sold)	Rs. 2 × 60,000 units = Rs. 120,000
Additional receivables = (Additional sales revenue)/(Receivable turnover for new customers)	Rs. 600,000/6 = Rs. 100,000
Investment in additional receivables = (Variable cost per unit/Sales price per unit) × (Additional receivables)	0.80 × Rs. 100,000 = Rs. 80,000
Additional Bad Debts Expense = (Additional Sales x Bad debts percentage)	600,000 x 10% = Rs. 60,000
Required before-tax return on additional investment = (Opportunity cost) × (Investment in additional receivables)	0.20 × Rs. 80,000 = Rs. 16,000
Total additional cost = Bad debts + Opportunity cost	60,000 + 16,000 = Rs. 76,000

Impact of Cash Discount in Credit Policy



Inventory Management



Part 1

Financial Management Module 156



Types of Inventory

1. Stock in Trade

- a) Raw Material
- b) Work in Process
- c) Finished

2. Stores, Spares, and Tools





Centralized view of stock



Cost control



Avoid stockouts



Planning and forecasting



Time saving

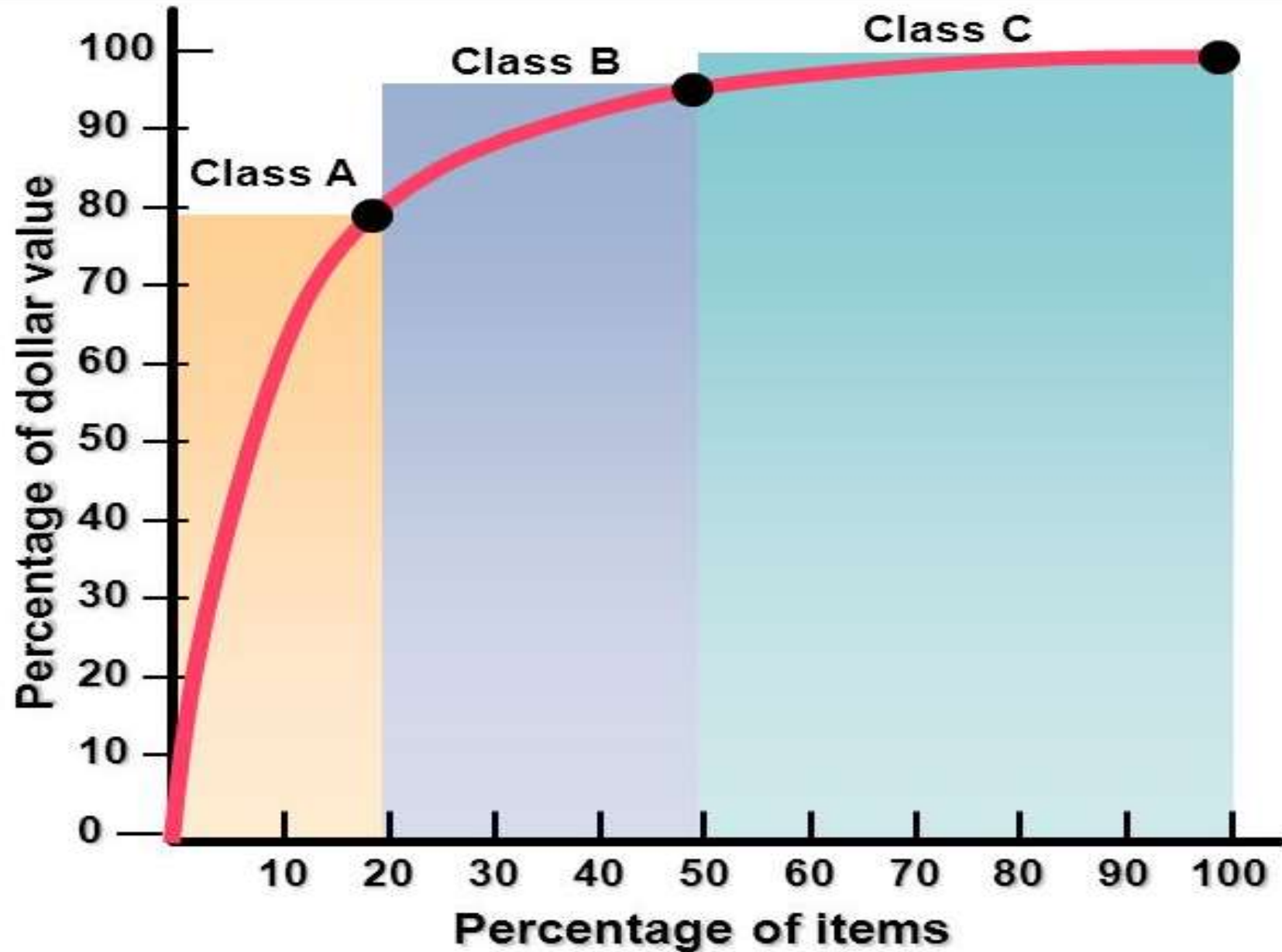
Inventory Management

Part 2

Financial Management Module 157

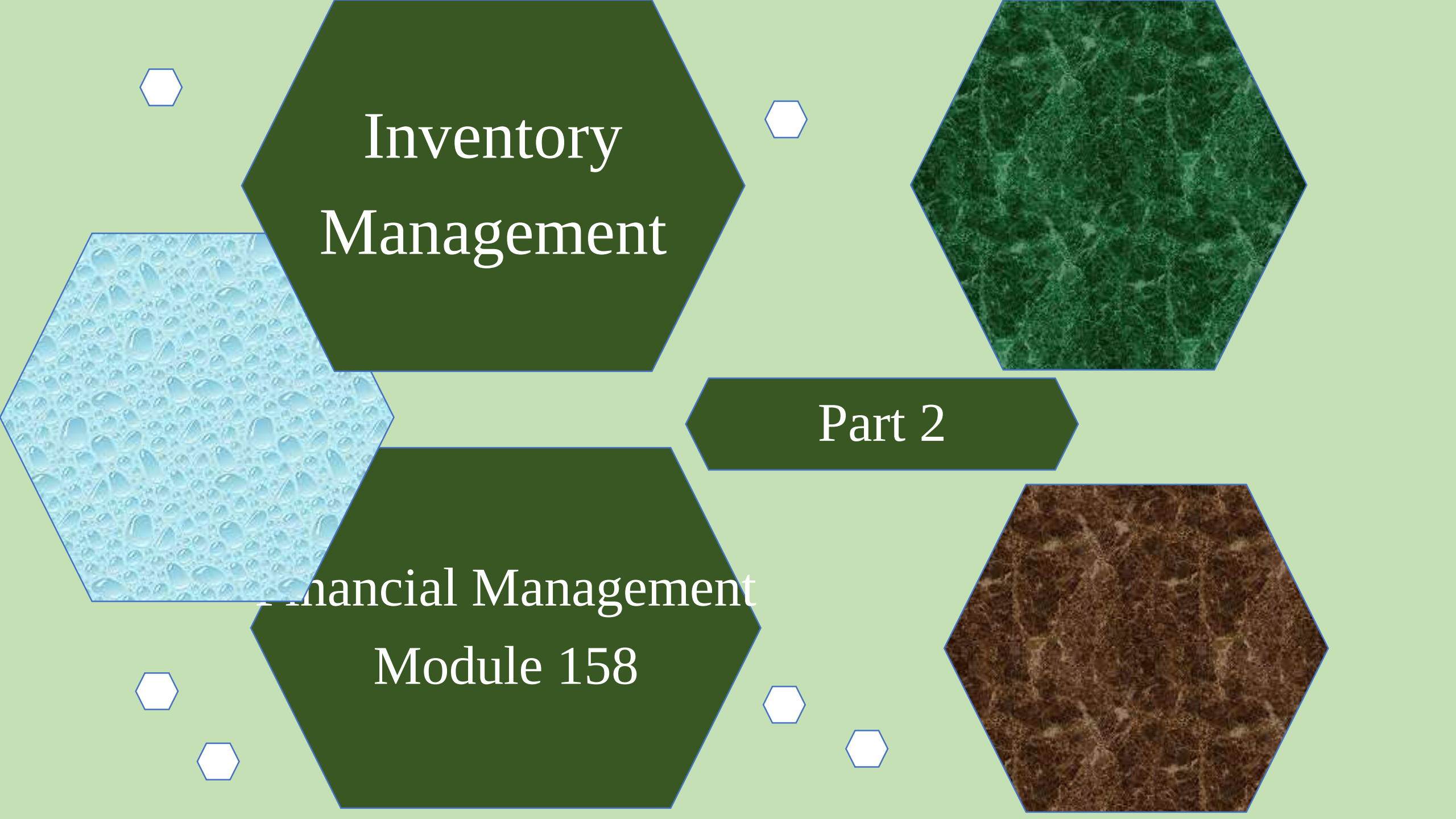


ABC Analysis



Safety Stock!





Inventory Management

Part 2

Financial Management Module 158

Lead Time

Lead Time?

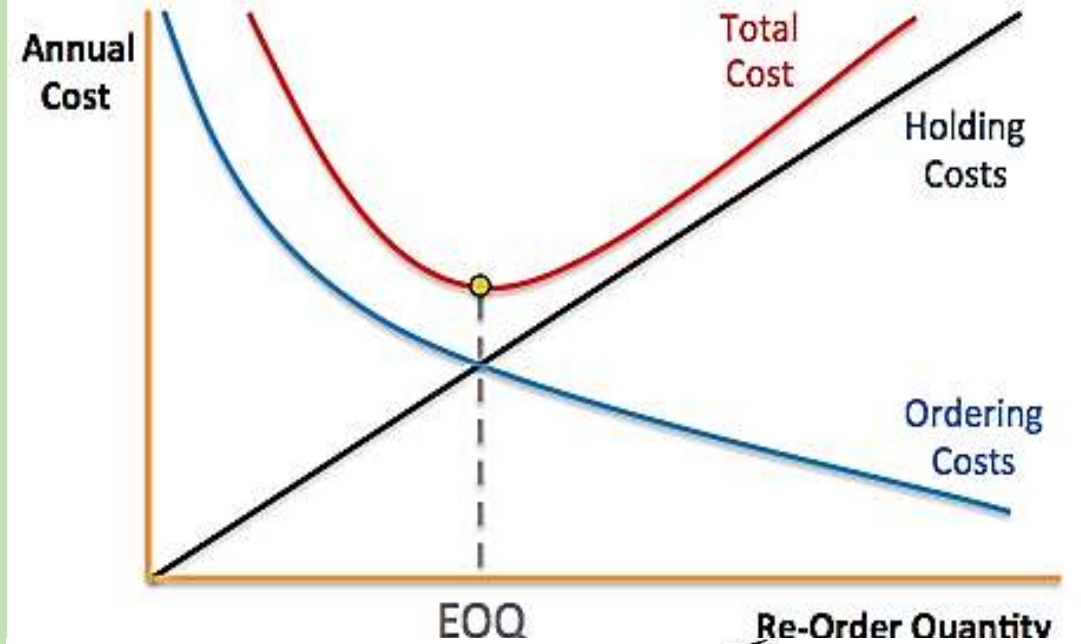


Lead Time Usage



Economic Order Quantity-EOQ

- Ordering Cost
- Carrying Cost
- Total Cost



EOQ =

$$\sqrt{2 \times (\text{Demand in units} \times \text{ordering costs}) / \text{Carrying costs per unit}}$$



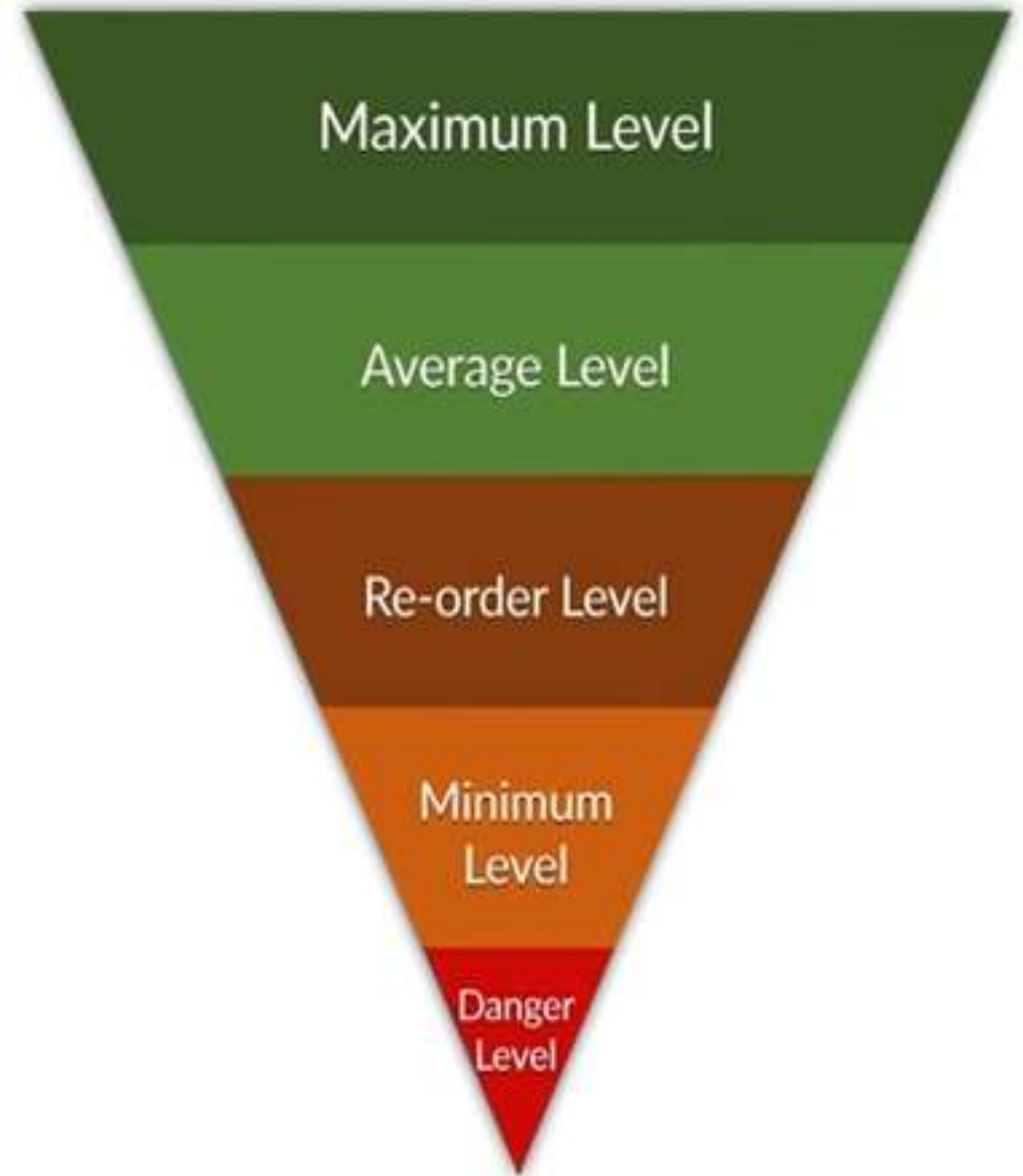
Inventory Management

Part 3

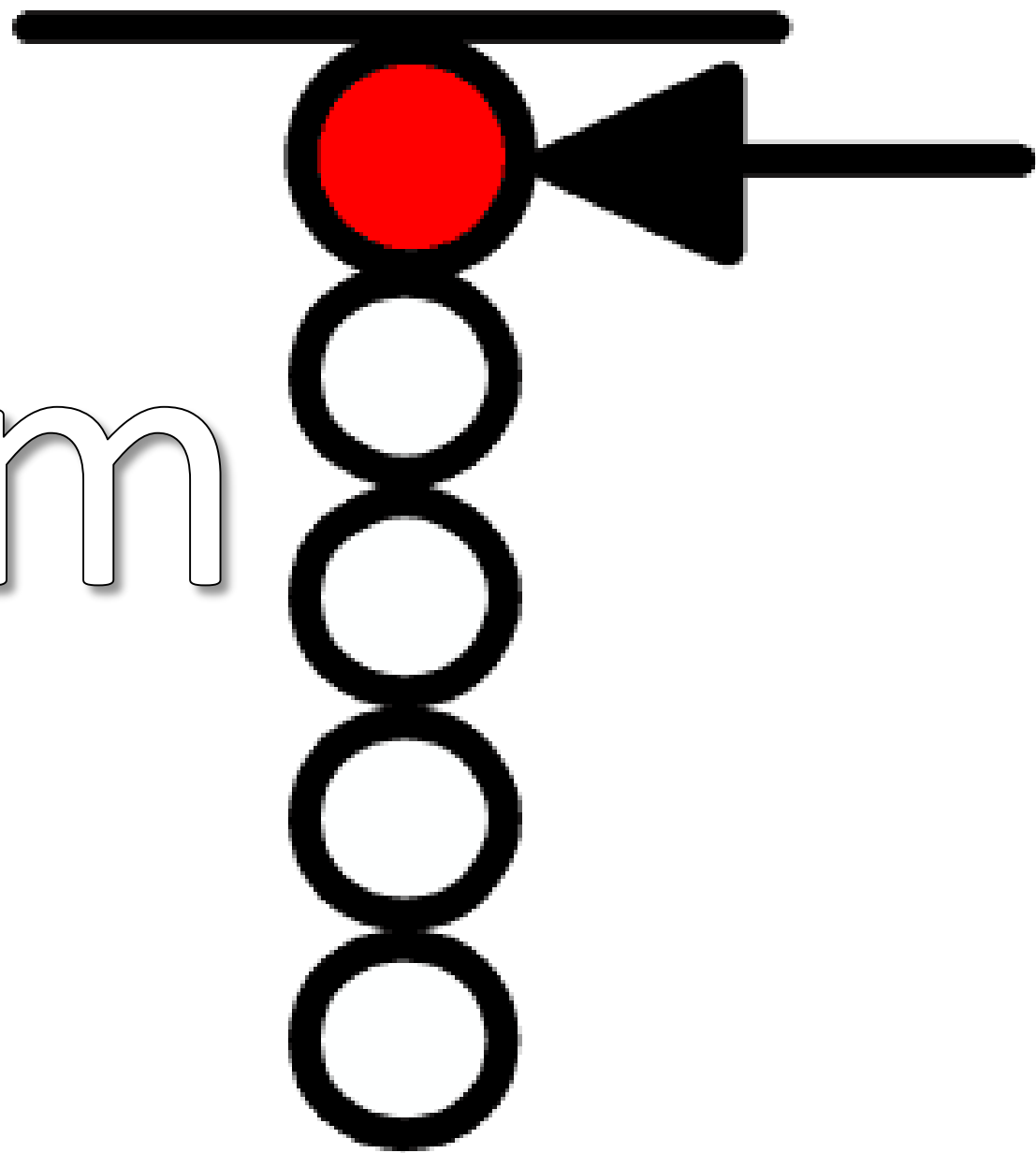
Financial Management Module 159



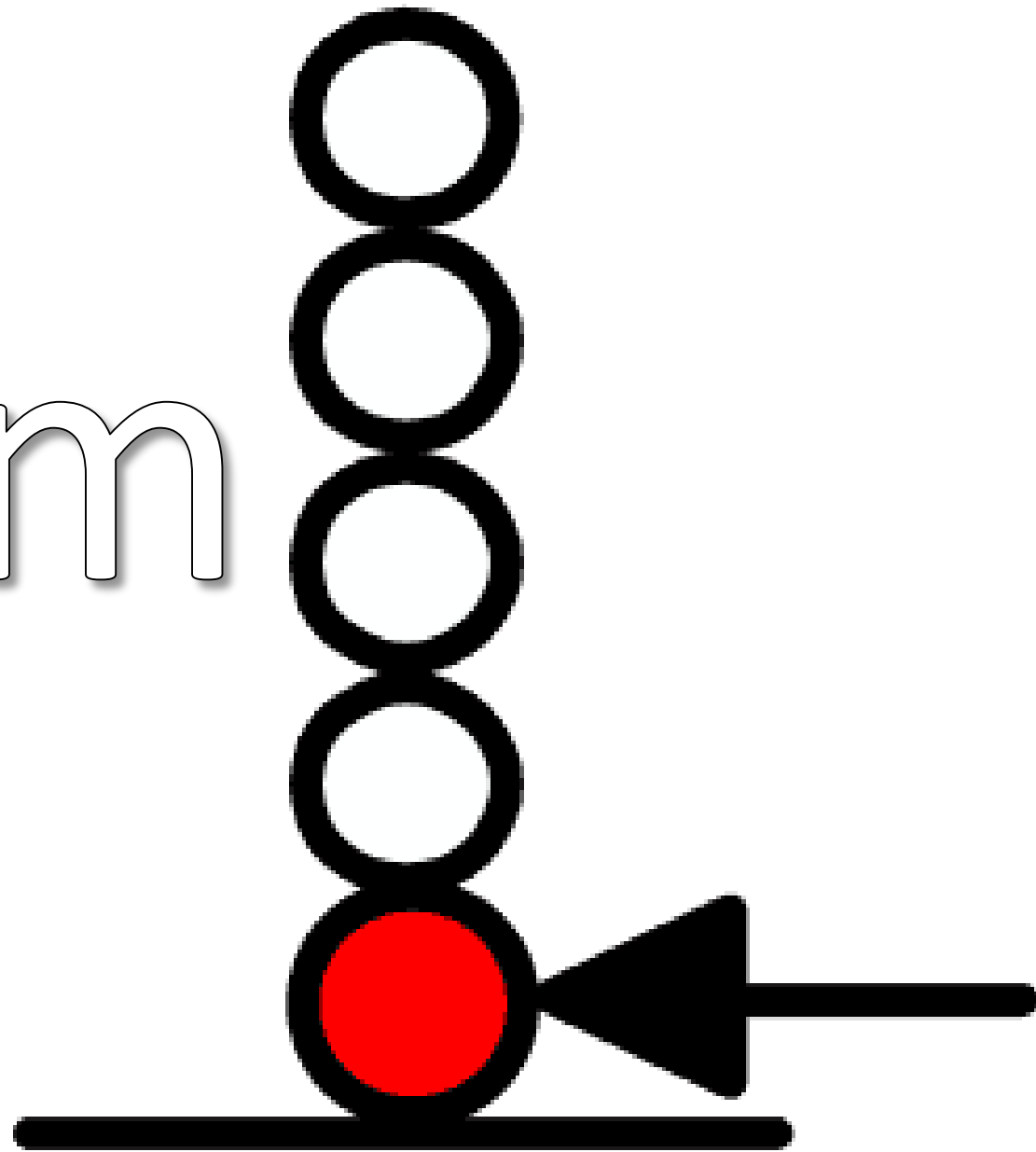
Inventory Levels



Maximum



Minimum



Danger Level



**Average
Level**



Reorder Level



Topic 160: Cash Management-1

Working Capital Management Module-8

- Cash in Hand
- Cash at Bank
- Cash Convertibles

Cash

Why We Need CASH?



Technical Insolvency



Topic 161: Cash Management-2

Working Capital Management Module-8

Motives of Holding Cash

- Transactional
- Speculative
- Precautionary



Cash Cycle



CASH CONVERSION CYCLE

**CCC = Days of Sales Outstanding PLUS
Days of Inventory Outstanding MINUS Days
of Payables Outstanding**

How to Improve Cash Cycle?



Topic 162: Cash Management-3

Working Capital Management Module-8

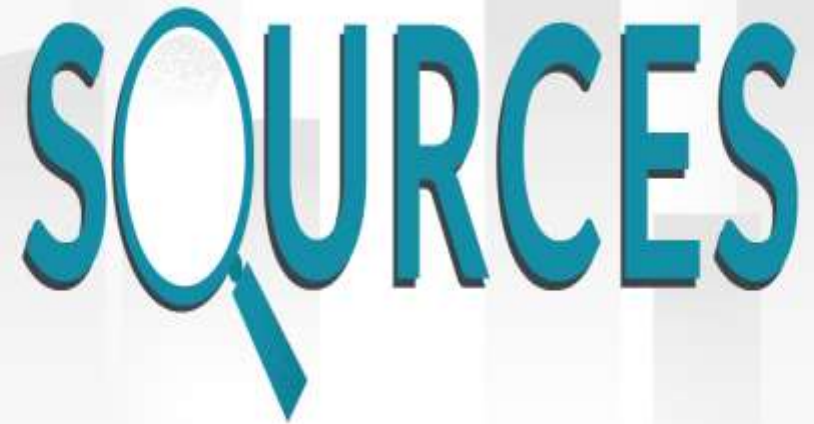
Cash Budget

- Opening Balance
- Receipts
- Payments
- Closing Balance
- Surplus or Deficit



Sources of Cash

- Sales/Revenue
- Loans/Debts
- Sale of Assets



SOURCES

Applications of Cash

- Purchases
- Expenses
- Repayment of Loan
- Investment



Deficit or Surplus

- Deficit
- Surplus



Topic 163: Capital Budgeting-1

Capital Budgeting Techniques Module-9

Capital Budgeting

- Concept
- Significance
- Techniques
- Criteria
- Cashflows

A photograph of a desk with various items: a wooden paperclip holder, a gold pen, and several financial documents. One document is titled 'FINANCE TRAILER'. Overlaid on the image is a white rounded rectangle containing the text 'Capital Budgeting' in a black, handwritten-style font.

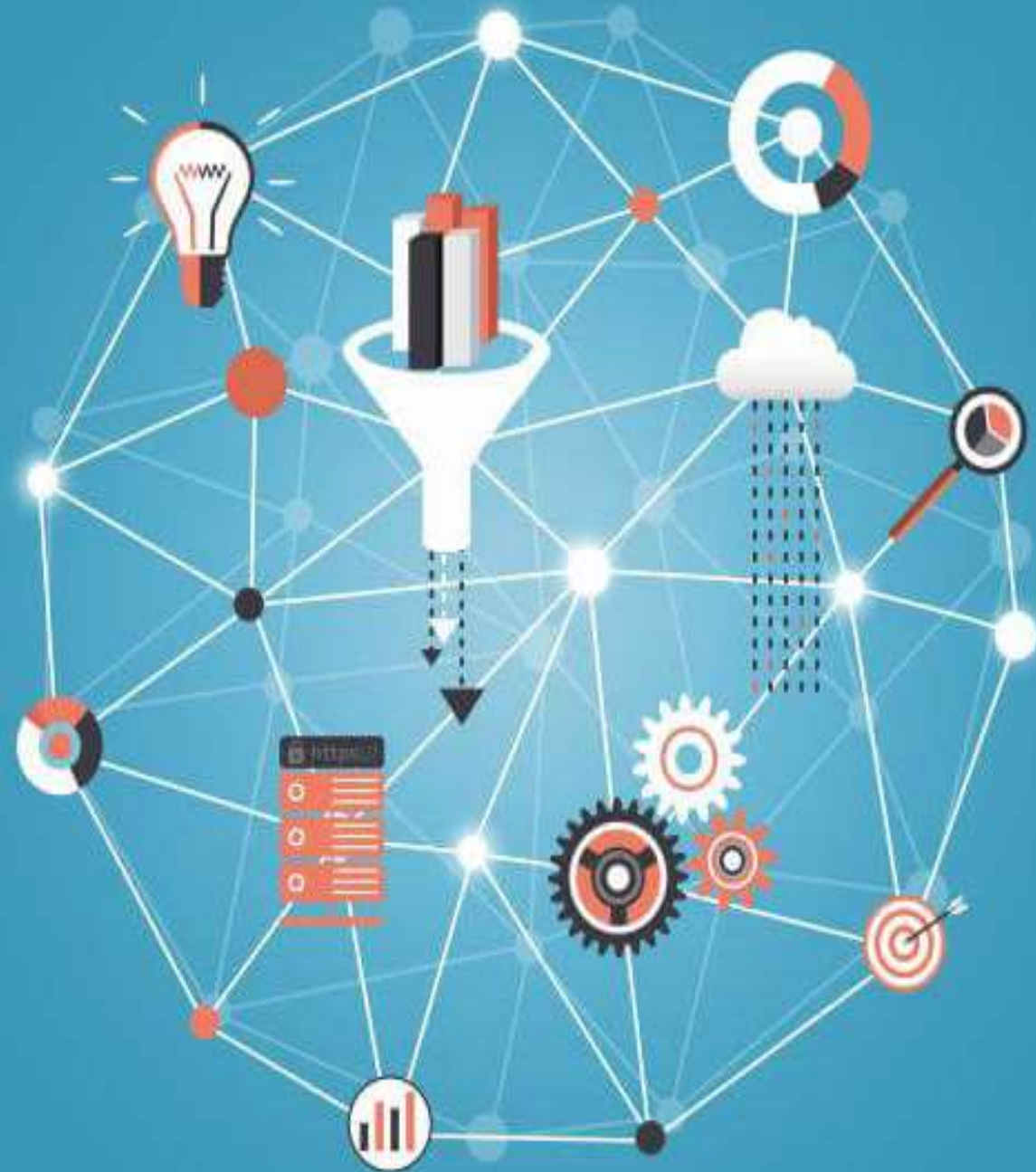
Capital
Budgeting

Topic 164: Capital Budgeting-2

Capital Budgeting Techniques Module-9

Capital Budgeting Techniques

- Payback Period
- Net Present Value
- Internal Rate of Return
- Profitability Index



Topic 165: Capital Budgeting-3

Capital Budgeting Techniques Module-9

Estimating Cash Flows

- Initial Cash Outflow
- Incremental Cash Inflows
- Terminal Cashflow



Topic 166: Capital Budgeting-4

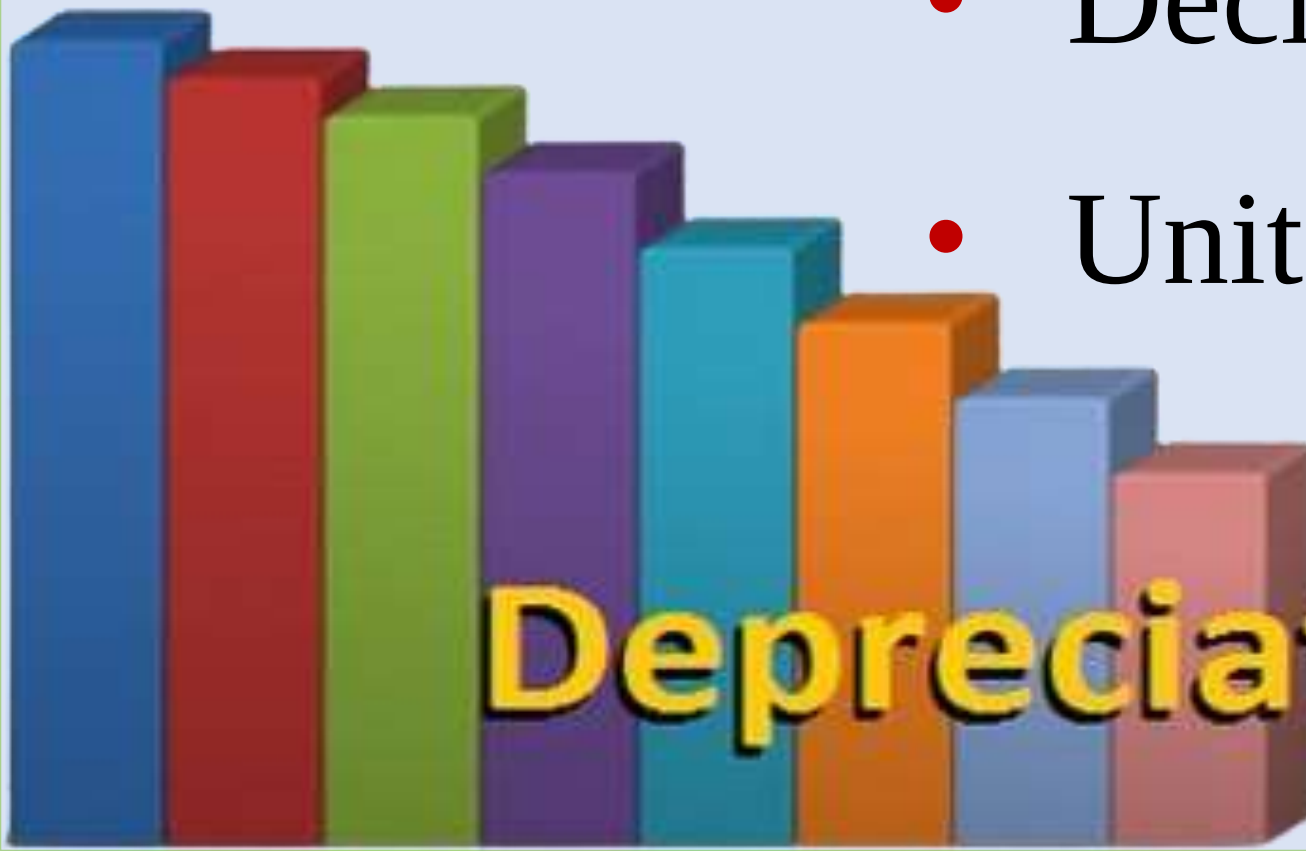
Capital Budgeting Techniques Module-9

Depreciation Methods and Cash Flows



Depreciation Methods

- Straight Line Method
- Declining Balance Method
- Units of Output Method



Depreciation

- Non-Cash Expense
- How to treat in Capital Budgeting?



Non-Cash Charge

FINANCIAL ANALYSIS

Topic 167: Capital Budgeting-5

Capital Budgeting Techniques Module-9

Initial Cash Outflow



1. Purchase Price

a. FOB

b. C & F

c. CIF



2. Direct Costs

- a. Freight
- b. Insurance
- c. Installation
- d. Setup
- e. Duties & Taxes



Topic 168:Capital Budgeting-6

**Capital Budgeting
Techniques
Module-9**

Incremental Cashflows



Incremental Cash Inflows		
a		Additional Revenue
b	-	Additional Depreciation
c	=	Net Change in Profit before Tax
d	-	Additional Income Tax
e	=	Net Change in Profit after Tax
f	+	Additional Depreciation
g	=	Incremental Cash Inflow

Topic 169: Capital Budgeting-7

Capital Budgeting Techniques Module-9

Incremental Cashflows-2



- Why deduct depreciation?
- Why add back depreciation?
- Difference between Incremental Cashflow and Profit
- Other non-cash expenses
 - Amortization
 - Depletion
 - Bad debts



Topic 170: Terminal Cash flow

**Capital Budgeting
Techniques
Module-9**

Cashflow in the last
year of useful life

THE END

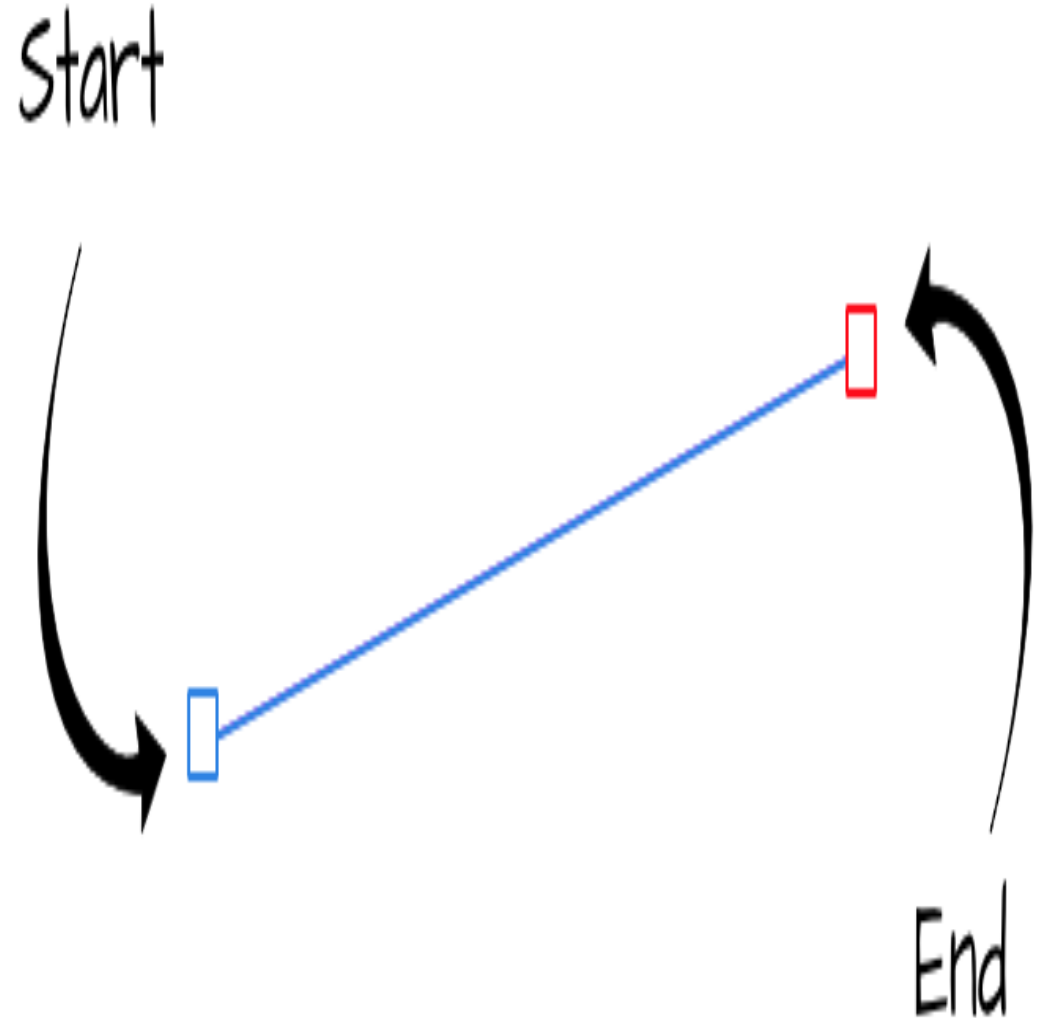
Terminal Cash Inflows		
a		Additional Revenue
b	-	Additional Depreciation
c	=	Net Change in Profit before Tax
d	-	Additional Income Tax
e	=	Net Change in Profit after Tax
f	+	Additional Depreciation
g	=	Incremental Cash Inflow of Last Year
h	+	Salvage Value of the Asset
i	-	Tax on Sale of the Asset
j	+	Release of Working Capital of the Project
k	=	Net Terminal Cashflow

Terminal Cash Inflows		
a		Additional Revenue
b	-	Additional Depreciation
c	=	Net Change in Profit before Tax
d	-	Additional Income Tax
e	=	Net Change in Profit after Tax
f	+	Additional Depreciation
g	=	Incremental Cash Inflow of Last Year
h	-	Cost to Dispose of Asset
i	+	Tax Saving on Disposal Cost
j	-	Increase in Net Working Capital
k	=	Net Terminal Cashflow

**Topic 171: Cash flows in Asset
Replacement Decisions**

**Capital Budgeting
Techniques
Module-9**

- Terminal Cashflow of Old Asset
- Initial Cashflow of New Asset



Cashflow of Replacement of Assets

a		Cost of New Asset (Outflow)
b	+	Capitalized Expenditure of New Asset
c	+	Taxes and Duties
d	+	Installation and Other Costs
f	+/-	Cost to Dispose of Old Asset/ Proceeds of Sale of the Old Asset
g	-/+	Tax Saving on Disposal Cost/ Tax on Proceeds of Salvage Value
h	+/-	Increase/Decrease in Net Working Capital
i	=	Net Cashflow of Replaced Asset

Capital Budgeting Techniques

Module-9

Topic 172: Discounted Cashflow Methods-DCFs

- Discounting/Present Value
- Discounted Cashflow Methods
 - Payback Period
 - IRR
 - NPV
 - PI



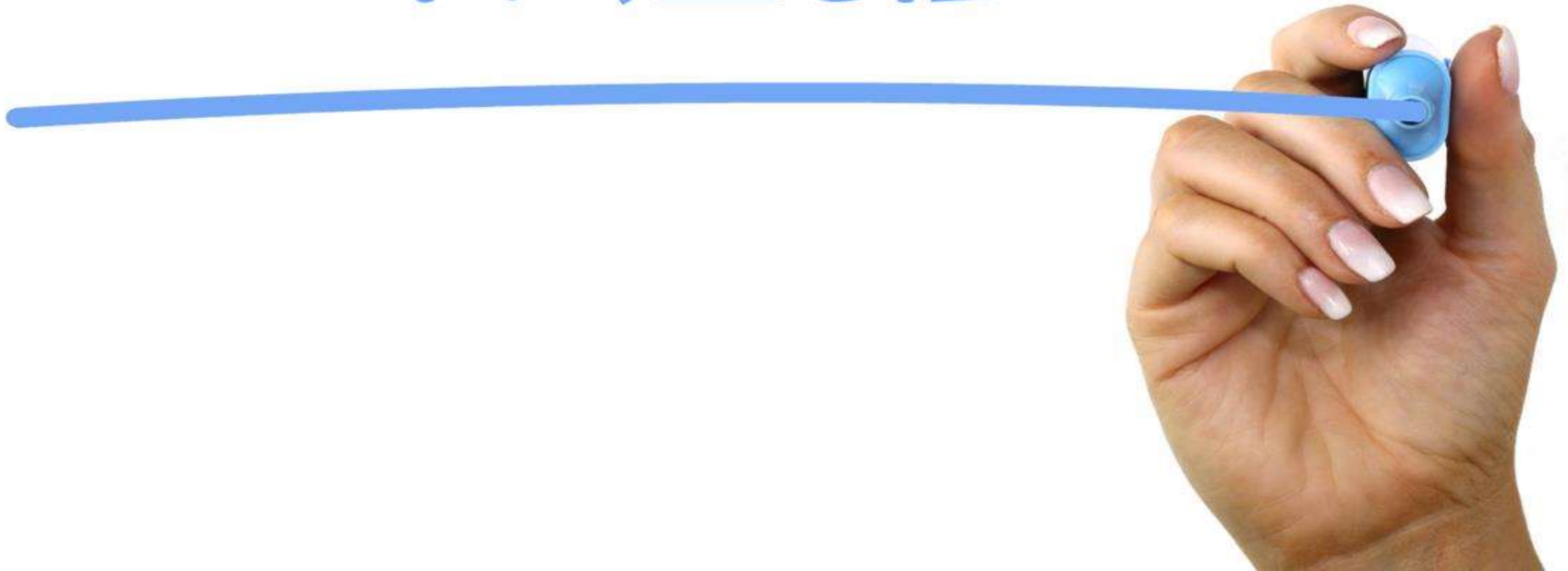
PAYBACK

A close-up photograph of a desk with various items. In the foreground, a white document is open, displaying the text 'IRR' in large, bold, black letters, followed by 'Internal Rate of Return' in a smaller, black font, underlined. To the left of the document is a stack of white papers. In the bottom left corner, the tip of a black pen with gold-colored accents is visible. In the background, a white calculator with grey buttons is partially visible, along with a black pen with a gold-colored clip. A yellow notebook with a red bookmark is also visible on the left side of the frame. The desk surface is dark and textured.

IRR

Internal Rate of Return

NET PRESENT
VALUE



PROFITABILITY INDEX (PI)

also known as prof

A hand holding a white marker is shown in the process of writing the text 'also known as prof'. The hand is positioned in the center-right of the frame, with the marker tip touching the text. The hand is wearing a dark blue, ribbed sweater sleeve. The background is plain white.

Capital Budgeting Techniques

Module-9

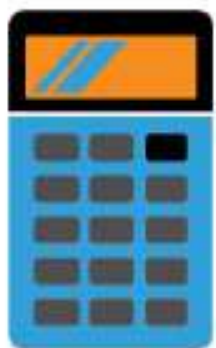
Topic 173: Payback Period Method-1

Payback Period

Years in which original investment is recovered



Payback Period Method		
Year	Cash Flow	Accumulated Cashflow
	Rs.	Rs.
0	(100,000)	
1	20,000	20,000
2	40,000	60,000
3	40,000	100,000
4	50,000	140,000
5	60,000	200,000



**Payback Period
Formula**

=

$$\frac{\text{Initial Investment OR Original Cost of the Asset}}{\text{Cash Inflows}}$$





ACCEPTANCE

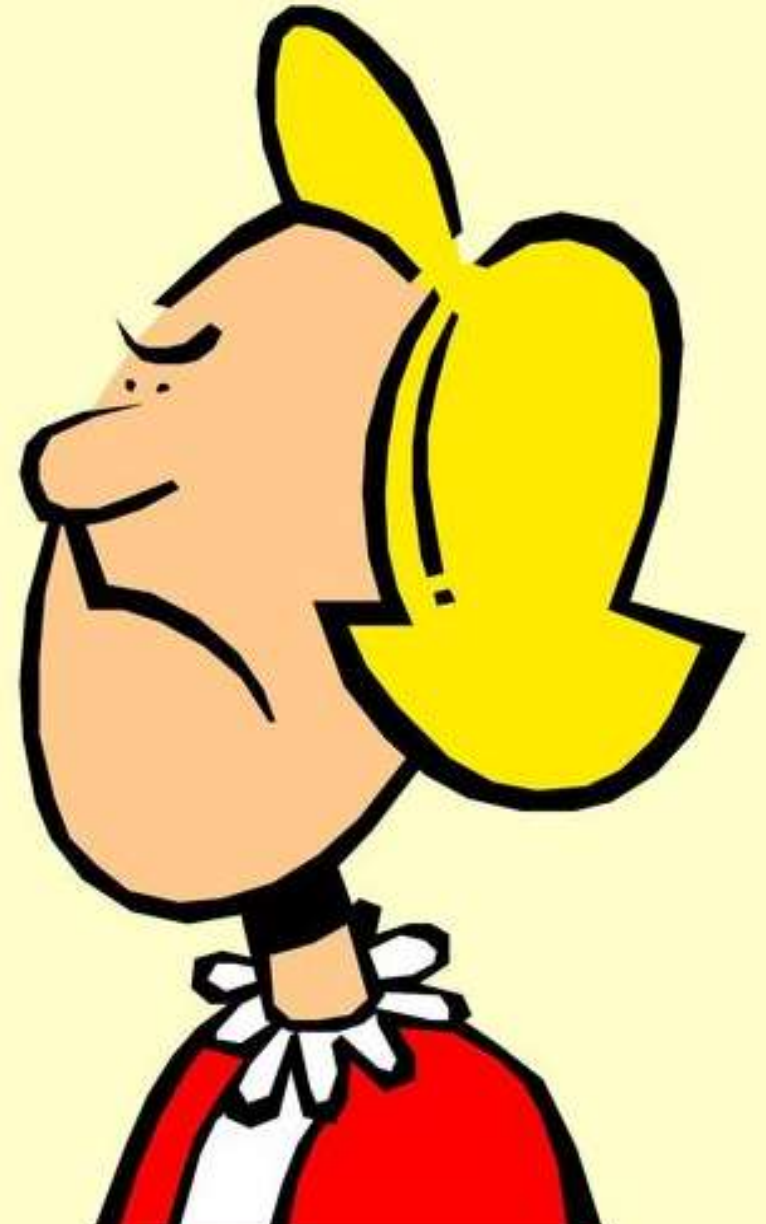
Capital Budgeting Techniques

Module-9

Topic 174: Payback Period Method-2

**Ignores the
time value
of money.**

**Ignores cash
flows after
the payback
period.**



Payback Period Method			
Payback Period Method using Present Value			
Year	Cash Flow	PVs of Cash Flow	Accumulated Cashflow
	Rs.	@ 10% Rs.	Rs.
0	(100,000)	(100,000)	
1	20,000	18,182	18,182
2	30,000	24,793	42,975
3	40,000	30,053	73,028
4	50,000	34,151	107,179
5	60,000	37,255	144,434

Comparison of Two Projects				
	Project-A		Project-B	
Year	Cashflow	Accumulated Cashflow	Cashflow	Accumulated Cashflow
	Rs.	Rs.	Rs.	Rs.
0	(100,000)		(100,000)	
1	20,000	20,000	20,000	20,000
2	30,000	60,000	30,000	60,000
3	40,000	100,000	30,000	90,000
4	50,000	150,000	60,000	150,000
5	60,000	210,000	60,000	210,000
6			60,000	270,000
7			60,000	330,000
8			50,000	380,000

Capital Budgeting Techniques

Module-9

Topic 175: Net Present Value-1

Net Present Value

The difference
between the sum of
PVs of cash inflows
and cash outflow



Net Present Value		
Year	Cash Flow	PVs of Cash Flow
	Rs.	@ 10% Rs.
0	(100,000)	(100,000)
1	20,000	18,182
2	30,000	24,793
3	40,000	30,053
4	50,000	34,151
5	60,000	37,255
		144,434

$$NPV = 144,434 - 100,000 = 44,434$$



Capital Budgeting Techniques

Module-9

Topic 176: Net Present Value-2

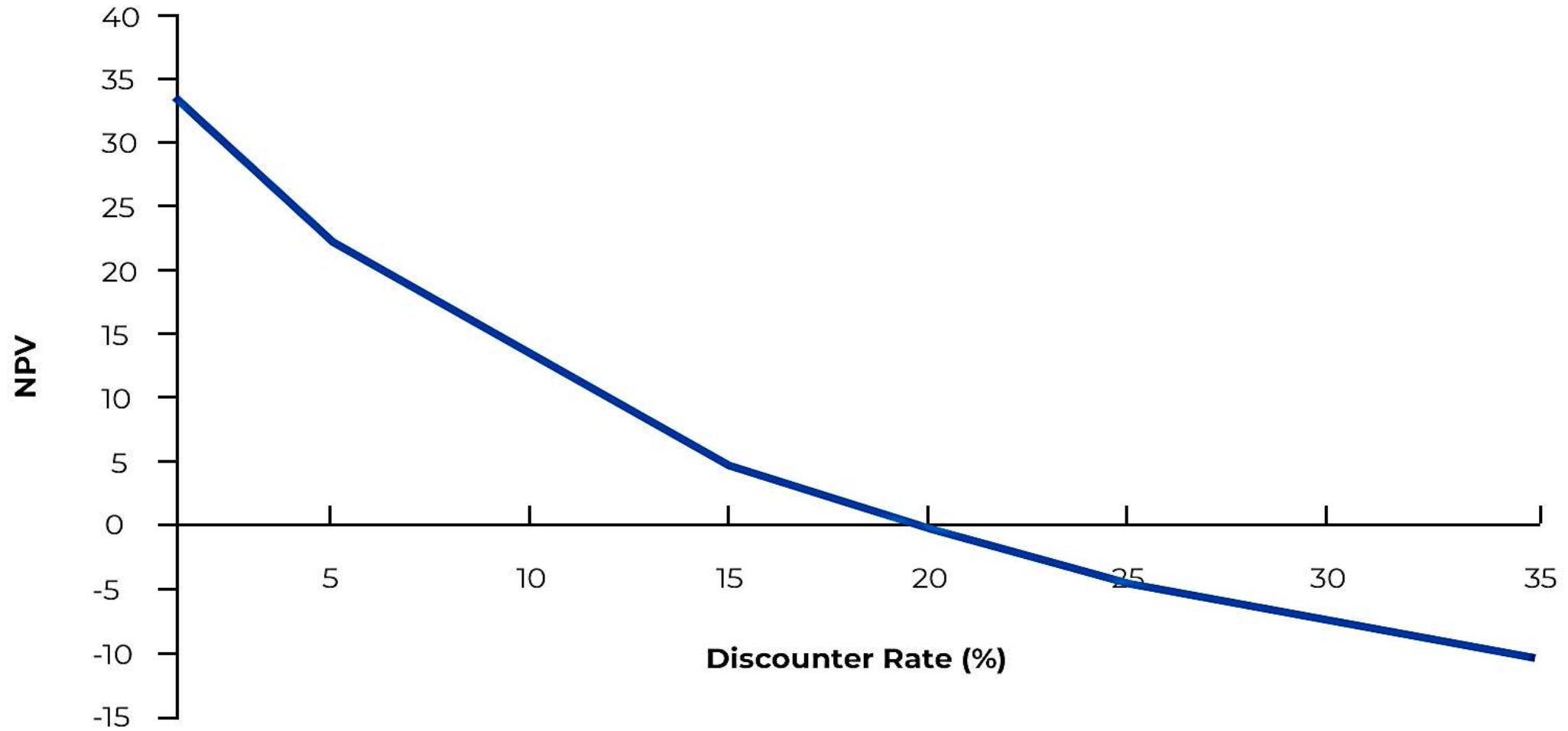
NPV and Discount Rate

Higher is the discount
rate lesser is the NPV





Example of an NPV Profile



Net Present Value		
Sr. No.	Rate	NPV
	%	Rs.
1	5%	21,000
2	10%	11,000
3	15%	5,000
4	20%	0
5	25%	(5,000)

Capital Budgeting Techniques

Module-9

Topic 177: Profitability Index

Profitability Index-PI

It is the ratio of sum of PVs of inflows and outflows



Profitability Index

Year	Cash Flow	PVs of Cash Flow
	Rs.	@ 10% Rs.
0	(100,000)	(100,000)
1	20,000	18,182
2	30,000	24,793
3	40,000	30,053
4	50,000	34,151
5	60,000	37,255
		144,434

$$\text{PI} = 144,434 / 100,000 = 1.44$$

Acceptance and Rejection Criteria

If PI is less than 1
REJECT



Capital Budgeting Techniques

Module-9

Topic 178: Internal Rate of Return-IRR Method-1

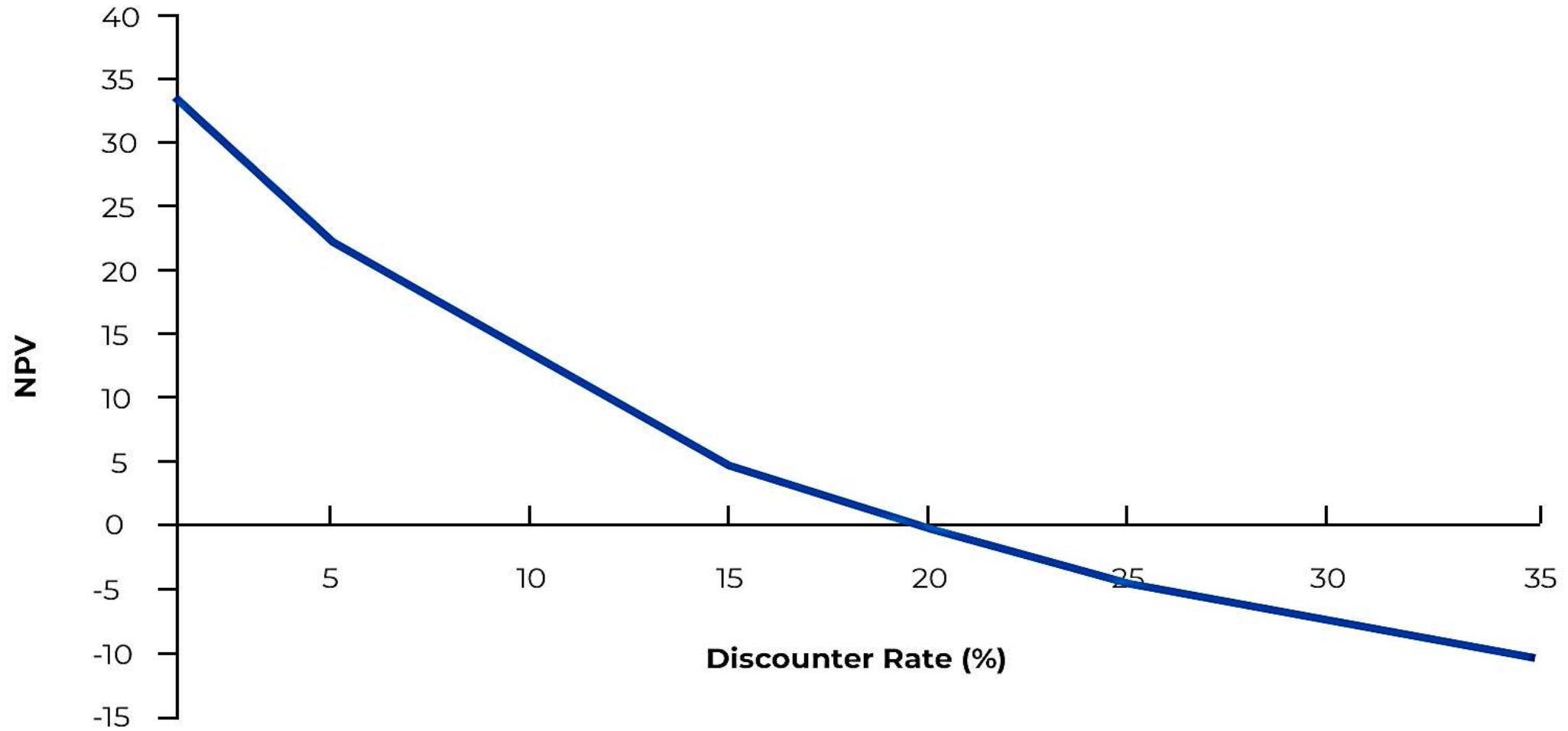
IRR

The rate of return for which the sum of PVs of inflows is equal to outflows





Example of an NPV Profile



Capital Budgeting Techniques

Module-9

Topic 179: Internal Rate of Return-IRR Method-2

Internal Rate of Return-IRR				
Year	Cash Flow	PVs of Cash Flow	PVs of Cash Flow	PVs of Cash Flow
	Rs.	@ 10% Rs.	@ 20% Rs.	@ 25% Rs.
0	(100,000)			
1	20,000	18,182	16,667	16,000
2	30,000	24,793	20,833	19,200
3	40,000	30,053	23,148	20,480
4	50,000	34,151	24,112	20,480
5	60,000	37,255	24,112	19,661
		144,434	108,872	95,821

Acceptance Criteria



Capital Budgeting Techniques

Module-9

Topic 180: Interpolation

Interpolation

To determine an
unknown rate lying
between two known
rates

what are other
words for
interpolate?



introduce, insert, interject,
interpose, add, inject,
intercalate, enter, insinuate,
include



Internal Rate of Return-IRR				
Year	Cash Flow	PVs of Cash Flow	PVs of Cash Flow	PVs of Cash Flow
	Rs.	@ 10% Rs.	@ 20% Rs.	@ 25% Rs.
0	(100,000)			
1	20,000	18,182	16,667	16,000
2	30,000	24,793	20,833	19,200
3	40,000	30,053	23,148	20,480
4	50,000	34,151	24,112	20,480
5	60,000	37,255	24,112	19,661
		144,434	108,872	95,821

$$\text{IRR} = 20\% + 5\%[(108,872 - 100,000)/(108,872 - 95,821)]$$

$$\text{IRR} = 0.2 + 0.034 = 0.234 = 23.4\%$$



ACCEPT



REJECT

Capital Budgeting Techniques

Module-9

Topic 181: Capital Rationing-1

Rationing

Controlled distribution of a scarce resource



Capital Rationing

When profitable investment options are more than the funds available



Project	Initial Cash Outflow	IRR %	NPV	PI
A	50,000	15	12,000	1.24
B	35,000	19	15,000	1.43
C	30,000	28	42,000	2.4
D	25,000	26	1,000	1.04
E	15,000	20	10,000	1.67
F	10,000	37	11,000	2.1
G	10,000	25	13,000	2.3
H	1,000	18	100	1.1

Project	IRR	NPV	Outflow
F	37%	11,000	10,000
C	28%	42,000	30,000
D	26%	1,000	25,000
		54,000	65,000

Project	NPV	NPV	Outflow
C	42,000	42,000	30,000
B	15,000	15,000	35,000
		57,000	65,000

Project	PI	NPV	Outflow
C	2.4	42,000	30,000
G	2.3	13,000	10,000
F	2.1	11,000	10,000
E	1.67	10,000	15,000
		76,000	65,000

Capital Budgeting Techniques

Module-9

Topic 182: Capital Rationing-2

Decision Making in Capital Rationing

1. Maximum NPV
2. Why on the basis of NPV?
3. Why the combination selected on the basis of PI gives maximum NPV?



Capital Budgeting Techniques

Module-9

Topic 183: Multiple IRRs

Multiple IRRs

In some unusual cash flows we may end up with multiple IRRs i.e. two IRRs for one project.



Multiple IRRs			
Year	Cash Flows	PVs @ 25%	PVs @ 400%
0	-1600	-1600	-1600
1	10000	8000	2000
2	-10000	-6400	-400
		0	0

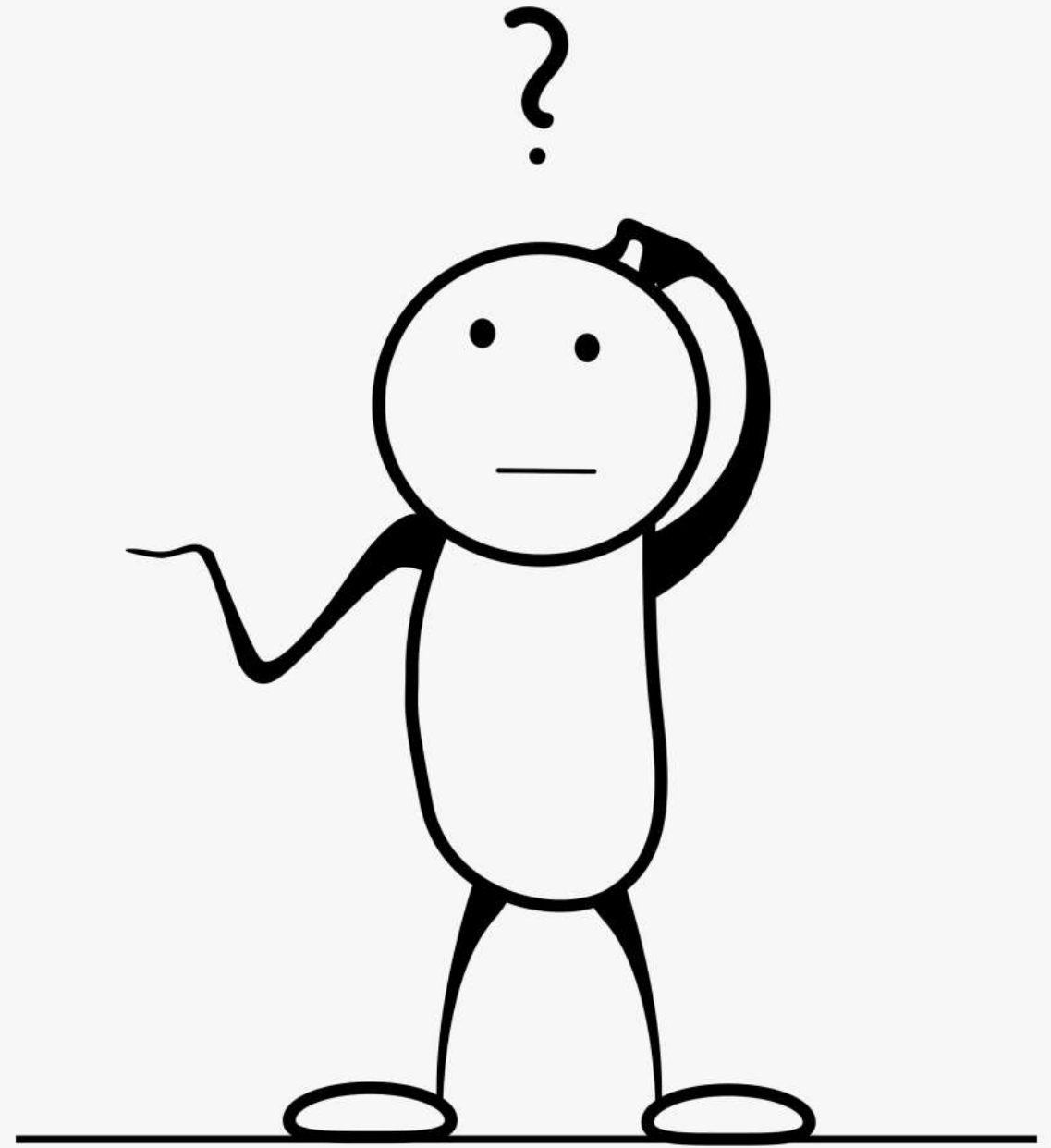
Capital Budgeting Techniques

Module-9

Topic 184: Sensitivity Analysis in Capital Budgeting

Sensitivity Analysis

What If? Things
Change



Sensitivity Analysis

The impact on the outcome on the basis of change in Point Estimates, like Initial Outflow, Net Operating Income, Salvage Value etc.



Sensitivity Analysis					
	-10%	-5%	Original Estimate	5%	10%
Initial Cash Outflow	90,000	95,000	100,000	105,000	110,000
Net Operating Income	22,500	23,750	25,000	26,250	27,500
Salvage Value	9,000	9,500	10,000	10,500	11,000

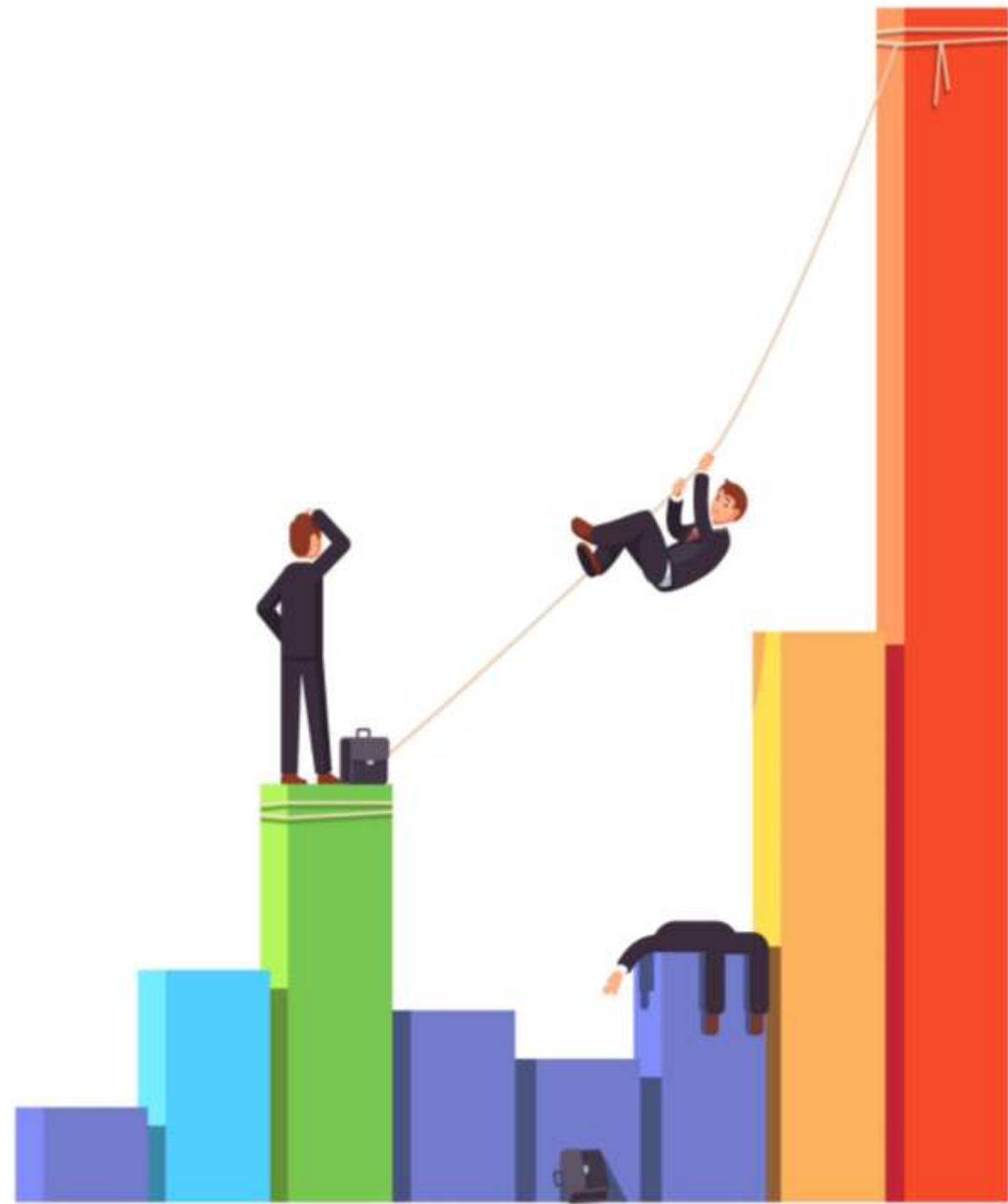
Capital Budgeting Techniques

Module-9

Topic 185: Risk and Capital Budgeting

Risk

Uncertainty: Chances of variation



Economy	Probability	Cashflow
		Rs.
High Boom	0.1	10,000
Boom	0.2	9,000
Normal	0.3	7,000
Recession	0.3	5,000
High Recession	0.1	4,000

Economy	Probability	Cashflow	CF x P	(CF-Ave)2 x P
		Rs.		
High Boom	0.1	10,000	1000	1,024,000
Boom	0.2	9,000	1,800	968,000
Normal	0.3	7,000	2,100	12,000
Recession	0.3	5,000	1,500	972,000
High Recession	0.1	4,000	400	784,000
			6,800	3,760,000
				SD = 1,939

Topic 186: Cost of Capital

**Cost of capital
Module-10**

Types of Capital

- Equity
- Debt



- Cost of Capital
- Required Rate of Return
 - Market Return
 - Opportunity Cost
 - Other Benchmarks



Cost of Capital

- Cost of Equity
- Cost of Debt



Topic 187: Cost of Equity Capital

**Cost of capital
Module-10**

Equity

- Common Stocks
- Preferred Stocks



Debt

- Bonds
- Fixed Income
- Securities



Market Price and Cost of Capital



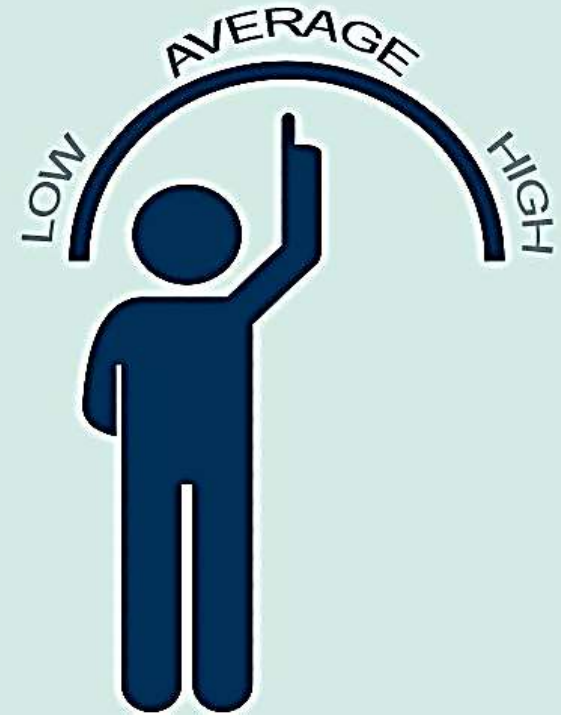
Topic 188: Weighted Average Cost of Capital WACC-1

**Cost of capital
Module-10**

WACC

Average cost of equity and debt calculated on the basis of percentage of each in total capital

Average



Why WACC is
Important?



The Formula



WACC
Formula



$$= [\text{Cost of Equity} \times \% \text{ of Equity} + \\ [\text{Cost of Debt} \times \% \text{ of Debt} \times \\ (1 - \text{tax rate})]]$$

Topic 189: Weighted Average Cost of Capital WACC-2

**Cost of capital
Module-10**

Weighted Average Cost of Capital			
Type of Capital	Amount	Cost of Capital	Cost of Capital
	Rs.	%	Rs.
Common Stocks	300,000	22%	66,000
Preferred Stocks	200,000	18%	36,000
Loans	100,000	14%	14,000
Bonds	150,000	17%	25,500
Lease	50,000	19%	9,500
Total	800,000		151,000

Weighted Average Cost of Capital

$$WACC = 151,000/800,000$$

$$WACC = 0.18875$$

$$WACC = 18.875\%$$

WACC Explanation

$$WACC = ???$$

If the company earns 18.875% on the entire capital of Rs. 800,000 it will be able to pay to all the capital holders

Topic 190: Cost of Preferred Stocks

**Cost of capital
Module-10**

Preferred Stocks get Preference in:

- Distribution of Dividend
- Claim on Assets in Case of Liquidation



talking about
preferences

The Formula

$$r = D/P$$

- r = Cost of Preferred Stocks
- D = Preferred Dividend
- P = Market Price of Preferred Stocks



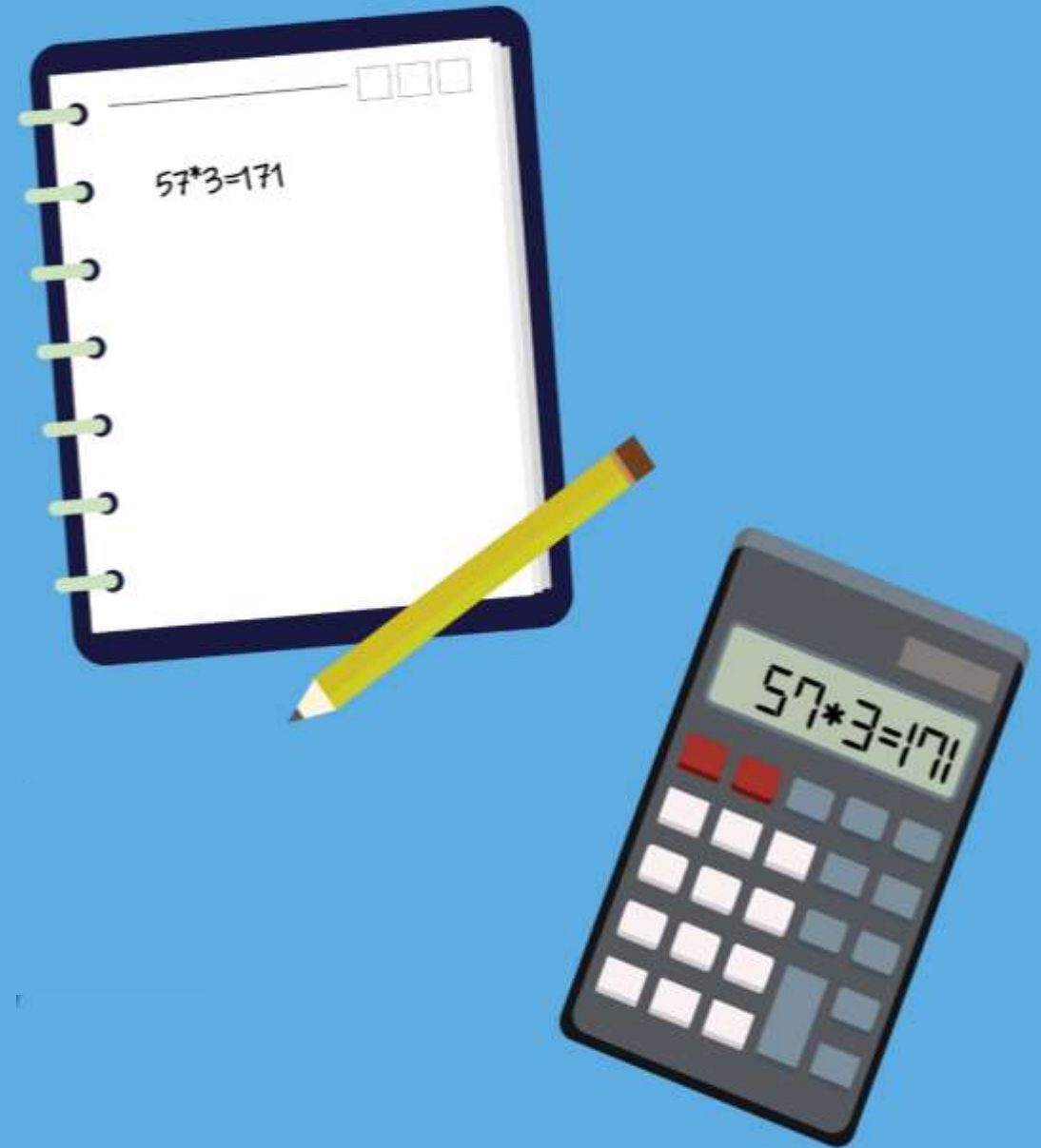
Calculations

- $r = ?$
- $D = \text{Rs. } 5$
- $P = \text{Rs. } 60$

$$r = D/P$$

$$r = 5/60$$

$$r = 8.33\%$$



Topic 191: Cost of Common Stocks-1

**Cost of capital
Module-10**

Common Stocks

Assumptions

- No Growth of Dividend
- Constant Growth of Dividend
- Growth in Stages



Why No
Growth of
Dividend?



Why Constant Growth of Dividend?



Why Growth in Stages?



Topic 192: Cost of Common Stocks-2

**Cost of capital
Module-10**

No. Growth of Dividend Formula

$$r = D/P$$

Where 'r' is the cost, 'D' is the dividend, and 'P' is the market price.



No Growth of Dividend *Calculations*

$$r = D/P$$

$$r = 5/50 = 10\%$$



Constant Growth of Dividend Formula

$$r = D_1/P + g$$

Where 'r' is the cost, 'D' is the dividend, 'P' is the market price, and 'g' is the growth rate.



Constant Growth of Dividend

Calculations

- $r = D_1/P + g$
- $r = 5.5/50 + 6\% = 17\%$



Topic 193: Cost of Common Stocks-3

**Cost of capital
Module-10**

Growth in Stages:

- *Two-Stage Growth Model*
- *Three-Stage Growth Model*



Growth in Stages:

- *Discounting dividends of each stage*
- *Last stage always goes till infinity*
- *Apply constant growth formula on the last stage*
- *Last stage's dividends value should be further discounted for the years from where the last stage starts*



Topic 194: Cost of Common Stocks For Specific Period

**Cost of capital
Module-10**

Effect of a Specific Period

- Buying and Selling Price
- Fixed Number of Dividends



How to Calculate Cost?

- Hit and Trial Method
- Discount Dividends for the Holding Period
- Discount Expected Price at the End of Holding Period



Topic 195: Cost of Bonds-1

**Cost of capital
Module-10**

What are Bonds?

- Public Debt
- Loan
- Secured and Unsecured



Cost of Bonds

- Cost of Bond Vs Loan Interest
- Cost and Bond Rating



Types of Bonds

- No Maturity Bonds
- Zero Coupon Bonds
- Coupon Bonds



What is Discounted?



- Interest
- Maturity Value

Topic 196: Cost of Bonds-2

**Cost of capital
Module-10**

Cost of Zero Coupon Bond

- No Instalment
- Discount the Maturity Value
- Formula
 - $V = MV(1+r)^{-n}$
 - $r = (P/MV)^{-1/n} - 1$

ZERO

Cost of Perpetual Bond

- Instalment
- No Maturity Value
- Formula
 - $V = I/r$
 - $r = I/P$



Cost of Coupon Bond

- Hit and Trial Method
- Discount Instalments
- Discount Maturity Value

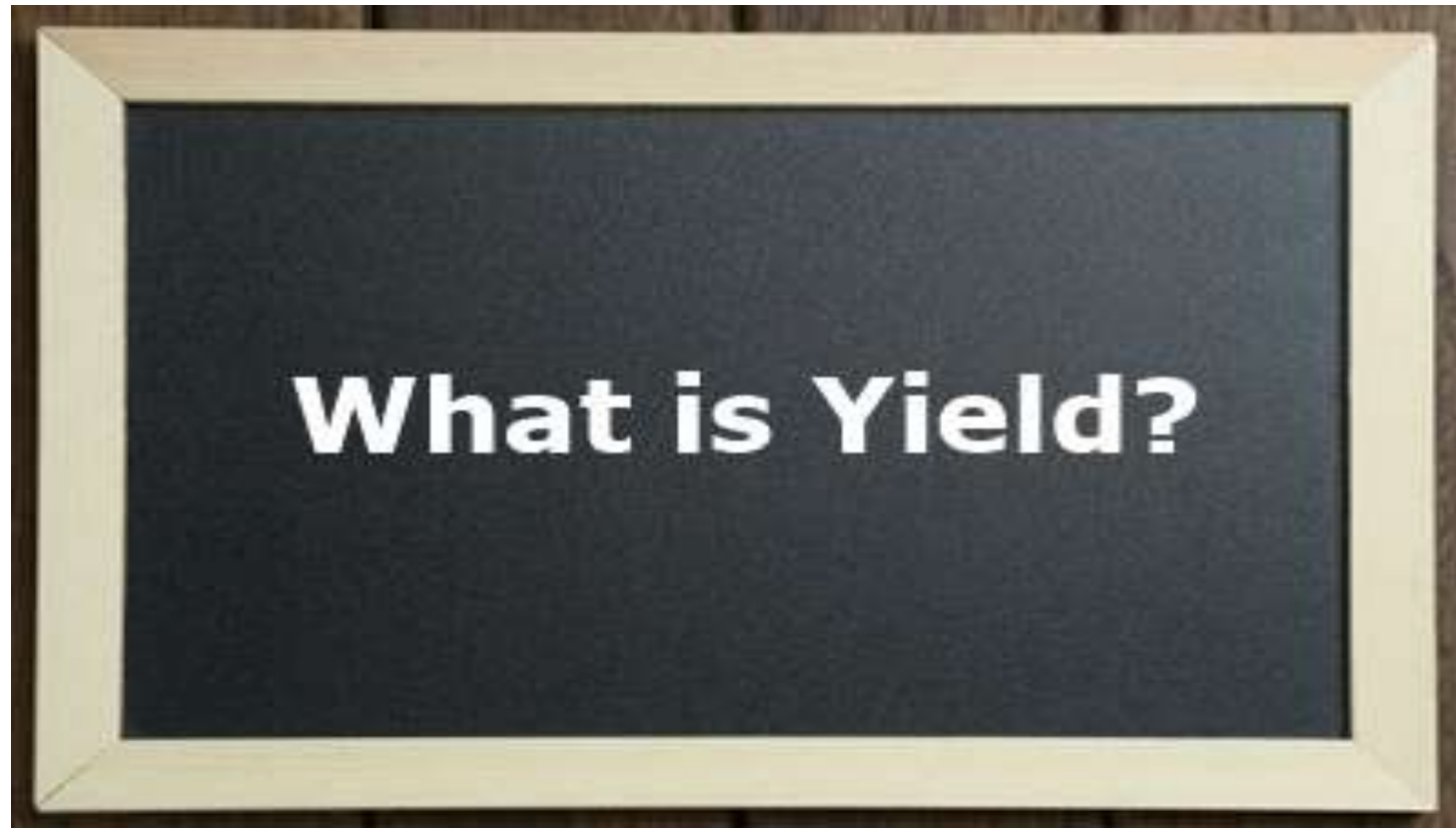


**Topic 197: Bond Yield And Yield to
Maturity**

**Cost of capital
Module-10**

Bond Yield

Return of the Bond holder on current market price



Yield to Maturity

Return of the Bond holder
including Interest and
Capital Gain



The Formula

$$\text{YTM} = \frac{C + \frac{\text{FV} - \text{PV}}{t}}{\frac{\text{FV} + \text{PV}}{2}}$$

$$\text{YTM} = \frac{150 + \frac{1,000 - 850}{7}}{\frac{1,000 + 850}{2}} = 18.53\%$$

Topic 198: Cost of Equity Using CAPM

**Cost of capital
Module-10**

Capital Asset Pricing Model

- Cost of Equity
- Risk Free Rate
- Market Rate
- Risk Premium
- Beta



CAPM Formula

Capital Asset Pricing Model Formula



$$R_e = R_f + \beta \times (R_m - R_f)$$

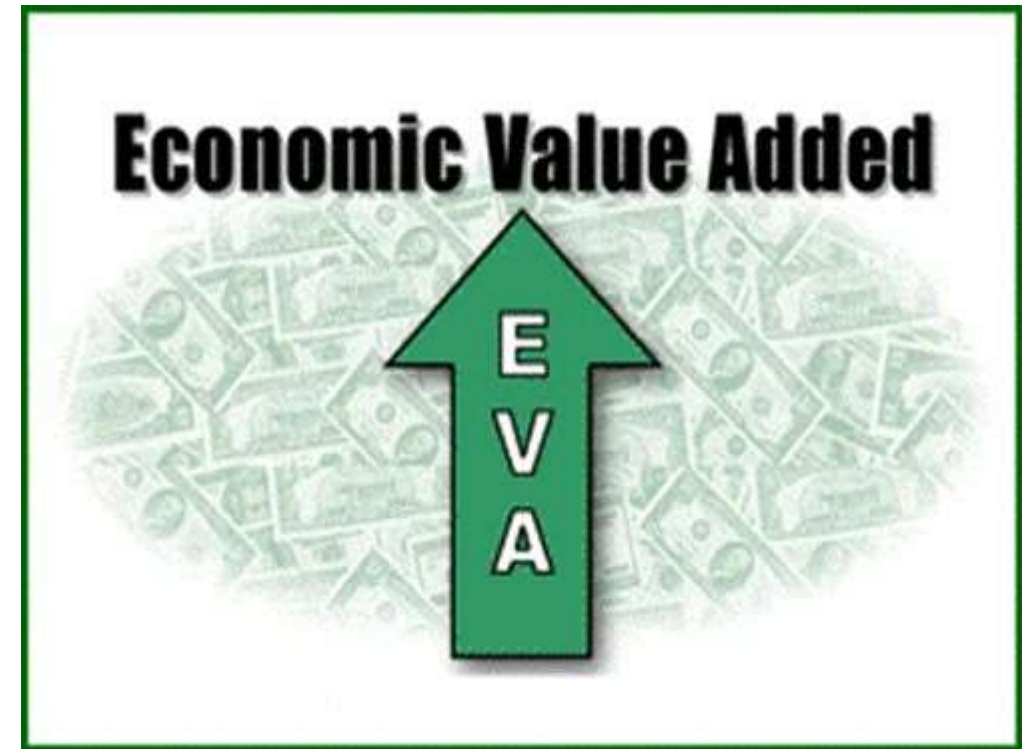


Topic 199: Economic VALUE ADDED

**Cost of capital
Module-10**

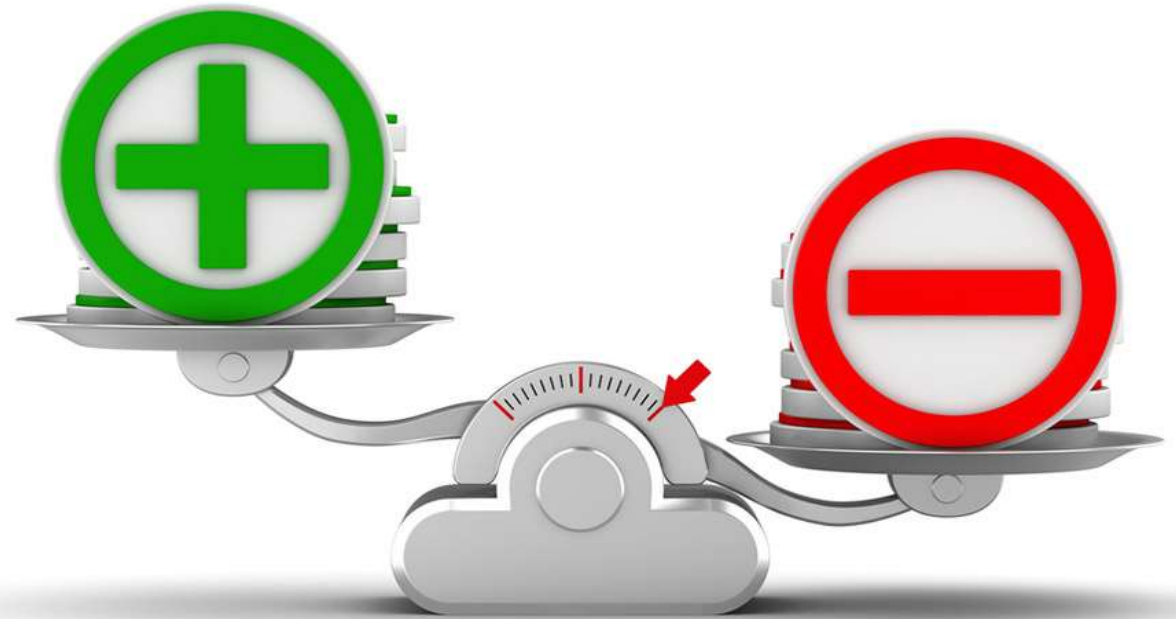
Economic Value Added-EVA

- Measure of Financial Performance
- $\text{Cost of Capital less Operating Profit}$



EVA

- What if EVA is negative or positive?
- Can it be Zero?



The Formula

$$\text{EVA} = \text{NOPAT} - (\text{Invested Capital} \times \text{WACC})$$



Topic 200: Capital Structure-Introduction

**Capital Structure
Module-11**

Capital Structure

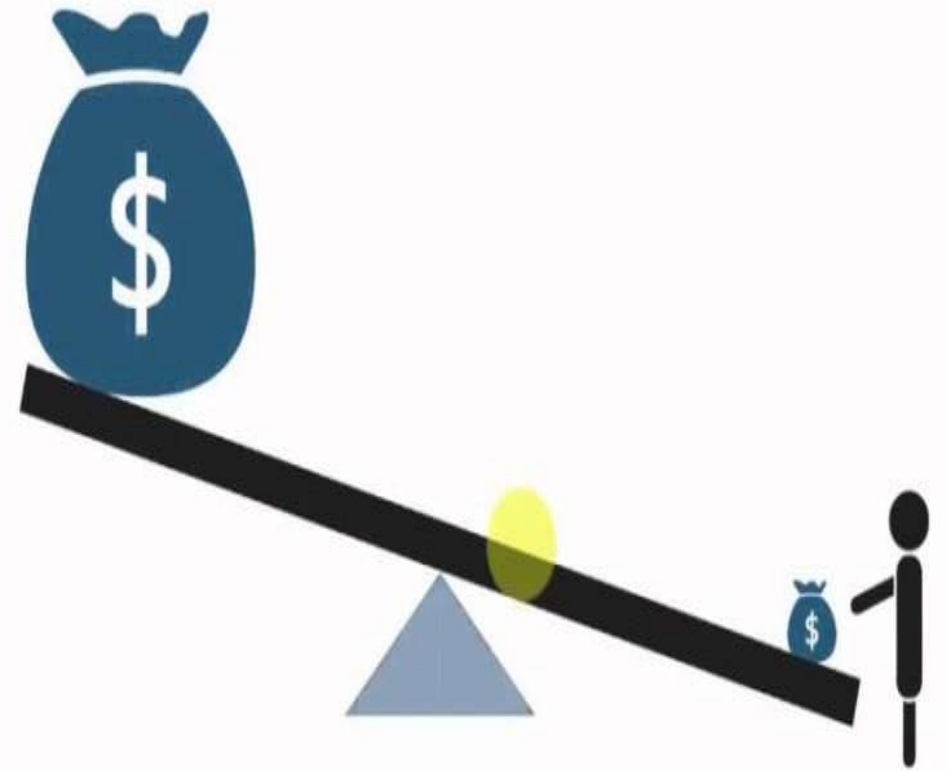
- Mix of Debt and Equity
- Financial Leverage



Components of Capital Structure

- Common Share Capital
- Preferred Share Capital
- Loans
- Bonds

Leverage



Topic 201: Goals of a Firm

**Capital Structure
Module-11**

Profit Maximization



Maximization of Wealth



Other Goals

- CSR (Corporate social Responsibility)
- Economic Uplift
- Employment Generation



**Topic 202: Value of Firm:
Capitalization Rate**

**Capital Structure
Module-11**

Value of Firm

- Market value
- Book value
- Other value (Going concern value, Intrinsic value etc.)



C Capitalization

Estimating the value by considering
the return

Capitalization Rate

The rate is used to
determine the value



Topic 203: Capital Structure Theories

Capital Structure Module-11



Theories

- Traditional Theory
- Net Operating Income Theory
- Miller & Modigliani Theory

Costs of Capital

- Cost of Debt
- Cost of Equity
- Overall Cost of Capital



Topic 204: Optimal Capital Structure

**Capital Structure
Module-11**

Concept of Optimal Capital Structure

- Value of firm is maximum
- Cost of capital is minimum



Topic 205: Traditional Theory of Capital Structure-1

Capital Structure Module-11

Leverage and Required Rate

- Earning Capacity
- Return on Equity
- Leverage and ROE
- Cost of Equity Capital



LEVERAGE

Topic 206: Traditional Theory of Capital Structure-2

Capital Structure Module-11

Return on Equity

- Depends on earning power
- Profit after Tax (PAT) / Equity
- Cost of debt and earning power



General Assumptions of Theories

1. The cost of debt is less than the cost of equity
2. There are no taxes
3. The risk perception of investors does not change by the use of debt

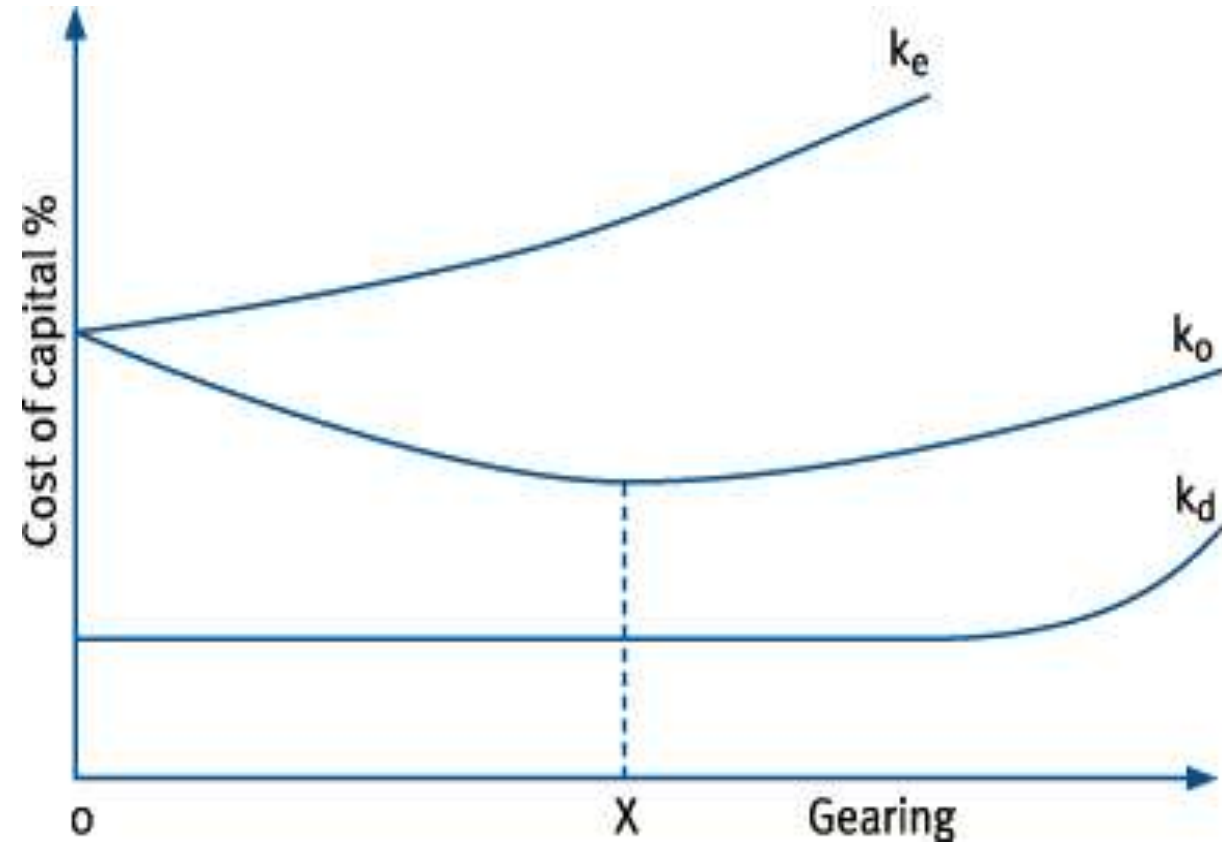


Topic 207: Traditional Theory of Capital Structure-3

Capital Structure Module-11

Traditional Theory

- With the increase in leverage the cost of equity increases
- Initially the overall cost goes down
- Later overall cost goes up
- At the point where cost is minimum Value is maximum



Traditional Theory

- How does cost of capital change with leverage?
- There is an optimal capital structure

Optimal Capital Structure



**Topic 208: Net Operating Income
Approach-1**

**Capital Structure
Module-11**

The Theory

- Capital structure has no impact on the value of firm
- Overall cost of capital does not change with the change in capital structure
- There is no optimal capital structure
- Value of firm is based on earnings

**Topic 209: Net Operating Income
Approach-2**

**Capital Structure
Module-11**

Assumptions of the Theory

1. The market capitalizes the value of the firm as a whole
2. The business risk remains constant at every level of debt equity mix
3. There are no corporate taxes



Value of Pie Remains the Same



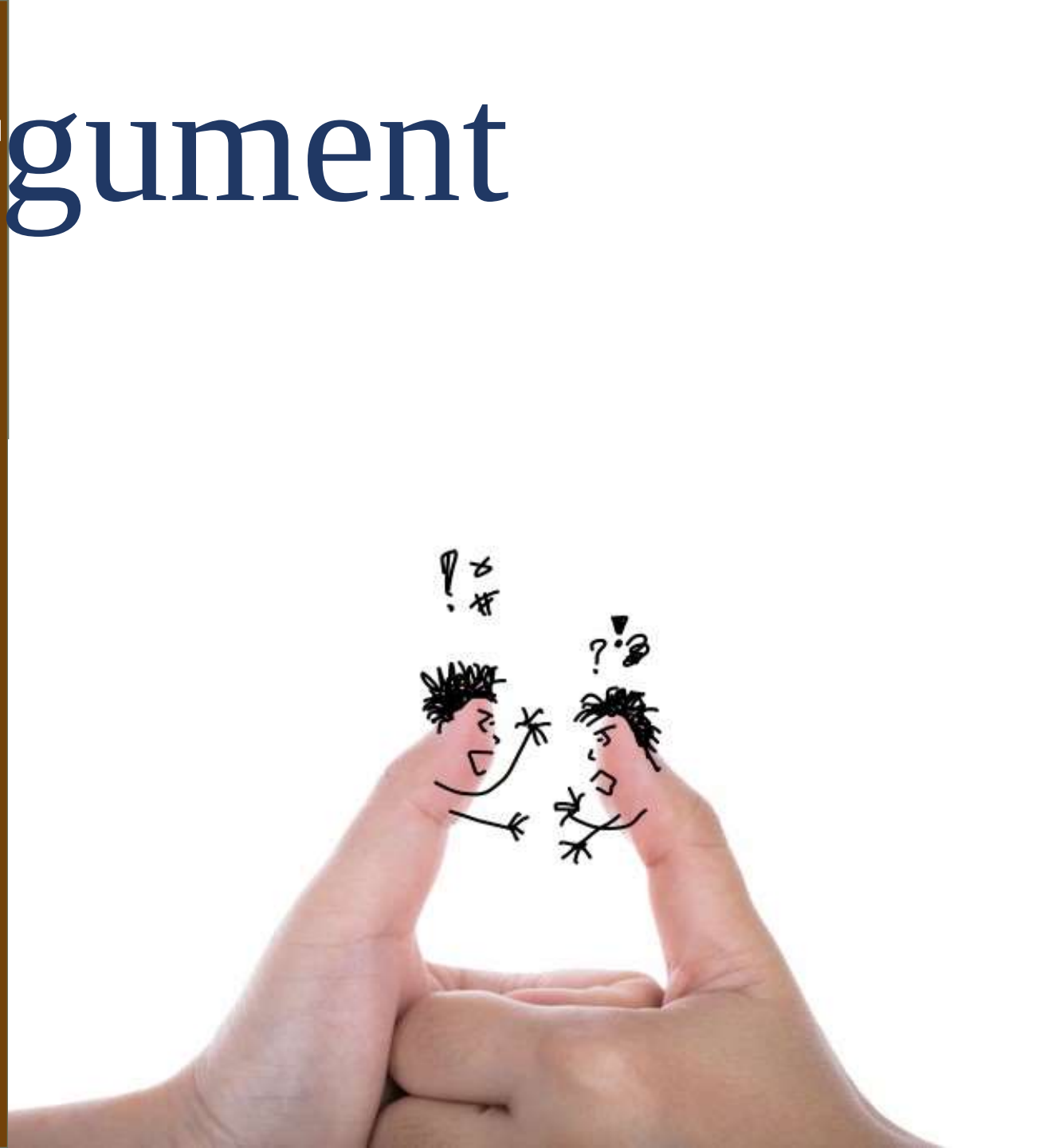
- Value is affected by profit not the capital structure
- No matter in how many pieces the pie is cut the total value does not change

**Topic 210: Net Operating Income
Approach-3**

**Capital Structure
Module-11**

The Argument

1. Impact of leverage on cost of equity
2. Rate of change of cost of equity
3. Impact on overall cost of capital
4. Impact on value of firm



Topic 211: Arbitrage Theory: Miller and Modigliani-1

Capital Structure Module-11

Irrelevance Theory

1. Capital structure is irrelevant w.r.t. value of a firm
2. If relevance theory (Traditional Theory) is accepted, Arbitrage will take place.



Arbitrage

The simultaneous buying and selling of assets in order to take advantage of differing prices for the same assets.



M&M Argument

1. Two same assets (stocks) cannot have different prices
OR arbitrage will take place.
2. The prices of identical firms' (other than capital structure) stocks cannot be different.



Topic 212: Arbitrage Theory: Miller and Modigliani-2

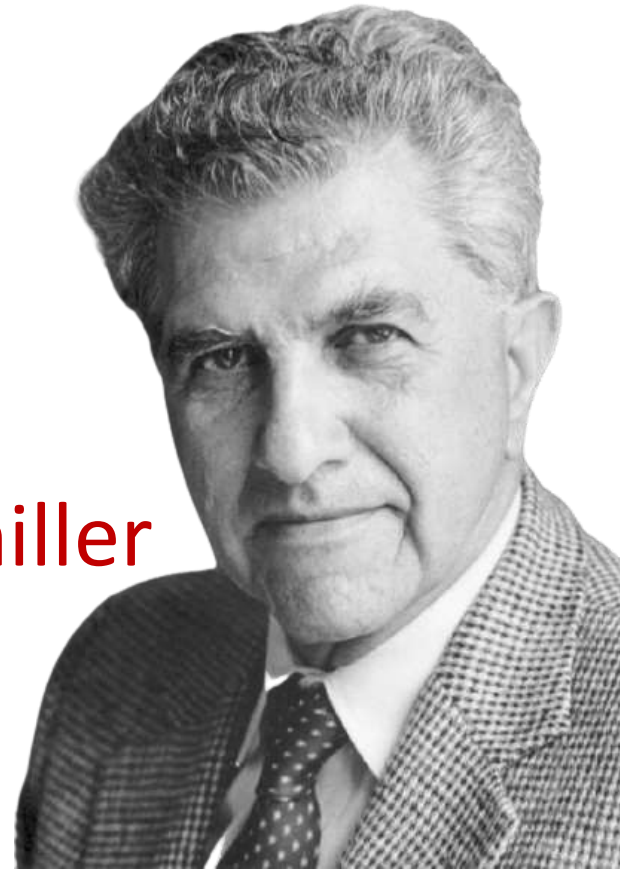
Capital Structure Module-11

M&M Argument

1. Personal leverage
2. Cost of debt is same for firm and individuals
3. Investors shift to the other firm with same return and value and yet save some money.



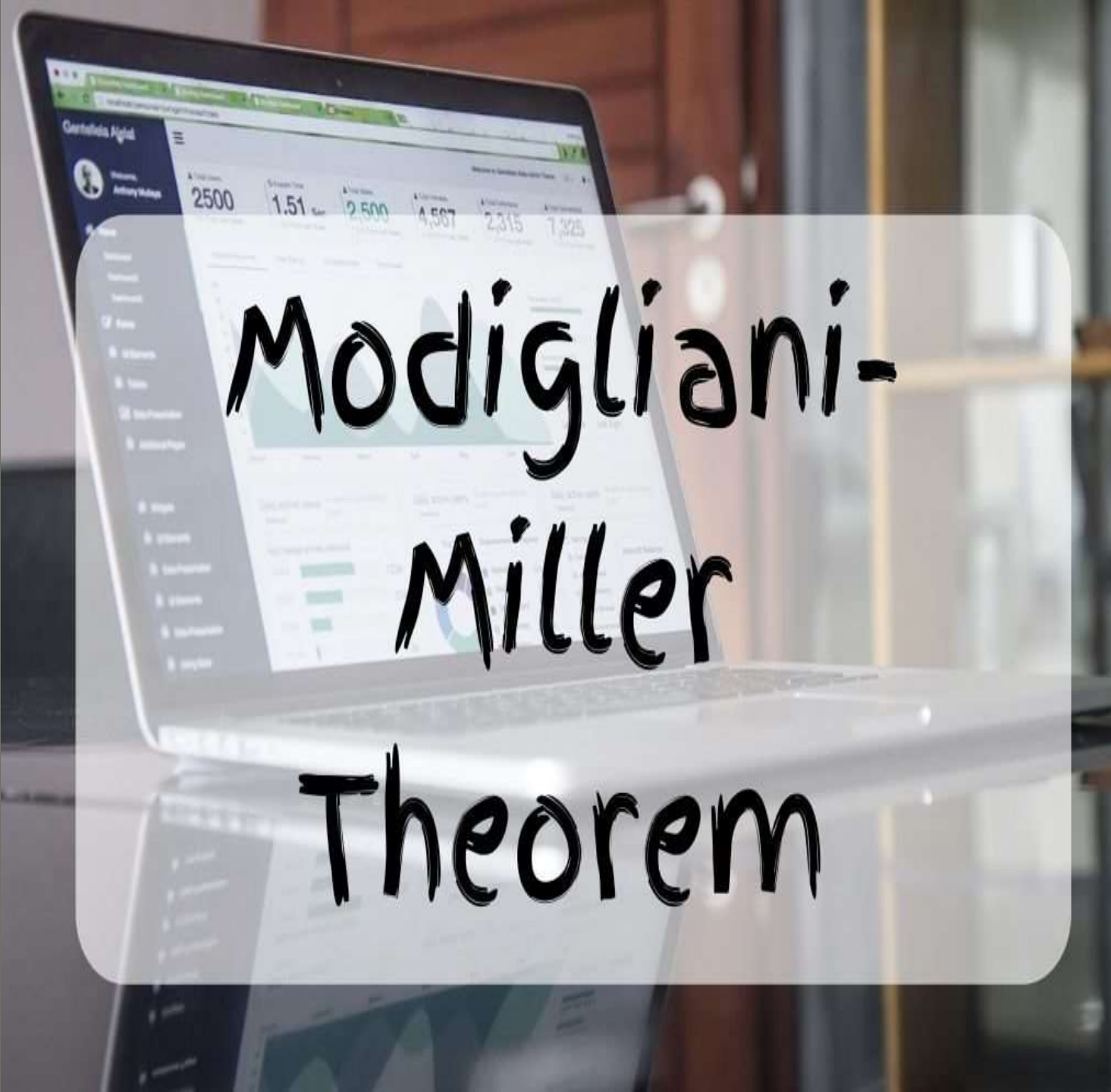
Franco Modigliani



Merton Miller

M&M Argument

1. Percentage holding will not change
2. Return will not change
3. There will be cash saving

A laptop is shown in the background, displaying a financial dashboard. The dashboard includes a sidebar with navigation options like 'Dashboard', 'Reports', 'Settings', 'Users', 'Groups', 'Accounts', 'Transactions', 'Invoices', 'Payments', 'Expenses', 'Budgets', 'Goals', 'Analytics', and 'Help'. The main content area shows a 'Portfolio' overview with a line chart and several data points: 'Current Value: 2500', 'Previous Value: 1.51', 'New Value: 2,500', 'Current Value: 4,507', 'Previous Value: 2,315', and 'New Value: 7,325'. The text 'Modigliani-Miller Theorem' is overlaid on the laptop screen in a large, black, handwritten font.

Modigliani-
Miller
Theorem

Topic 213: Arbitrage Theory: Miller and Modigliani-3

**Capital Structure
Module-11**

General Assumptions

1. Firms can be identical
2. External factors do not affect
3. No corporate taxes
4. Perfect market
5. Rational behavior of the investor
6. Identical risk characteristics
7. All earnings are paid as dividend



Topic 214: Market Imperfections and Issues

**Capital Structure
Module-11**

Perfect Market

- A theoretical market
- No product differentiation
- Large buyers and sellers
- No entry or exit barriers
- Perfect information
- Rationality
- No monopoly or other manipulations



Impact of Market Imperfections

- All real market are imperfect
- Manipulation
- No possibility of Arbitrage
- M&M theory does not hold good in imperfect markets



Issues in M&M Argument



M& M does not consider:

- Bankruptcy cost
- Agency cost
- Leverage increases efficiency
- Institutional restrictions
- Transaction costs

Conclusion

“In an imperfect market capital structure may affect the value of a firm”



Topic 215: Income Tax Considerations

**Capital Structure
Module-11**

Tax

- Tax charged on the income
 - Corporate taxes
 - Individual taxes
- Tax Shield
- Uncertainty of Tax Shield



Topic 216: Value of a Firm in the Presence of Income Tax

**Capital Structure
Module-11**

Value of a Firm in a Taxable World

Value of levered firm is higher in a taxable world.

Value of a Levered Firm	=	Value of an Unlevered Firm	+	Tax Shield Benefit
----------------------------	---	-------------------------------	---	-----------------------

Value of a Levered Firm	=	500,000 + 150,000 = Rs. 650,000
----------------------------	---	---------------------------------

Value of Tax Shield

- $\text{Tax Shield Benefit} = \text{Debt} \times \text{Tax Rate}$
- $\text{Tax Shield Benefit} = 500,000 \times 30\% = \text{Rs. } 150,000$
- $\text{Tax Shield Benefit} = \text{Tax Saving} / \text{Discount Rate}$
- $\text{Tax Shield Benefit} = 22,500 / 15\% = \text{Rs. } 150,000$

Topic 217: Concept of leverage

**Financial and
Operating Leverage
Module-12**

Leverage

- Operating and Financial Leverage
- Use of fixed cost to enhance the return
- Leverage is due to Debt
- Debt generally has fixed cost



Topic 218: Fixed and Variable Costs

**Financial and
Operating Leverage
Module-12**

Fixed Cost

- The cost that does not change with the change in the level of operations.
- Period cost
- Per unit fixed cost changes



Variable Cost

- The cost that changes with the level of operations.
- Marginal cost
- Per unit variable cost does not change



**Topic 219: Contribution Margin and its
Significance**

**Financial and
Operating Leverage
Module-12**

Contribution Margin

- Difference of Revenue and Variable Cost
- Formula:

$$CM = \text{Revenue} - VC$$



Contribution Margin Formula = Net Sales - Total Variable Expenses

Significance of CM

- Relevant for Decision Making
- Additional profit for additional activity



CM Ratio

- The ratio of CM and Revenue/Sales
- Formula:

$$\text{CM Ratio} = \text{CM} / \text{Sales}$$

$$\text{CM Ratio} = 200 / 500$$

$$\text{CM Ratio} = 40\%$$



CM and Variable Cost

$$\text{VC Ratio} = 1 - \text{CM Ratio}$$

$$\text{CM Ratio} = 1 - \text{VC Ratio}$$



Topic 220: Break Even Point

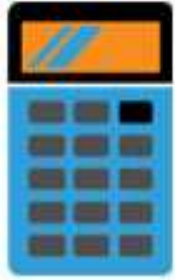
**Financial and
Operating Leverage
Module-12**

Breakeven

- No profit and no loss
- Self-sustainability
- Contribution is equal to fixed cost



Break-Even Sales Formula



**Break-Even
Sales**

=



Fixed Costs

Contribution Margin Percentage

**Break-Even
Sales**

=

Fixed Costs x Sales

Sales – Variable Costs





**Break even
point in units**

=



Fixed Costs

Contribution Margin



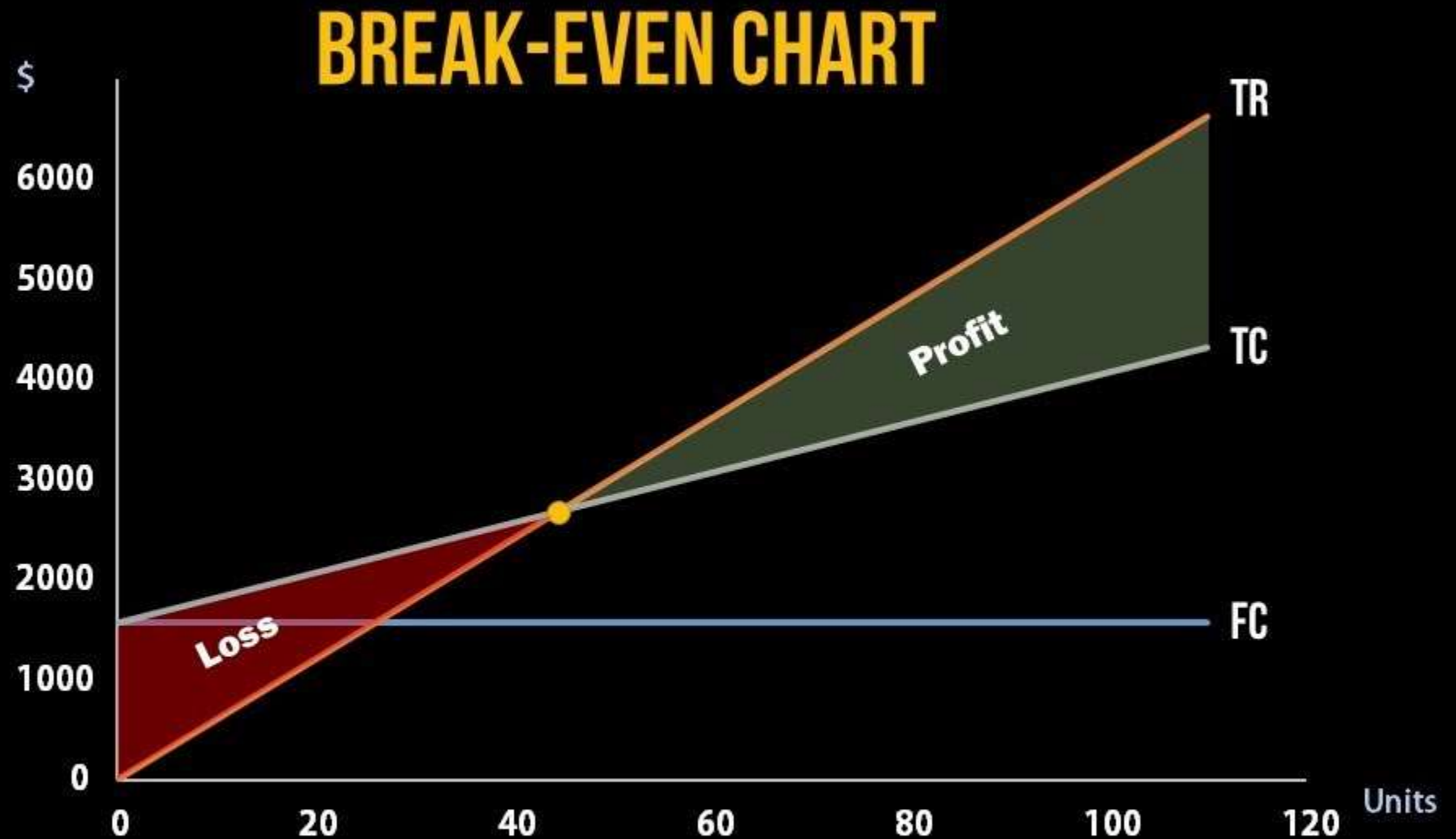
SELLING PRICE = 60

VARIABLE COST = 25

FIXED COST = 1575

BEP (UNITS) = 45

BEP (\$) = 2700



**Topic 221: Cost Volume Profit
Analysis-1**

**Financial and
Operating Leverage
Module-12**

Cost, Volume, & Profit

- Cost is the amount incurred to produce
- Volume is the number of units produced
- Profit is the difference of cost and revenue



CVP Analysis

Cost-volume-price (CVP) analysis is a way to find out how changes in variable and fixed costs affect a firm's profit





CVP Assumptions

- Fixed selling price
- Constant fixed cost
- Per unit variable cost is constant

**Topic 222: Cost Volume Profit
Analysis-2**

**Financial and
Operating Leverage
Module-12**

Profit Equation

- $R - TVC - FC = P$

Where:

R = Revenue

TVC = Total variable cost

FC = Total Fixed Cost

P = Profit





Profit Equation Application

- $(SP \times Q) - (PUVC \times Q) - FC = P$
- $(50Q) - (30Q) - 200,000 = P$

$$(50 \times Q) - (30 \times Q) - 200,000 = 100,000$$

$$(50Q) - (30Q) = 300,000$$

$$Q = 300,000/20 = 15,000 \text{ units}$$

EXAMPLE

Example: Breakeven

- $(50 \times Q) - (30 \times Q) - 200,000 = 0$
- $(50Q) - (30Q) - 200,000 = 0$
- $Q = 200,000/20 = 10,000$ units



Topic 223: Operating Leverage

Financial and Operating Leverage Module-12

Leverage Defined

Use of Fixed Cost to enhance the return/profit



Operating Cost

- The cost incurred to operate or run a business, including:
 - Cost of Sales
 - Sales, General & Administrative-SG&A Expenses
- Does not include **Financial Cost**



Operating Leverage-OL

The use of fixed operating cost to enhance profit



How does OL Work?

- Fixed cost is the period cost
- When production increases per unit fixed cost decreases
- Higher component of Fixed Cost creates higher OL



Topic 224: Financial Leverage

Financial and Operating Leverage Module-12

Financial Leverage

Use of fixed financial cost to enhance the return of equity holders



How does it Work?

- Higher proportion of debt creates financial leverage
- When ROI is more than cost of debt ROE goes up



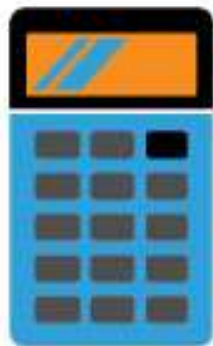
**Topic 225: Degree of Operating
Leverage**

**Financial and
Operating Leverage
Module-12**

How to Measure Operating Leverage?

- Degree of Operating Leverage-DOL
- Higher is the DOL, higher is the risk





Degree of Operating Leverage Formula

$$= \frac{\% \text{ Change in EBIT}}{\% \text{ Change in Sales}}$$



DOL Formula

$$\text{DOL (In Rupees)} = \frac{\text{EBIT} + \text{Fixed Cost}}{\text{EBIT}}$$

$$\text{DOL} = \frac{500,000 + 800,000}{500,000} = 2.6$$

DOL Formula (Units)

$$\text{DOL (In Units)} = Q / (Q - Q_{\text{BE}})$$

$$\text{DOL} = 5000 / (5000 - 4000)$$

$$\text{DOL} = 5$$

Where:

Q = Quantity Produced

Q_{BE} = Breakeven Quantity

**Topic 226: Degree of Financial
Leverage**

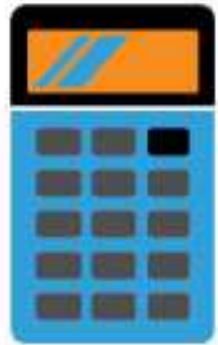
**Financial and
Operating Leverage
Module-12**



How to Measure Financial Leverage?

- Degree of Financial Leverage-DFL
- Higher DFL shows higher risk

Degree of Financial Leverage Formula



Degree of Financial Leverage

$$= \frac{\% \text{ Change in EPS}}{\% \text{ Change in EBIT}}$$



Degree of Financial Leverage

$$= \frac{\text{EBIT}}{\text{EBIT} - \text{Interest}}$$



DFL Formula

$$\text{DFL (In Rupees)} = \text{EBIT} / \text{EBIT} - \text{I} - [\text{PD}/(1-t)]$$

$$\text{DFL} = 500,000 / 500,000 - 200,000 - [100,000/(1-40\%)]$$

$$\text{DFL} = 3.75$$

Topic 227: Degree of Total Leverage

**Financial and
Operating Leverage
Module-12**

What is Total Leverage?

- Change in EPS with
reference to change in Sales

The image shows the Greek letter Sigma (Σ), which is a bold, black, serif capital letter. It is positioned on the left side of the slide.

Sigma

The image shows the Greek letter Pi (Π), which is a bold, black, serif capital letter. It is positioned on the right side of the slide.

Pi

How to Measure Total Leverage?

- Combined effect of Operating and Financial Leverage
- Product of DOL and DFL



DTL Formula

$$\text{DTL} = (\% \text{age Change in EPS}) / (\% \text{age Change in Sales})$$

$$\text{DTL} = \text{DOL} \times \text{DFL}$$

$$\text{DTL} = 2.6 \times 3.75 = \mathbf{9.75}$$

$$\text{DTL (In Rupees)} = (\text{EBIT} + \text{FC}) / \text{EBIT} - \text{I} - [\text{PD} / (1 - t)]$$

$$\text{DTL} = 500,000 + 800,000 / 500,000 - 200,000 - [100,000 / (1 - 40\%)]$$

$$\text{DTL} = \mathbf{9.75}$$

Topic 228: Risk and Leverage

Financial and Operating Leverage Module-12

- Uncertainty
- Probable deviation from expected results

R I S K

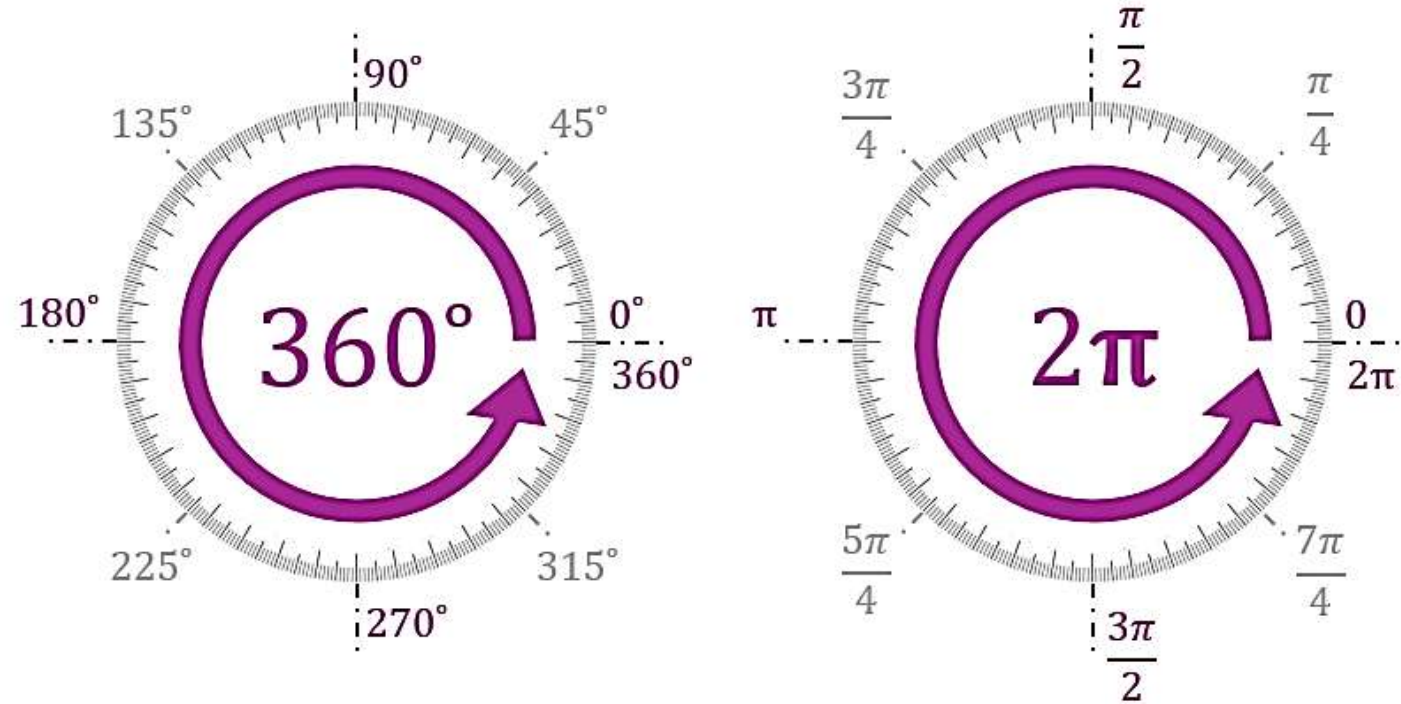
Risk & Leverage

- Leverage enhances return
- Return can be Positive or Negative

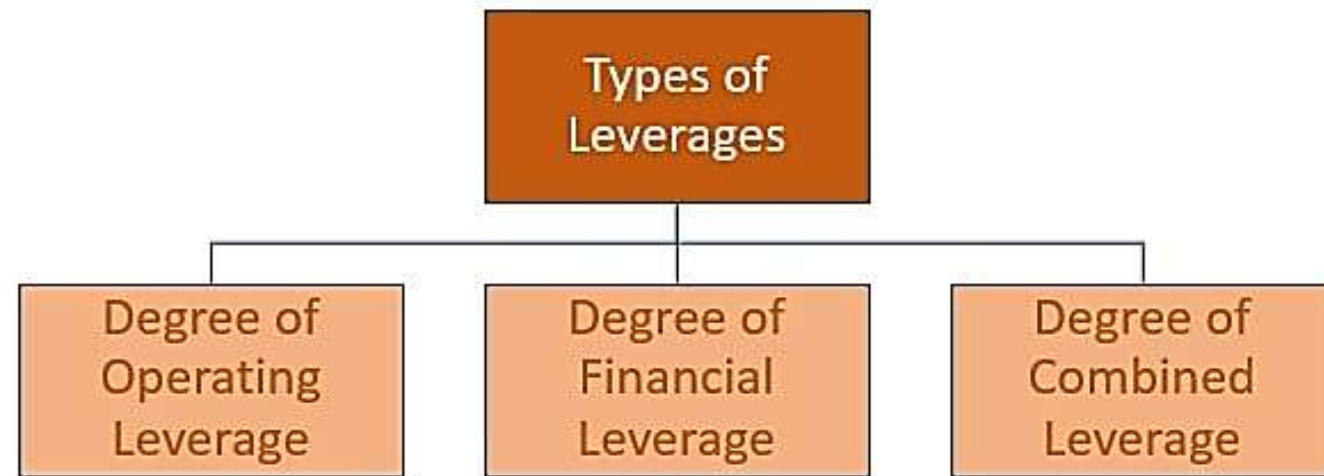


Degree of Leverage Measures Risk

- High DOL, DFL, and DTL indicate higher risk



- DOL: Small change in sales can cause big change in Profit
- DFL: Small change in profit can cause big change in EPS
- DTL: Small change in sales can cause big change in EPS



Topic 229: Concept of dividend

**Dividend Policy
Module 13**

Dividend

- Profit distributed among shareholders
- Paid out of Retained Earnings
- Annual dividend
- Interim dividend



Why Pay Dividend?

- Return of investor
 - Dividend
 - Capital Gain
- Common share dividend
- Preferred share dividend



Topic 230: Dividend policy

**Dividend Policy
Module 13**

Policy:

A system of principles to guide decisions and achieve rational outcomes



Dividend Policy:

- The structure of dividend payouts OR retention of profit
- Dividend Policies
 - Irrelevance of dividend
 - Relevance of dividend



Topic 231: Dividend payout and retention

Dividend Policy Module 13

Dividend Payout:



- The percentage of profit distributed as dividend
- $DPO = DPS/EPS$

Where:

DPO = Dividend Payout

DPS = Dividend per Share

EPS = Earnings per Share

Retention:

- The percentage of profit not distributed as dividend
- Also called Plough back
- $RR = (1 - DPS)/EPS$

Where:

RR = Retention Ratio



Effect of DPO and RR

- Higher DPO reduces growth
- Higher RR increases future growth



Topic 232: Dividend Yield

**Dividend Policy
Module 13**



- Term is also used for output
- Here yield means Return
- The return of investor on the current market price of investment

Stock Yield

The total return of a shareholder with reference to the current market price



Dividend Yield

- Dividend is declared as %age of Par Value
- Dividend Yield is relating DPS to per share market price



Topic 233: Passive and Active Dividend Policies

Dividend Policy Module 13

Dividend Passive Residual

- Irrelevance of Dividend
- Passive Residual refers to leftover profit (having no investment opportunity)



Question 1:

Should policy be
active or passive?

Active Dividend Policy:

- Dividend is more than a passive residual
- Investors want to have dividend
- What percentage to be paid?
- How to balance DPO and growth?



Topic 234: Dividend Stability

**Dividend Policy
Module 13**

Stable Dividend

- Constant payout or fixed percentage of profit
- A fixed amount of dividend
- Stability is dependent on current and accumulated profits



Payout

- In stable dividend DPO is not constant
- Amount of Dividend is Constant
- In long-run DPO may be the same



Stable Dividend Preference

- Shareholders prefer fixed dividend over time
 - Regular source of income
 - Certainty
 - Informational content (image of the company)
 - Institutional preference



Topic 235: Why Dividends are Paid?

**Dividend Policy
Module 13**

Why Dividend?

- To satisfy shareholders
- Regulatory compliance
- No investment opportunity



Why Dividends are not Paid?

- Investment opportunity
- Liquidity
- Control may be diluted if further shares are issued
- Binding covenants

**Don't
ask why,
ask why
not.**

Topic 236: Types of Dividends

**Dividend Policy
Module 13**

Dividend Types

- Cash Dividend
- Stock Dividend/Scrip Dividend
- Annual and Interim Dividend



Cash Dividend

- Investors prefer cash
- Liquidity is required
- May lead to borrowing
- Reduces market price of the stock



Stock Dividend/Bonus Share

- Payment of company's shares as dividend
- Increases the issued capital
- Percentage holding does not change
- No cash outlay
- Market price goes down



Topic 237: Stock Split and Reverse Split

Dividend Policy Module 13

Stock Split

- Dividing the shares: 2 for 1
- Book value reduces
- Par value reduces proportionally but Book Value does not
- Number of shares increases
- Makes it convenient for investors
- Share capital does not change

Reverse Split

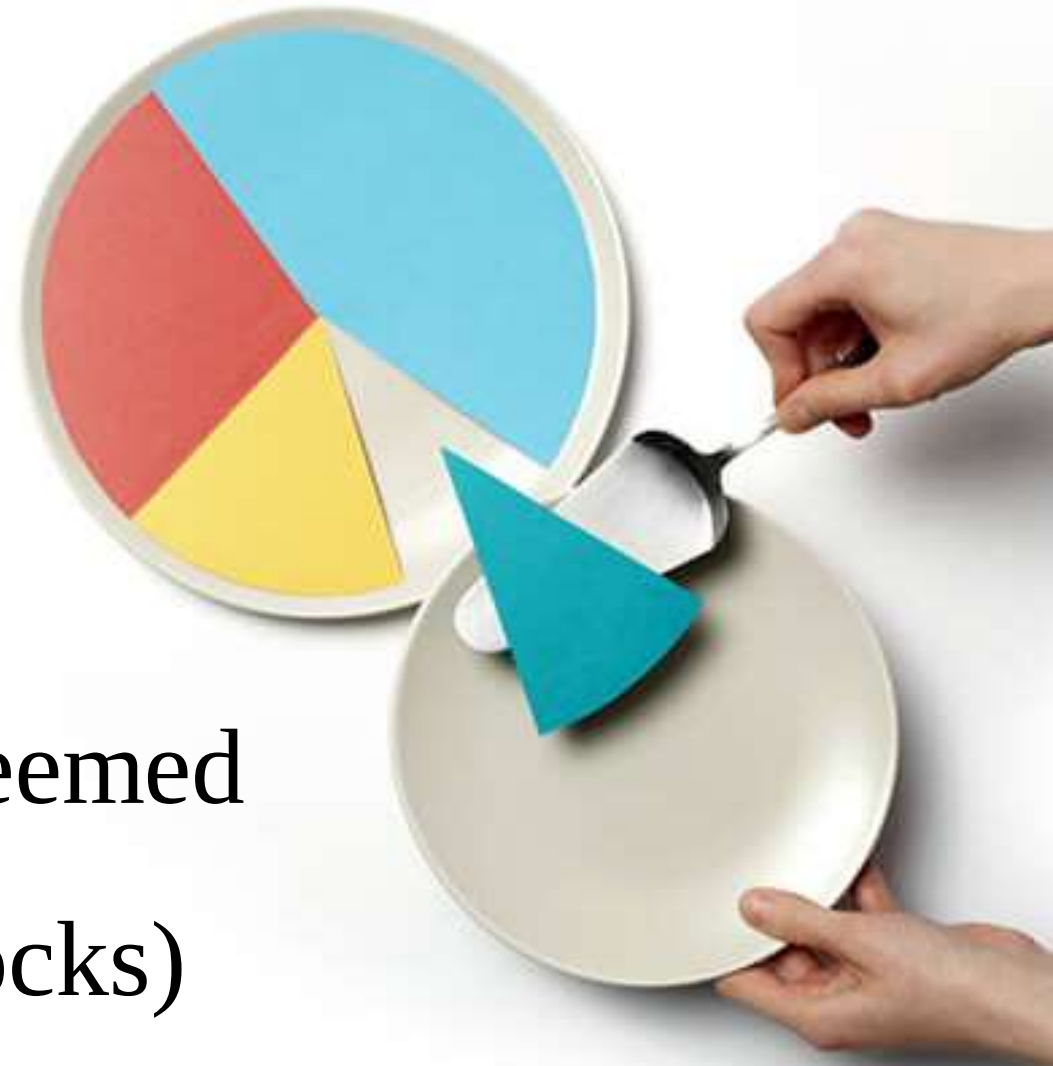
- Combining the shares: 1 for 2
- Par value increases
- Number of shares decreases
- High par value requires more investment
- Share capital does not change
- Book value is increased but not proportionally
- Reverse Split signals financial difficulties

**Topic 238: Stock Repurchase: Treasury
Stock**

**Dividend Policy
Module 13**

Stock Redemption

- Common stocks cannot be redeemed
- Going Concern principle
- Bonds can be redeemed
- Preferred stock can be redeemed
(Redeemable Preferred Stocks)



Treasury Stock

- Purchase of company's own stocks
- Issued capital does not decrease
- Outstanding shares decrease
- Reduces the value of Shareholders'

Equity



Why Treasury Stock?

- Increases the return of the remaining shareholders
- Distribution of funds when there are not attractive investment options
- May have a positive signaling effect
- Affects the percentage interest in the company



Topic 239: Retentions or Plough Backs

**Dividend Policy
Module 13**

Retention

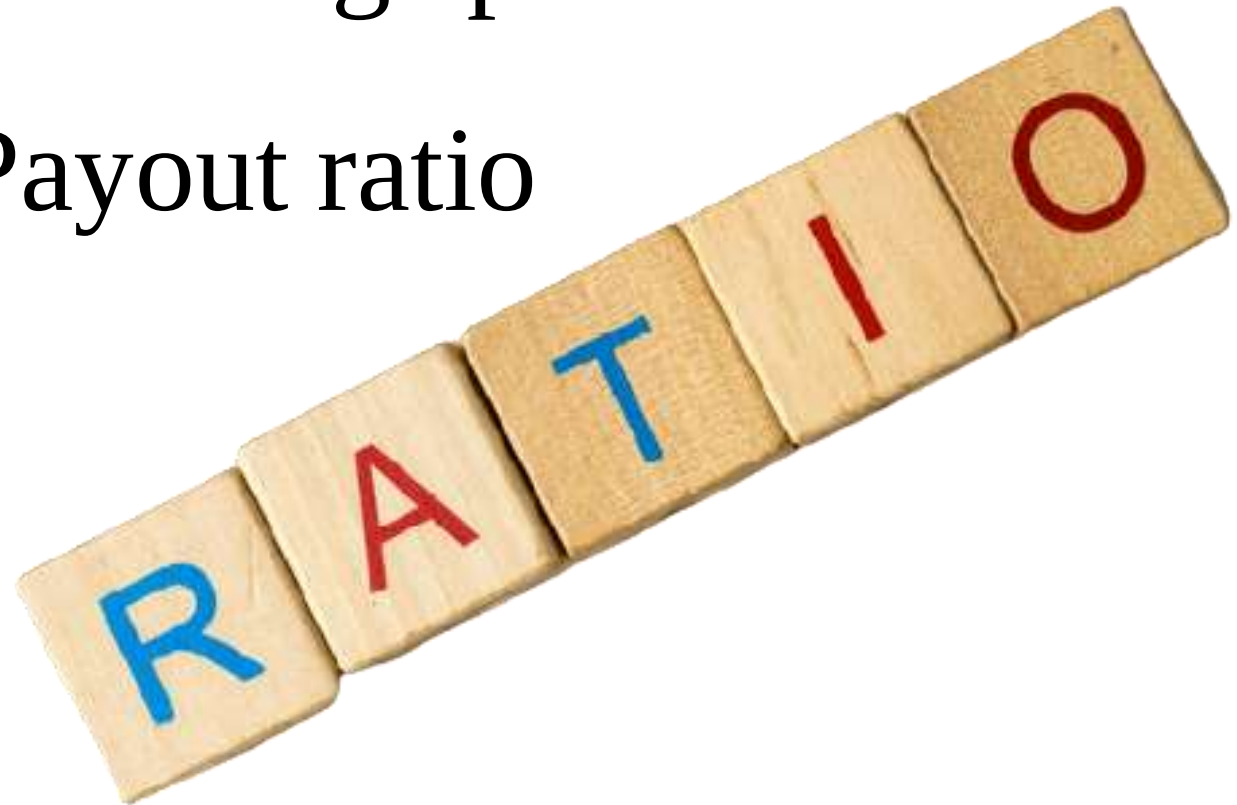
- Portion of profit not distributed as dividend
- Profit is an internal source of financing
- Return of shareholders should be maximized
 - By distributing
 - By retaining
- Investment Opportunity



Retention Ratio

- Profit retained to profit earned
- Retention per share to Earnings per share
- One minus Dividend Payout ratio

$$RR = 1 - DPO$$



Topic 240: Investment Opportunity

**Dividend Policy
Module 13**

Why Invest?

- To earn profit
- Preference is to get maximum profit
- Company should make prudent decisions for maximum return to shareholders





Investment Opportunity

- How much shareholders will get if profit is distributed as dividend?
- Does the company has an opportunity to investment the profit at a higher rate?

Excess Return

- Market Rate = 14%
- Return from reinvestment in the Business = 19%
- Profit may be reinvested



Topic 241: Considerations for Dividend

**Dividend Policy
Module 13**

Dividend Considerations

- Investors' preference
- Investment opportunity
- Availability of profit
- Taxes (Double taxation)
- Financing cost
- Regulatory requirements
- Liquidity



Topic 242: Relevance of Dividend

**Dividend Policy
Module 13**

Relevance of Dividend

- Preference of cash dividend
(Bird in hand concept)
- Regular income
- Tax consideration
- Financial signaling (market price)
- Home-made dividend is not practical



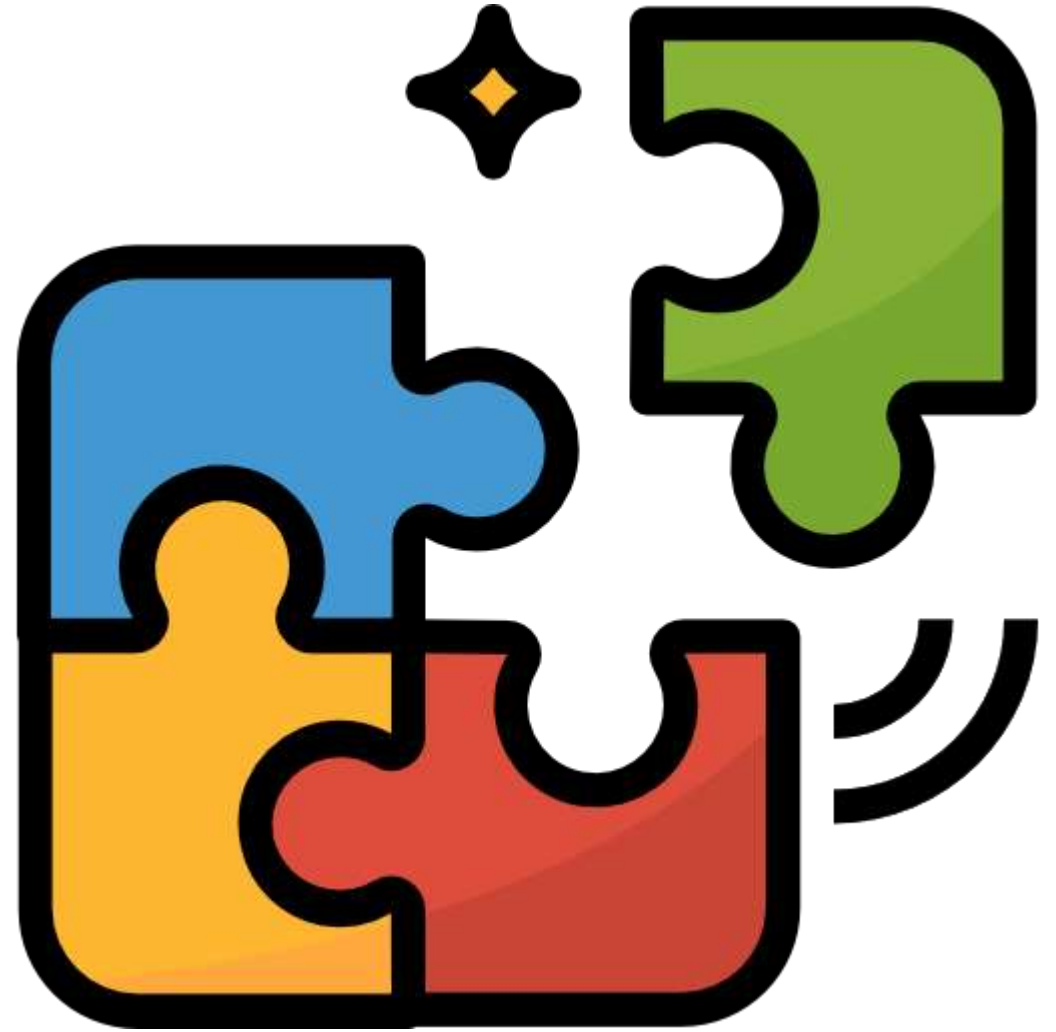
A word cloud centered around the word "relevant". The word "relevant" is the largest and most prominent, rendered in a large, orange, lowercase font. Surrounding it are numerous other words in various sizes, colors (including orange, blue, green, and red), and orientations. These words include: pertinence, propriety, germane, congruous, applicable, pertinent, fitting, suitable, apt, fit, right, licit, suit, material, relevance, befitting, apposite, felicitous, proper, corrective, acknowledgement, rightness, seemly, opportune, and admissible. The words are scattered in a cloud-like pattern, with some appearing more frequently or in larger fonts than others.

**Topic 243: Theory of Relevance of
Dividend-1**

**Dividend Policy
Module 13**

Theory of Relevance of Dividend

- The relevance theory of dividend argues that dividend decision affects the market value of the firm and therefore, dividend matters



Theory of Relevance of Dividend

- Dividend decision affects the market value of the firm
- Investors are generally risk averse
- Timing of return matters
- A bird in the hand is better than two in the bush
- Market response to dividend news
- Decrease in share price is less than the amount of dividend
- Market adjusts the price soon after the payment of dividend



Walter's Approach

- If ROI is greater than ROE then retain earnings
- If ROI is less than ROE then do not retain earnings
- If the two rates are the same then the company should be indifferent



Walter's Formula for the Value of Equity

$$VE = D/(KE - g)$$

Where,

VE = market value of equity shares

D = initial dividend

KE = costs of equity and

g = expected growth rate of earnings



**Topic 244: Theory of Relevance of
Dividend-2**

**Dividend Policy
Module 13**

Gordon's Approach

- Shareholders are risk-averse
- Today's cash is better than future capital gains, thus a “bird in the hand is worth more than two in the bush”



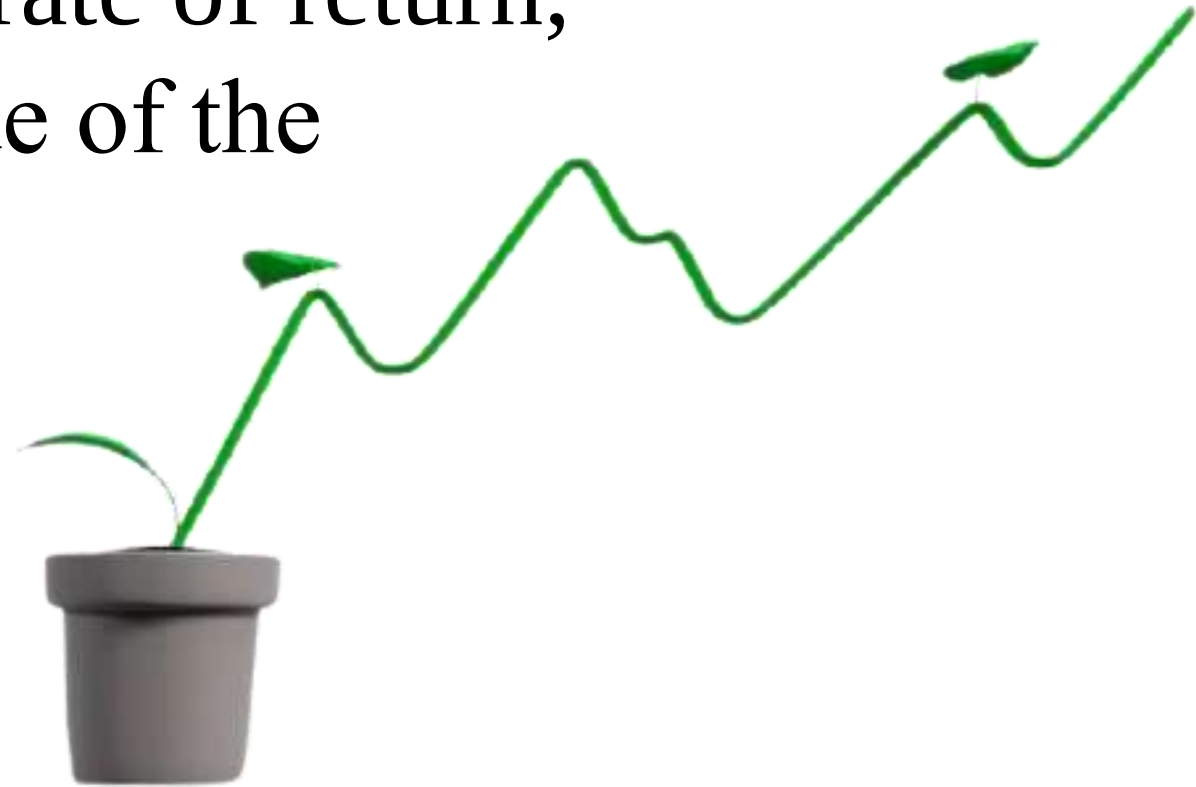
Gordon's Approach

- Resolves investor uncertainty (future is uncertain)
- Investors prefer even less but certain income today than more but less certain income in the future



Gordon's Approach

Due of the less risky nature dividends, investors will discount the firm's dividend stream at a lower rate of return, 'r', thus increasing the value of the firm's shares.



- All equity firm
- No external financing but RE
- Internal rate of return of the firm remains constant
- Cost of capital remains the same regardless the risk complexion
- Going Concern
- The retention ratio is constant
- Corporate tax does not exist



Gordon's Model: Growth Rate

Thus the growth rate (g) is also constant:

- Growth rate = retention ratio $\times$ cost of capital
- $g = b \times r$
- $g = 40\% \times 15\% = 6\%$



**Topic 245: Theory of Relevance of
Dividend-3**

**Dividend Policy
Module 13**

Dividend Capitalization Model

- The market value of a share is equal to the present value of the future streams of dividends
- Hence the value of a firm is based on dividends and so it is relevant



The Formula

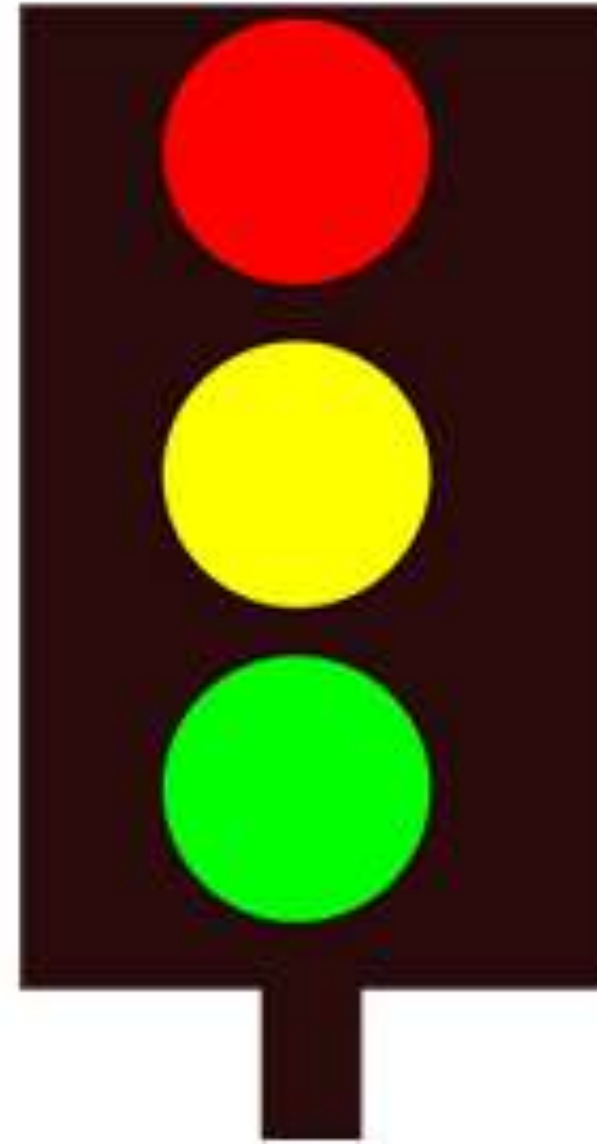
$$P = E (1 - b) / KE - br$$

Where:

- P = Price of a share
- E = Earnings per share
- b = Retention ratio
- $1 - b$ = Dividend payout ratio
- KE = Cost of capital or the capitalization rate
- br = Growth rate

Signaling Theory

- Payment of dividend sends signals of strength and future growth of the company
- Change in the dividend policy sends a signal:
 - Positive signal
 - Negative signal



Topic 246: Bird in Hand Theory

**Dividend Policy
Module 13**

What about Bird in Hand Theory

- Today's return is better than that of tomorrow
- Dividends make up about 40% of total stock market returns
- A bird in hand is better than two in the bush: Lesser amount in hand today is better than more with the company



Criticism

- First, reinvesting dividends (they invest the money they get)
- Second, tax minimization (tax on dividend today reduces capital)
- Third, by asking for dividend they do not act rationally



Topic 247: Dividend a Passive Residual

**Dividend Policy
Module 13**

The Residual Theory of Dividends

- Dividend paid by a firm is viewed as a residual
- The amount remaining or leftover after all acceptable investment opportunities



Stages of Dividend:

1. First determine optimal level of capital expenditures
2. Estimate equity needed to support the expenditure
3. Reinvested profits is utilized to meet the equity requirement
4. If there is a surplus that is, the residual, it is paid as dividends



The Residual Theory

Thus, the formula of dividend in Residual Theory is:

Dividend = Net income –
Additional equity needed



Example

- A project required Rs. 120 million. Equity ratio is 70% and profit is Rs. 95 million. How much Dividend can be paid?

Solution:

$$\text{Additional Equity} = 120 \times 70\% = 84 \text{ M}$$

$$\text{Dividend} = \text{Net income} - \text{Additional equity needed}$$

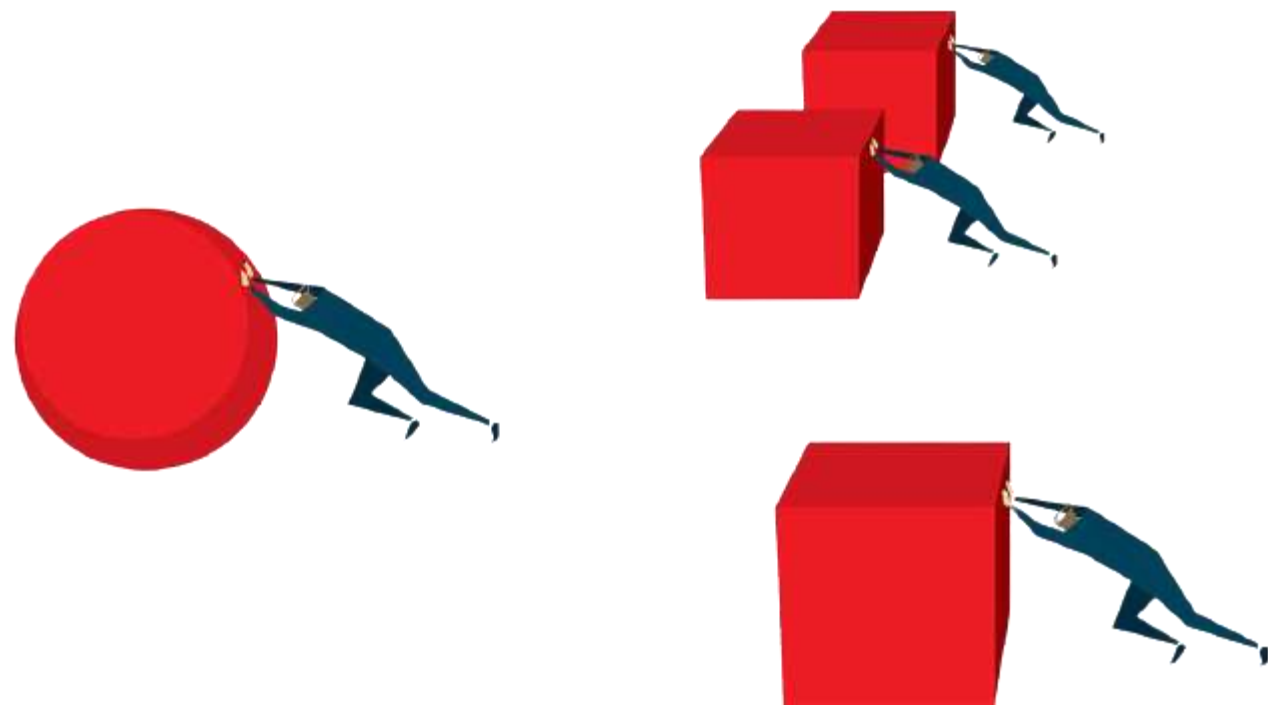
$$\text{Dividend} = 95 - 84 = 11 \text{ M}$$

**Topic 248: Theory of Irrelevance of
Dividend-1**

**Dividend Policy
Module 13**

The Irrelevance Theory

- Dividend is irrelevant
- A company's ability to earn and grow determines its market value and drives the stock price; not the dividend payments



The Irrelevance Theory

- There is no impact of payment of dividend on the stock price
- Dividend does not affect a company's growth
- Cash dividend is offset by external financing



The Irrelevance Theory

- The wealth of an investor is not affected by the payment or non-payment of dividend
- So, the split of Dividend and Retained Earnings has no impact



**Topic 249: Theory of Irrelevance of
Dividend-2**

**Dividend Policy
Module 13**

Dilution Defined

The decrease in the proportional claim on earnings and assets of a share of common stock because of the issuance of additional shares



Dilution Does Not Affect

The sum of the discounted value per share of common stock after financing plus current dividends paid is exactly equal to the market value per share of common stock before the payment of current dividends



Result of Dilution

Shareholders are indifferent between receiving dividends and having earnings retained by the firm



Homemade Dividend

- If dividends are lower sell some shares to get desired cash distribution.
- If dividends are higher purchase additional shares of stock in the company



**Topic 250: Theory of Irrelevance of
Dividend-3**

**Dividend Policy
Module 13**

Criticism

- Fractional shares cannot be sold
- Brokerage commissions
- The percentage holding reduces with the sale of shares
- Institutions with large holding will not make homemade dividend
- There are other forces of the market affecting price
- Tax rates may be different on dividends and capital gains



More Criticism

- Retained earnings have no cost of financing, whereas others have floatation cost
- Dividend send positive financial signals
- Laws and regulations
- Lack of trust on directors



Example

Example 1: Earnings are all paid as dividend

- Current position: Earnings = \$0.8 per share (all paid out as dividend); RE = 20%, the price per share. would be
- $P = 0.8/0.2 = \$4$ (the PV of constant dividends received in perpetuity).



Example

Example 2: Earnings are reinvested at the cost of equity. Half earnings retained RE = 0.2

- $P = E1 (1 - b)/(RE - bR)$
- $P = 0.8(1 - 0.5)/(0.2 - 0.5 \times 0.2) = \4

No change in the share value, and so the dividends are irrelevant.

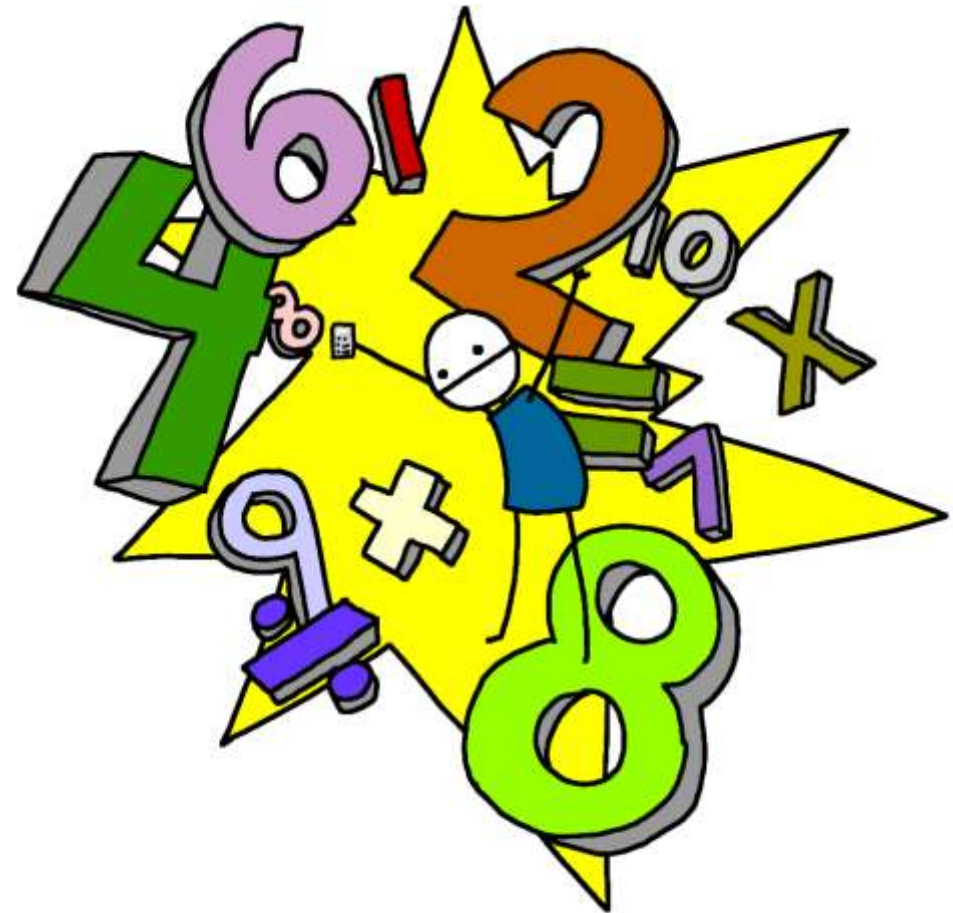


- Example 3: earnings are reinvested at more than the cost of equity
- $RE = 0.2$ (as before) and R (Cost of Capital) $= 0.3$
- $P = 0.8(1 - 0.5)/(0.2 - 0.5 \times 0.3) = \8
- In this case, the share price rises because the extra earnings retained have been invested in a particularly valuable way

EXAMPLE

Examples

- Example 4: earnings are reinvested at less than the cost of equity
- $RE = 0.2$ (as before) and $R = 0.1$
- $P = 0.8(1 - 0.5)/(0.2 - 0.5 \times 0.1) = \2.67



Topic 251: Miller and Modigliani's Argument

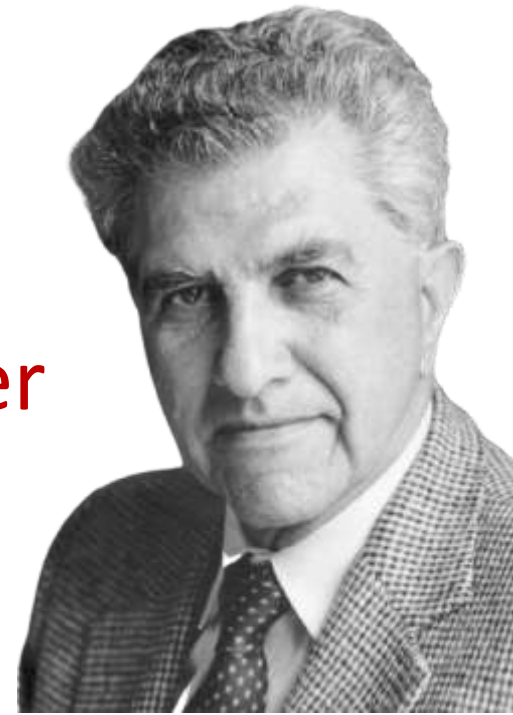
**Financial
Management
Module-251**

The M&M Dividend Policy

- The crux of M&M's position is that the effect of dividend payments on shareholder wealth is exactly offset by other means of financing
- The assumptions are certainty and perfect capital markets



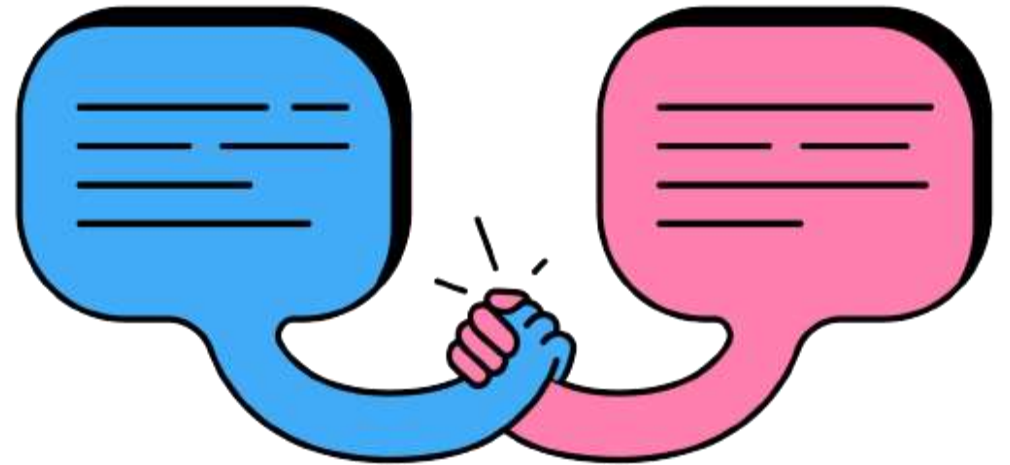
Franco Modigliani



Merton Miller

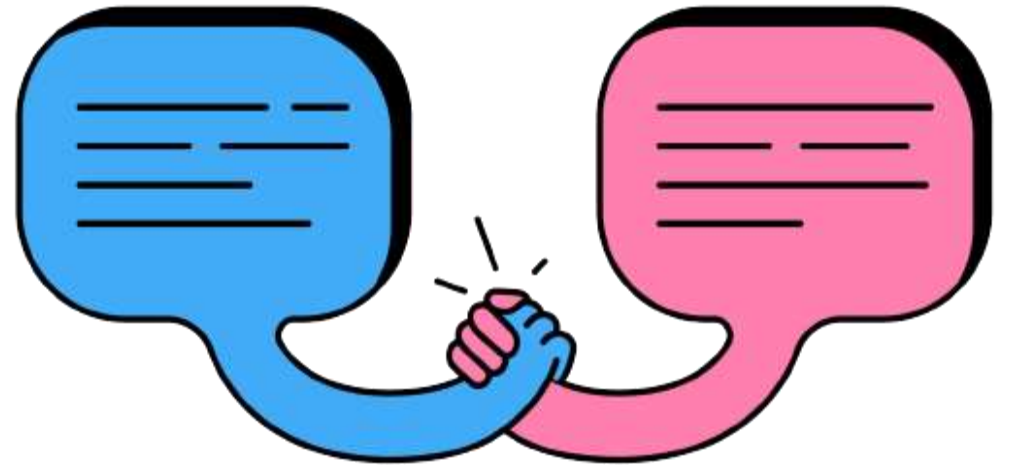
The M&M Argument for Shareholder

- The retention of profit does not affect shareholders as they can make home made dividend



The M&M Argument for the Firm

- The firm is not affected by the payment of dividend as it can be offset by the external financing.



**Topic 252: Issues of Dividend
Consideration-1**

**Dividend Policy
Module 13**

Conservation of Value

- Firm cannot create value by altering the dividend payout
- Investors are able to replicate any dividend stream
- **Lower Dividend:** sell some shares and get desired cash
- **Higher Dividend:** buy additional shares of the company
- Thus they can make “homemade” dividends



Issues of Dividend Consideration-2

**Financial
Management
Module-253**

Tax Consideration

- Personal tax rate on capital gains is less than that on dividend income (better not to get dividend)
- The capital gains tax is deferred until the actual sale of stock (better not to get dividend)
- Paying Capital Gain tax and not Tax on Dividend gives timing option
- Corporate and individual investors have different tax rates
- No impact on tax exempted individuals



Topic 253: Issues of Dividend Consideration-2

**Dividend Policy
Module 13**

Floatation Cost



The costs associated with issuing securities, such as underwriting, legal, listing, and printing fees.

Floatation Cost: Formula

$$DGR = [D/P(1-F)] + g$$

Where:

- D = Dividend
- P = Issue price of share
- F = Floatation cost to Issue price ratio
- g = growth rate



Floatation Cost: Formula

$$\text{Cost of New Equity} = [D/P(1-F)] + g$$

$$\begin{aligned} 1. &= [1/10 \times (1 - 7\%)] + 10\% \\ &= 20.7\% \end{aligned}$$

$$\begin{aligned} 2. &= [1/10 \times (1 - 0\%)] + 10\% \\ &= 20\% \end{aligned}$$



**Topic 254: Issues of Dividend
Consideration-3**

**Dividend Policy
Module 13**

Transaction Cost and Share Divisibility

- Brokerage fee reduces the return/cash
- Brokerage is paid differently on Small and Large value stocks
- Shares are generally sold in Round Lots
- Shares are not divisible



**Topic 255: Issues of Dividend
Consideration-4**

**Dividend Policy
Module 13**

Institutional Restrictions

- Companies which pay regular dividend
- Investments from Endowments are restricted for Capital Gains
- Restrictions on Liquidation of Capital



Topic 256: Issues of Dividend Consideration-5

**Dividend Policy
Module 13**

Financial Signaling

- Financial signaling is based on market imperfections
- Dividends send signals about a firm's profitability
- Dividends are better than announcements of profit/growth
- Consistent payout signals: better than what it looks
- Share price responds to signals

